

Gas Prices and the Macroeconomy ^{*}

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Abstract

We identify supply and demand shocks to the real price of natural gas leveraging exogenous temperature variation and high-frequency price movements around gas-supply events. Gas demand is less elastic than supply, leading to larger price fluctuations following supply shocks. These shocks have notable macroeconomic effects: strongly inflationary in the Euro Area, amplified by inventory dynamics and financial volatility, suggesting a transmission channel driven by expectations and uncertainty. They were a major driver of the post-pandemic inflation surge, though effects were more muted in the U.S. Aggregate real effects appear short-lived, though we document notable sectoral heterogeneity and a substantial decline in German activity.

JEL classification: C32, E31, Q41, Q43.

Keywords: Business cycles, gas price shocks, gas supply, gas demand, elasticities, external instruments, temperature deviations, inflation.

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1 Introduction

Russia’s invasion of Ukraine in 2022 drove natural gas prices to unprecedented levels, especially in Europe, as the surge in gas and broader energy prices coincided with a global rise in inflation (see Figure 1). More recently, major disruptions to gas flows through the Strait of Hormuz revived concerns about future gas availability and supply security (IEA, 2026). These episodes raise a central question: how costly are natural gas shocks, and through which channels do they transmit to the broader economy? Answering this question requires understanding how quickly economies adjust to supply disruptions through changes in consumption, alternative gas sourcing, or substitution toward other energy sources (Moll et al., 2023); whether gas prices primarily reflect fundamental supply and demand forces or other market factors, such as speculative activity (Knittel & Pindyck, 2016); and how much gas price shocks contributed to the inflation surge relative to other drivers. Addressing these issues is challenging, as energy prices are endogenous and respond to economic conditions, while the assessment of macroeconomic effects is further complicated by concurrent macroeconomic, financial, and geopolitical shocks. Nevertheless, this assessment is crucial for understanding the macroeconomic transmission of energy price shocks and their role in shaping inflation and output dynamics.

This paper develops a new strategy to identify structural supply and demand shocks in natural gas markets. We identify demand shocks using exogenous variation in temperatures, and supply shocks using high-frequency movements in gas prices around gas-market-relevant events. Employing the resulting series as instruments in a Bayesian VAR framework, we analyze gas-market dynamics and their broader macroeconomic transmission in the Euro Area and the United States. This yields responses of prices and quantities that are consistent with, but not constrained by, standard theoretical predictions. For the Euro Area, where gas supply is largely imported and exposed to geopolitical risk, the identification strategy further separates realized flow-supply shocks from supply-security news shocks, allowing us to distinguish current disruptions from news about expected future availability. To the best of our knowledge, we provide the first macro-level estimates of gas demand and supply elasticities based on instrumented variation. Building on these estimates, we trace the transmission of gas price shocks to the broader economy and quantify their contribution to the inflation surge. Many of the cross-region differences are consistent with disparities in gas balances and supply structures, but the results also point to distinct propagation mechanisms. In particular, Euro Area supply-security news shocks are amplified through inventories and financial-market volatility, revealing a transmission channel driven by expectations and uncertainty. We also examine the implications of gas price shocks for fuel switching and CO₂ emissions.

Preview of results. Gas price shocks have economically meaningful effects. In both regions, gas demand is less elastic than gas supply: short-run demand and supply elasticities are -0.08 and 0.32 , respectively, in the Euro Area and -0.04 and 0.23 , respectively, in the United States. This asymmetry implies that supply shocks account for a larger share of gas-price fluctuations.

In the Euro Area, flow-supply and supply-security news shocks transmit differently. Flow-supply shocks generate short-lived real effects: industrial production falls on impact and recovers within a few months, while pass-through to consumer prices is modest. Supply-security news shocks are more inflationary and persistent: consumer-price pass-through rises from about 2% on impact to roughly 5% over the medium run, with only muted effects on industrial production. These news shocks are accompanied by increases in inventories and financial-market volatility, consistent with a transmission channel operating through expectations and uncertainty. Although aggregate real effects are short-lived, they vary across sectors. Energy-intensive industries are most affected, though rising costs are passed downstream over time, limiting output losses but sustaining inflationary pressures, which materialize more rapidly in goods-producing sectors than in services. In Germany, however, gas supply shocks have sizable real effects, consistent with the view that the disruption in Russian gas flows contributed to the country’s recent economic stagnation.

In the United States, by contrast, greater energy security limits the macroeconomic impact of gas supply shocks, whose pass-through to consumer prices is close to 1%. Gas demand shocks are expansionary, reflecting increased activity in the domestic energy sector. Consistent with these regional differences, we find that gas price shocks were a key driver of the inflation surge in the Euro Area, while in the United States oil price shocks played a larger role. Finally, our results have implications for the green transition. Because gas demand is inelastic, demand and flow-supply shocks raise CO₂ emissions through short-run switching toward more carbon-intensive fuels. By contrast, supply-security news shocks, which lead to persistent gas-price increases, lower emissions at longer horizons.

A comprehensive set of sensitivity analyses confirms the robustness of our results across several dimensions, including model specification, estimation method, and instrument construction. The findings hold when using informationally robust instruments that account for observable confounding information, and when relaxing key identifying or orthogonalization assumptions. The results also remain consistent when impulse responses are estimated using local projections or frequentist VARs; moreover, the instruments are not systematically correlated with standard measures of other macroeconomic and financial shocks.

Related literature and contribution. A first contribution of this study is the estimation of elasticities of gas demand and supply in both the United States and the Euro Area using instrumental variables (Lunsford, 2015; Stock & Watson, 2018). This connects with a long literature estimating oil-market elasticities and assessing the relative importance of supply and demand forces (Kilian, 2009; Kilian & Murphy, 2012; Baumeister & Hamilton, 2019; Caldara et al., 2019; Kilian, 2022; Baumeister & Hamilton, 2024b). The limited number of studies estimating gas-market elasticities has largely drawn from this literature. Some contributions rely on recursively identified VAR models (Nguyen & Okimoto, 2019; Rubaszek & Uddin, 2020), while others adapt the approach of Baumeister and Hamilton (2015, 2019), incorporating zero, sign, and magnitude restrictions while integrating prior beliefs (Wiggins & Etienne, 2017; Rubaszek et al., 2021; Casoli et al., 2024). By constructing new instruments for natural gas demand and supply, we estimate both elasticities without relying on sign or zero restrictions. The region-specific instruments also allow us to address fragmentation in natural gas markets, which implies that global elasticity estimates would mask important cross-region heterogeneity (Szafranek & Rubaszek, 2024).

This work relates to the literature on the macroeconomic transmission of commodity price shocks, which has traditionally focused on oil in the United States (Hamilton, 1983, 2003; Känzig, 2021). The literature on natural gas shocks is more recent and remains comparatively smaller. The 2022 Russian invasion of Ukraine intensified concerns over the economic consequences of the war, including through expectations and uncertainty (Gorodnichenko et al., 2025). Fears of gas supply disruptions prompted an intense policy debate and a series of theoretical studies on output losses (Deutsche Bundesbank, 2022; Auclert et al., 2023; Moll et al., 2023; Bachmann et al., 2024; Di Bella et al., 2024; Albrizio et al., 2025), followed by a smaller empirical literature focused mostly on the Euro Area and identified primarily via sign restrictions (Adolfson et al., 2024; Boeck & Zörner, 2025). Closest to our approach, Alessandri and Gazzani (2025) construct a high-frequency instrument for gas supply shocks in the Euro Area. Relative to their study, we explicitly disentangle demand forces, separate current supply disruptions from news about future availability, and extend the analysis to the United States. Our findings indicate that gas price shocks played a central role in the inflation surge in the Euro Area, whereas in the United States oil price shocks were more important. These findings also contribute to the literature on the drivers of the inflation surge (Bańbura et al., 2023; Baumeister, 2023; Giannone & Primiceri, 2024; Bernanke & Blanchard, 2025; Coibion & Gorodnichenko, 2025). Existing estimates suggest that energy price shocks have limited effects on core inflation (Gao et al., 2014; Kilian & Zhou, 2022, 2023). We show that in the United States, gas demand shocks generate the larger headline-price response, while gas supply shocks pass through only modestly; in the Euro Area, where

gas prices transmit strongly to electricity prices, supply shocks also pass through to core inflation.

The starting point of this paper is to isolate exogenous variation to construct instruments for natural gas demand and supply, which serve as the foundation for the broader empirical analysis. For the demand instrument, we build on the literature examining temperature as a key determinant of natural gas prices (Mu, 2007; Nick & Thoenes, 2014; Wang et al., 2019; Chen et al., 2023). While temperatures have been used to estimate natural-gas demand elasticities in microeconomic settings (Davis & Muehlegger, 2010; Hausman & Kellogg, 2015), to the best of our knowledge we are the first to use them to identify aggregate gas demand shocks and trace their macroeconomic effects. Similarly, a substantial body of research applies event-study techniques to assess how gas prices respond to market-relevant announcements and disruptions (Gay et al., 2009; Halova et al., 2014; Bartelet & Mulder, 2020; Goodell et al., 2023, 2024). We integrate this event-study literature with VAR analysis by constructing gas supply instruments from market-relevant supply events. Methodologically, we employ a high-frequency identification strategy developed in the monetary policy literature (e.g. Kuttner, 2001; Gertler and Karadi, 2015; Altavilla et al., 2019) and subsequently applied to other domains (Wu & Cavallo, 2012; Känzig, 2021, 2023). We also construct informationally robust versions of the instruments to account for observable confounding information (Romer & Romer, 2004; Nakamura & Steinsson, 2018; Miranda-Agrippino & Ricco, 2021).

Finally, this paper contributes to the literature on the environmental effects of energy shocks. Previous studies find that, in the short run, lower gas prices reduce emissions through coal-to-gas substitution (Knittel et al., 2015; Cullen & Mansur, 2017; Brehm, 2019), while over longer horizons higher gas prices can promote cleaner investment and reduce emissions (Stock & Zaragoza-Watkins, 2024). This tension underlies the “gas trap” discussed by Harstad and Holtsmark (2024): while gas can reduce emissions in the near term, its continued use may hinder the transition to cleaner energy sources (Acemoglu et al., 2023). Our findings speak directly to this trade-off. Because gas demand is inelastic, demand and flow-supply shocks raise emissions through short-run switching toward more carbon-intensive fuels. Supply-security news shocks, by contrast, lead to more persistent gas-price increases and lower emissions at longer horizons. These asymmetric responses imply that gas-specific price increases are not equivalent to a broad carbon price.

Layout. The remainder of the paper is structured as follows. Section 2 provides background on natural gas markets. Section 3 details the identification strategy for gas demand and supply shocks. Section 4 outlines the econometric framework and estimation approach.

Section 5 presents the main empirical findings, and Section 6 concludes. Additional material is provided in the online Appendix, which contains further details on the data, the Bayesian estimation framework, supplementary results, and sensitivity analyses.

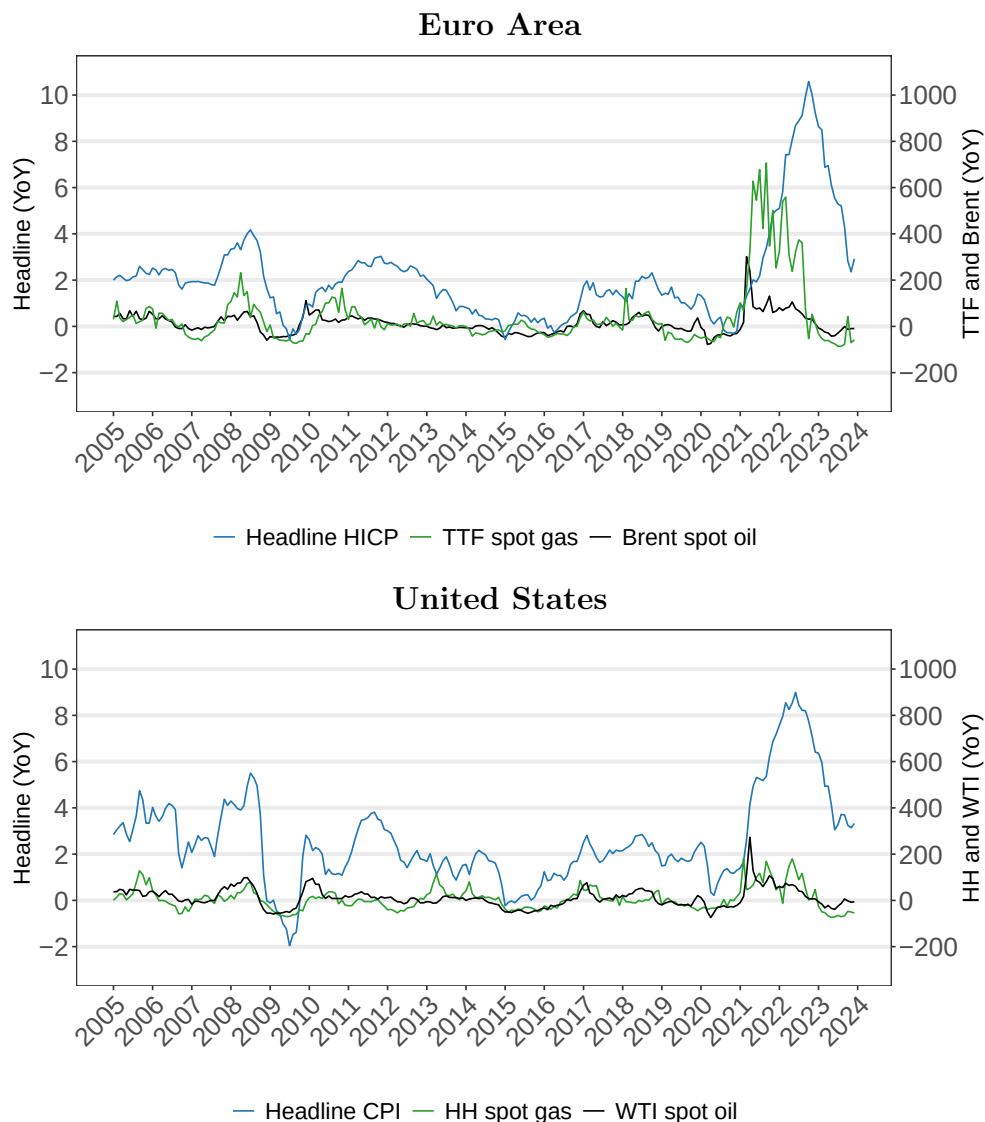


Figure 1: *Inflation and energy prices*

Notes: Year-on-year (YoY) headline inflation and YoY changes in benchmark gas and oil spot prices, 2005–2023. The benchmarks are the Dutch TTF and Brent for the Euro Area (top panel) and the NYMEX Henry Hub and WTI for the United States (bottom panel). Gas and oil spot prices are referenced to the right-hand scale.

2 Natural gas markets

This section outlines key characteristics of the natural gas markets in the Euro Area and the United States, emphasizing important regional differences. These distinctions motivate the use of a region-specific approach to analyzing natural gas shocks and provide the foundation for our identification strategy. Additional descriptive evidence is reported in online Appendix B.

Natural gas is a critical energy source, accounting for about one quarter of total energy supply in both the Euro Area and the United States. It plays a crucial role as a primary fuel in residential and commercial heating, and it also serves as an important input for industrial activities and electricity generation. In industry, natural gas is used not only as a fuel for heat and steam generation, but also as a direct production input in gas-intensive sectors such as chemicals, ceramics, paper, and basic metals. Residential consumption represents a substantial share of total natural gas demand, 35% in the Euro Area and 22% in the United States, primarily for heating purposes. Gas demand is highly sensitive to temperature fluctuations, particularly in winter (Chen et al., 2023). Consequently, anomalous weather-driven variability can be exploited as a source of exogenous variation to study the effects of gas demand shocks. Despite accounting for a sizeable share of total energy use, natural gas represents a relatively small expenditure item at the aggregate level: final gas consumption accounts for approximately 0.5% of nominal GDP in the Euro Area and 0.4% in the United States.¹

While the demand side of the natural gas market is broadly comparable across the two regions, the supply side exhibits substantial differences. European gas supply relies overwhelmingly on imports. Monthly domestic production has fallen steadily and now stands below 100 petajoules per month. As a result, import dependence has risen sharply, increasing from approximately 50% of total consumption in the early 1990s to over 90% in 2019, with imports concentrated among a limited set of major suppliers. Before the sharp reorientation of import flows after 2022, these included Russia, Norway, Algeria, and the United States (European Council, 2023). This dependence on external suppliers means that both disruptions to current gas flows and news about future availability or supply security are closely monitored by financial markets and can result in significant price fluctuations. The price fluctuations that follow market-relevant events can be leveraged to study the effects of

¹Consumption shares refer to average residential natural-gas consumption over 2004–2022 and are based on own calculations using IEA data (2024a, 2024b); see online Appendix Figure B21 for a detailed breakdown by end use. Expenditure shares are averages over 2004–2022 and are computed by valuing total final natural-gas consumption at TTF prices for the Euro Area and Henry Hub prices for the United States, divided by nominal GDP from the IMF and FRED.

gas price shocks using high-frequency identification techniques.

The United States presents a very different supply structure: it is the world’s largest natural gas producer. Domestic production has grown substantially, doubling from approximately 2000 PJ per month in the early 2000s to 4000 PJ in 2023. This remarkable expansion has been primarily driven by advancements in shale gas extraction (Hausman & Kellogg, 2015; Acemoglu et al., 2023; Stock & Zaragoza-Watkins, 2024), with the United States gradually emerging as a net exporter, particularly of LNG to European and Asian markets. While natural gas prices in the United States exhibit lower volatility than in the Euro Area, they remain sensitive to domestic production disruptions.

The global natural gas market remains regionally segmented, primarily due to supply-side constraints and limited fungibility. As a result, prices can diverge substantially across regions, despite the expansion of global LNG trade. For instance, the United States is less exposed to gas price shocks originating abroad, owing to its relative self-sufficiency and constraints on LNG export capacity, which limit the ability of domestic production to reach international markets (IMF Blog, 2023). This degree of market segmentation highlights the importance of a region-specific analysis of gas price shocks, beginning with the identification of the relevant regional price benchmarks. The United States pioneered natural gas market liberalization in the 1970s, adopting a gas-on-gas pricing model, with the Henry Hub (HH) serving as the benchmark price since 1990. In Europe, although liberalization occurred more recently, the Dutch Title Transfer Facility (TTF) has emerged as the benchmark for gas prices and is recognized as the most liquid trading hub. Online Appendix B.3 presents supporting evidence of price integration within the European gas market. Because futures contracts on these benchmarks are liquid and actively traded, intraday movements in HH and TTF futures prices provide the high-frequency variation used in our identification strategy.

3 Identification strategy

Our identification strategy is motivated by the features of the natural gas markets discussed in the previous section. The objective is to construct monthly proxies for gas demand and gas supply to serve as instruments for identifying structural gas price shocks. Demand shocks are identified by exploiting exogenous variation arising from deviations in temperature from seasonal norms. Supply shocks are identified using market-relevant news combined with high-frequency data on natural gas futures prices, distinguishing between events that affect current gas flows and, for the Euro Area, news about future availability and supply security.

3.1 Demand shocks

We identify natural gas demand shocks using exogenous variation in temperature. Specifically, we construct a proxy for heating demand based on deviations of temperature from its seasonal norm. Unusually warm weather lowers heating demand and tends to reduce natural gas prices, whereas unusually cold weather raises heating demand and prices (see, e.g., Colombo and Ferrara, 2025 for evidence on temperature effects on energy production and prices).

We construct the monthly proxy as follows. First, we build a monthly temperature series as a gas-consumption-weighted average of local temperatures, using predetermined weights based on past gas consumption.²

Second, we remove regular seasonal patterns by subtracting, for each calendar month, the average temperature for that month computed over the full sample. Equivalently, the monthly series can be interpreted as the average of daily temperature deviations from the relevant seasonal benchmark within each month. Third, we retain only unusually large deviations from these seasonal norms by setting small fluctuations to zero. This reduces noise and isolates months in which temperature movements are most likely to generate meaningful shifts in heating demand. Moreover, unlike typical seasonal temperature fluctuations, large deviations from average temperatures are less likely to be anticipated by economic agents. As a result, these deviations are less likely to be incorporated into trading decisions and affect gas prices primarily through the heating-demand channel, supporting the validity of the instrument.³ Formally, the monthly temperature index $TI_{y,m}$ is defined as

$$TI_{y,m} = T_{y,m}^{SA} \mathbf{1}\{|T_{y,m}^{SA}| > \sigma_T\}, \quad T_{y,m}^{SA} = T_{y,m} - \bar{T}_m, \quad (1)$$

where $T_{y,m}$ denotes the regional monthly temperature in year y and month m , and $\bar{T}_m$ is the sample average for calendar month m . The term σ_T denotes the standard deviation of the seasonally adjusted temperature series over the full sample. Additionally, we set the index to zero in summer months (April to September), as the relationship between temperatures and gas prices during warmer months is ambiguous: higher temperatures may

²Using lagged consumption weights helps ensure that the aggregation is not driven by contemporaneous demand conditions. Specifically, for year y , the weight for country i is based on a three-year moving average of natural gas consumption over years $y - 3$ to $y - 1$. Since gas traded at the TTF serves multiple countries, for Europe, we construct an aggregate temperature measure based on the average temperatures of Belgium, Germany, France, Luxembourg, and the Netherlands—countries that predominantly rely on TTF gas. For the United States, we construct the corresponding series as a gas-consumption-weighted average across states.

³Temperature forecasts deteriorate significantly as the forecast horizon extends, remaining relatively unreliable even with the most advanced prediction methods. See, for example, Lopez-Gomez et al. (2023). Figure C29 provides evidence that anticipation effects are likely limited in both magnitude and timing, supporting the suitability of our approach within a monthly estimation framework.

either increase electricity demand for air conditioning or reduce heating demand.⁴ Finally, we exclude months affected by the COVID-19 pandemic. In practice, this adjustment only removes a single spike in the U.S. series, in March 2020, and has no material effect on the results. Online Appendix C.1 provides additional details on the data sources and alternative constructions of the series. Figure 2 presents the resulting index for each region.

Validation of the demand channel. Positive spikes in the index tend to coincide with unexplained negative spikes in the price of gas, and conversely. More generally, the series exhibits a strong negative correlation with the real price of gas, even after controlling for relevant macroeconomic variables. This pattern is consistent with the proposed demand channel and supports the relevance of the instrument. If the primary mechanism through which temperature fluctuations affect natural gas prices is heating demand, we would expect the strongest correlations to occur during months when absolute temperatures necessitate heating. Unexpectedly cold temperatures during typically warm periods should have a minimal impact on gas prices, as absolute temperatures would remain too high to justify widespread heating use. To assess this, we examine *cooling degree days* (CDD) and *heating degree days* (HDD).⁵ When restricting the sample to months with low HDD, the correlation between the temperature series and the residual gas price is substantially weaker compared to months with high HDD. This indicates that temperature fluctuations primarily affect gas prices when they lead to the activation or deactivation of heating systems. Conversely, when the sample is limited to months with high CDD, the correlation remains lower than in months with low CDD, suggesting that cooling-related energy demand has a relatively smaller impact on natural gas prices. A similar pattern holds when the sample is restricted to either winter or summer months, reinforcing the choice to focus on temperature deviations during the winter for instrument construction. This approach reduces noise and improves the instrument’s relevance. Moreover, we construct an analogous index using heating degree days rather than temperatures. Once appropriately rescaled, this alternative series is very similar to the baseline measure, as shown in Figure 2. This is unsurprising, since HDD is directly constructed from temperature and therefore captures much the same variation in heating demand. Consistent with this, online Appendix E shows that replacing the temperature-based demand

⁴An argument can be made that this choice is more relevant in the case of the United States, where cooling demand during the summer months constitutes an important driver of electricity consumption (Sailor & Pavlova, 2003; Bartos et al., 2016). However, as shown in Section 5.3, including summer months in the demand instrument does not materially affect the results.

⁵CDD and HDD serve as proxies for the energy required to heat and cool buildings. For the precise definitions, see Eurostat (2023). The data are available at European Commission (2025). To ensure consistency with the temperature measure, we compute weighted averages of CDD and HDD using the same gas-consumption weights as in the baseline temperature index, at the country level for Europe and at the state level for the United States.

instrument with an HDD-based instrument leaves the impulse responses nearly unchanged.

3.2 Supply shocks

To identify gas supply shocks, we employ a high-frequency identification strategy, drawing on methodologies originally developed in the monetary policy literature (Cochrane & Piazzesi, 2002; Nakamura & Steinsson, 2018; Altavilla et al., 2019) and more recently adapted to crude oil and other commodity markets (Wu & Cavallo, 2012; Känzig, 2021, 2023). Specifically, we analyze changes in natural gas prices around market-relevant supply events. Our baseline instruments are based on flow-supply events, which reveal realized or near-realized changes in the physical availability of natural gas. These events provide a natural source of identifying variation: they include, for example, unplanned outages, pipeline disruptions, force majeure declarations, production shut-ins, and unexpected restorations. For these events, the identifying requirement is that the disruption or restoration is not itself driven by contemporaneous macroeconomic conditions, and that the event-window price response is not primarily confounded by other energy-market developments.

For the Euro Area, we also construct an additional instrument based on future-supply news. This reflects the import-dependent structure of European gas markets, where prices respond not only to current disruptions but also to news about transit routes, geopolitical developments, and expected future availability. These events need not affect current gas flows immediately. Instead, they reveal new information about expected gas availability and supply security, so the corresponding instrument captures gas supply-security news shocks rather than pure contemporaneous physical supply shocks. As with flow-supply events, identification requires that markets react to the relevant information within a narrow event window. This allows the resulting price surprises to capture information not yet incorporated into gas prices. The narrow window also limits reverse causality from broader economic conditions, which are typically already priced and unlikely to change discretely within the event window. For this instrument, the relevant exclusion restriction is that the price reaction is primarily driven by gas-supply information.

Unlike monetary policy announcements or OPEC meetings, gas-supply news is dispersed across producers, pipeline and LNG infrastructure operators, regulators, governments, and geopolitical developments. We therefore construct the event set from financial newswires rather than from a single announcement calendar. Specifically, we collect gas supply-related news via Factiva and carefully assess each event.⁶

⁶The Factiva search uses the keywords “Natural Gas” and “LNG” in Reuters Newswires, Dow Jones Newswires, and S&P Global Insight Daily Analysis. Searches are geographically restricted to Europe and North America, respectively. To capture region-specific institutional news, we additionally include

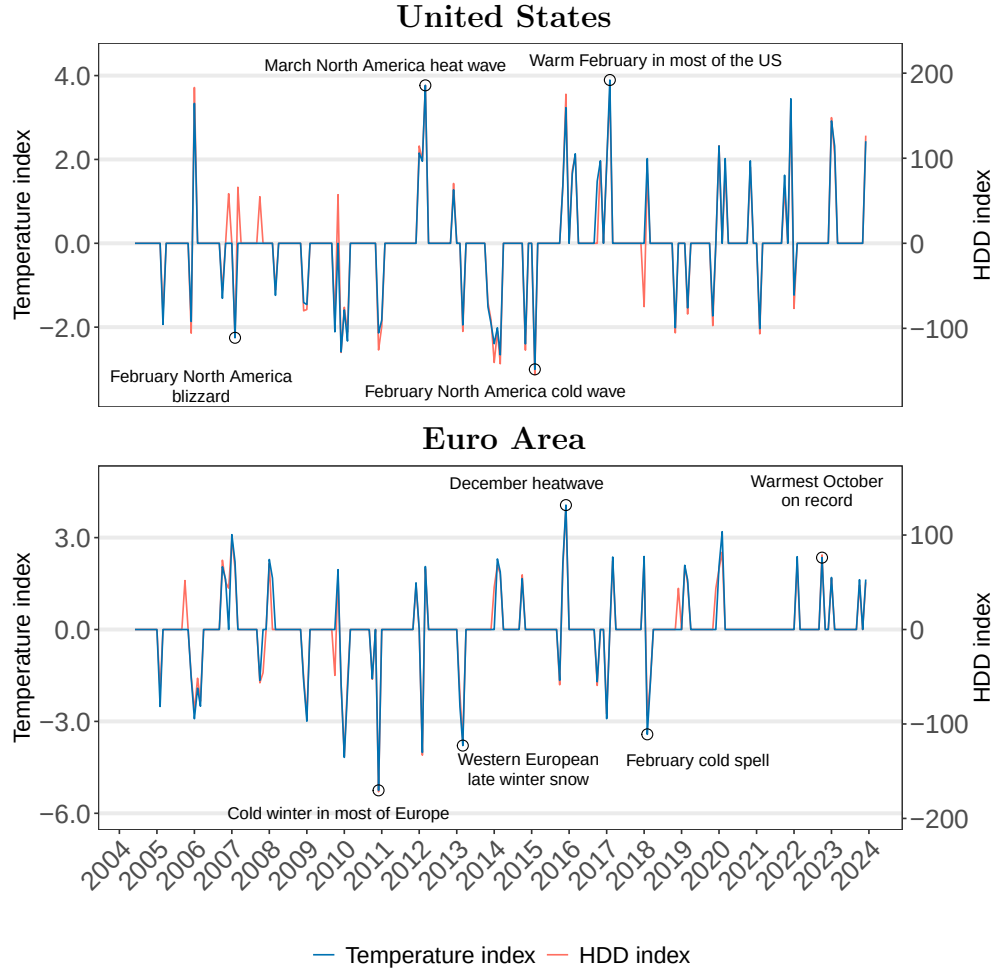


Figure 2: *Temperature proxy series for the United States (top panel) and Europe (bottom panel). In each panel, the red line denotes the baseline temperature index, while the blue line denotes the analogous index constructed from heating degree days.*

Notes: This figure displays the temperature index, constructed as a proxy for gas demand, for the United States and Europe. Circles highlight significant temperature-related events in the gas market. For the United States, these are: 2007M2, a winter storm affecting much of eastern North America; 2012M3, record-breaking heat across many U.S. states; 2015M2, a severe cold wave particularly impacting the eastern United States; and 2017M2, a late-winter heat wave across the central and eastern United States. For Europe, these are: 2010M12, the coldest December in 100 years and the coldest winter month since February 1986; 2013M3, a late-season snow event that impacted Western Europe; 2015M12, the warmest December on record for several countries; 2018M2, the cold spell commonly referred to as “The Beast from the East,” characterized by cold winds and low temperatures; and 2022M10, the warmest October on record in Europe.

Event selection and classification. The sample covers the 20-year period from January 1, 2004 to December 31, 2023, starting with the first full year after the establishment of the TTF and the first year for which daily European gas-price series are consistently available. We use Henry Hub and TTF spot and front-month prices from Datastream to construct daily percentage price changes. These maturities are used to screen for candidate event days because they are liquid and highly responsive to news about current or near-term gas-market conditions. We exclude days on which the relevant price changes cannot be computed because of missing observations.⁷ To construct a clean set of candidate event days, we first apply mechanical ex ante filters that remove dates likely to be confounded by forces unrelated to gas supply. We exclude months in which the temperature instrument records demand-driven movements, OPEC and monetary-policy announcement days, and March–May 2020, the onset of the COVID-19 pandemic. For the United States, we also exclude days with large same-direction movements in oil and gas prices, since such co-movements may reflect shocks common to domestic oil and gas production.⁸

We then retain only dates on which the absolute gas-price surprise exceeds 2% in either the spot or front-month contract.⁹ This conservative threshold removes background price noise while avoiding the premature exclusion of potentially relevant supply news. For the remaining dates, we use the Factiva search described above to identify those with natural-gas-related news. Dates without any such news are discarded, leaving 2,590 U.S. dates and 2,434 Euro Area dates for manual review. We then carefully read the corresponding articles one by one to identify clean gas-supply events and classify the associated news.

In the manual review, we retain flow-supply events such as explosions, unplanned outages, force majeure declarations, and disruptions to pipeline or LNG infrastructure.¹⁰ We exclude anticipated maintenance and voluntary production adjustments, which are more likely to be already priced or endogenous to contemporaneous market conditions. For Euro Area future-supply news, we retain gas-specific sanctions, transit disputes, and import-infrastructure decisions, while excluding broad geopolitical or macroeconomic events whose effects extend well beyond natural gas. We also assess same-day confounders beyond the mechanical fil-

“Gazprom” for the Euro Area and “FERC” for the United States.

⁷When only one of the two maturities is missing, we retain the day and set the corresponding price change to zero; this affects 45 U.S. dates and 436 Euro Area dates.

⁸We define these as days on which the WTI spot or front-month price change exceeds one standard deviation and oil and gas prices move in the same direction.

⁹Section 5.3 shows that the results are robust to stricter 3% and 4% thresholds, with the latter being close to a one-standard-deviation daily movement in the post-filtering sample and therefore providing a considerably more conservative event screen.

¹⁰Weather events are generally excluded because they may combine gas-demand effects with direct macroeconomic disruption. We retain only U.S. offshore storm events that do not make landfall, where the relevant news concerns offshore production shut-ins rather than direct onshore economic disruption.

ters. Minor unrelated news does not disqualify a date if the gas-supply event is clearly the dominant driver of the price movement, but we exclude otherwise valid supply events when demand-side developments or other energy-market shocks appear to dominate, or materially contribute to, the gas-price movement.

As an additional check, we audit the retained events with a large-language-model that re-applies the selection criteria to the underlying news articles and verifies each event’s classification and dating; flagged cases are re-examined manually. Additional details on the event-selection procedure, illustrative examples of retained and excluded candidates, and the audit are reported in online Appendix [C.2](#).

Events characterization. The above procedure identifies 25 clean events for the United States and 63 for the Euro Area. The Euro Area set comprises 28 realized or near-term flow disruptions and 35 broader supply-security news events about expected gas availability. The full list of retained events is reported in Table [1](#). In the United States, retained events mostly reflect disruptions to domestic production and infrastructure, consistent with the market’s reliance on domestic supply. These include pipeline and transmission disruptions, facility outages, storage events, Gulf of Mexico production shutdowns, and regulatory or restoration events. Some events, such as LNG export-terminal disruptions, are classified according to their effect on gas available to the domestic market. For the Euro Area, flow events are instead centered on import-relevant supply disruptions, including outages at gas fields, processing plants, pipelines, and import terminals, as well as unexpected changes in Russian and Norwegian flows. Future-supply news includes sanctions, transit and payment disputes, LNG contracts and cargo competition, infrastructure and storage decisions, reverse-flow capacity, and supply-diversification agreements relevant for future gas availability. We do not identify a separate category of future-supply news for the United States, because the relevant events in our sample are overwhelmingly realized domestic supply disruptions, and announcements with comparable forward-looking content are both less frequent and harder to distinguish cleanly from broader energy-market news.

Figure [3](#) illustrates the two types of events used in the supply instruments. The left panel shows a U.S. flow-supply event: the October 12, 2017 Texas Eastern Transmission force majeure, which sharply restricted Appalachian gas flows and was followed by a clear increase in Henry Hub spot and front-month prices over the daily event window. The right panel shows a Euro Area supply-security news event: the March 23, 2022 announcement that Russia would require payments for gas in rubles. This announcement did not immediately curtail physical flows, but it sharply repriced longer-dated TTF futures, consistent with a revision in expected gas availability and supply security. A Euro Area flow-supply event is

Date	Event	% ΔP
Panel A: United States — flow-supply events		
2005-08-16	Operational anomalies on the Amarillo Mainline force unplanned maintenance and reduce pipeline capacity.	2.10
2006-03-23	El Paso unscheduled outage on the Breton Island pipeline due to a compressor problem.	3.51
2006-07-14	Unscheduled maintenance at the Clarence gas compressor station in Louisiana cuts pipeline capacity by 100 MMcf/d.	6.41
2006-12-01	Kinder Morgan unexpectedly lifted force majeure on NGPL Segment 25 one day after announcing it.	-1.21
2007-01-09	Florida Gas Transmission halves Station 6 throughput for unscheduled Texas pipeline examination.	3.50
2007-01-30	Pipeline degradation forces Williams Transco to lower pressure in New Jersey, cutting available gas-transport capacity at Linden.	4.71
2007-09-17	Cheyenne Plains compressor outage triggers force majeure and restricts Rocky Mountain gas flows.	4.55
2008-01-24	Tennessee Gas Pipeline declares force majeure after offshore Louisiana leak, disrupting Gulf flows.	1.16
2008-04-09	Leak on the Independence Trail pipeline shuts the connected Independence Hub, halting about 850–900 MMcf/d.	2.47
2009-08-05	Enterprise HIOS platform fire shuts Gulf gas mainline, cutting ANR receipts by about 227 MMcf/d.	2.30
2009-08-13	Enterprise resumes partial HIOS service sooner by bypassing the fire-damaged compressor platform, despite ongoing force majeure.	-2.37
2010-06-07	Enterprise's 36-inch North Texas gas pipeline ruptures and explodes, curtailing about 250 MMcf/d of flows.	2.34
2010-07-01	Precautionary Gulf gas shut-ins ahead of Hurricane Alex keep 14% of production offline, though storm did not strike U.S. facilities.	2.44
2011-08-31	Fluid buildup disrupts gas flows in the offshore Gulf, leading BP to declare service disruptions on the Destin pipeline.	4.15
2013-02-16	Drilling kick and underground gas leak force Apache Corporation to shut in a Gulf of Mexico gas well.	2.69
2013-10-05	Tropical Storm Karen precautionary shut-ins keep 1.4 Bcf/d of Gulf gas offline, though the storm never made landfall.	2.69
2015-10-26	Leak at Aliso Canyon storage facility forces pressure-reducing withdrawals, increasing gas available to the market.	-8.31
2016-04-14	Tennessee Gas Pipeline lifts force majeure at HPL/Sabine, restoring receipt capacity after uncertainty from Sabine River flooding.	-4.81
2017-10-12	Texas Eastern force majeure after soil movement cuts southbound Appalachian gas flows by nearly 1.8 Bcf/d.	3.21
2020-08-18	EPNG declares force majeure after a gas release and fire near Line 3125, cutting Midland-area capacity to zero.	5.95
2021-07-30	Texas Eastern announces earlier-than-expected line restoration, returning 0.5 Bcf/d ahead of the September schedule.	-3.49
2022-06-08	Fire at the Freeport LNG terminal halts operations, leaving more gas in the U.S. market.	-1.65
2022-10-31	Cove Point LNG resumes near-full operations earlier than expected after month-long outage.	5.01
2023-07-27	Fire on TC Energy's Columbia Gas pipeline in Virginia cuts deliveries to Cove Point LNG, leaving more gas in the U.S. market.	-7.22
2023-08-19	Tennessee Gas Pipeline declares force majeure after an equipment-failure explosion and fire at the Centerville compressor station.	6.62
Panel B: Euro Area — flow-supply events		
2004-08-13	Faulty compressor prolongs Britannia outage, keeping North Sea output below normal.	1.82
2004-11-28	A gas leak shuts Marathon's Brae field in the U.K. North Sea, cutting about 400 mmcf/d of gas supply.	4.96
2007-07-02	CATS pipeline shut unexpectedly after condensate contamination forces emergency restrictions, blocking U.K. J-Block gas flows.	16.66
2007-07-20	Elgin-Franklin generator fault briefly halts output while unresolved CATS repairs keep North Sea gas flows offline.	5.36
2007-11-30	StatoilHydro shuts Norway's Ormen Lange gas field after a reported gas leak.	6.94
2008-08-20	Leak on the Kvittebjørn-Kollsnes pipeline shuts Kvittebjørn and cuts nearby Visund output.	9.88
2008-08-22	Technical problems shut Norway's Snoehvit LNG plant, with restart timing unclear.	5.15
2009-04-01	Moldovan transit pipeline blast cuts gas flows to Bulgaria by about 60% and reduces supplies across the wider Balkans.	8.09
2009-05-23	Kollsnes resumes gas processing sooner than expected after a condensate leak, restoring Norwegian gas flows.	-5.28
2009-07-06	Ormen Lange returns after Nyhamna maintenance, with Shell having withheld restart timing and volume-impact details.	-4.27
2010-04-14	Unexpected outages at Norway's Kollsnes and Kaarstoe plants cut gas flows to the U.K. and Northwest Europe.	3.85
2010-04-26	Unplanned Elgin-Franklin field outage cuts North Sea gas supply to Northwest Europe.	6.82
2010-06-08	Technical issue shuts Total's Elgin field overnight, cutting SEAL flows into Bacton by about 15 mcm/day.	5.07
2010-06-29	Gas leak shuts Troll A and cuts Kvittebjørn/Visund output, reducing Norwegian flows to Europe.	3.62
2011-10-12	Statoil shuts Snoehvit LNG plant for technical repairs, reducing near-term Norwegian LNG supply.	1.67
2013-12-03	Unexpected Ormen Lange outage cuts Langede flows to Britain, tightening Northwest European gas supply.	2.35
2014-07-25	Unplanned outage at Centrica's Morecambe field reduces North Sea gas flows.	7.38
2014-08-29	Gassco extends Troll maintenance by one week, prolonging a 46 mcm/day cut in Norwegian gas output.	4.37
2014-11-01	Kollsnes corrective maintenance and Skarv process problems cut Norwegian gas capacity to Britain and Northwest Europe.	8.56
2016-09-19	UK and Norwegian production outages tighten gas supply, with flows down at St Fergus, Theddlethorpe, and Easington Dimlington.	10.76
2018-11-07	Unplanned Kollsnes outage cuts Norwegian gas processing capacity for several days.	4.47
2019-04-09	Gassco extends the previous day's Aasta Hansteen outage from days to weeks, cutting Norwegian gas output by about 15.9 mcm/d.	7.35
2021-07-26	Unplanned Troll, Visund, Kvittebjørn and Kollsnes outages cut over 65 mcm/day, reducing gas flows to Britain and Europe.	4.64
2021-07-28	Unplanned Troll and Cygnus outages plus slower-than-expected Bacton SEAL ramp-up reduce gas supply through UK.	5.90
2021-08-05	Fire at Russia's Novy Urengoy plant cuts Yamal-route flows into Germany.	2.34
2021-11-15	Troll compressor failure cuts 34 mcm/day of Norwegian gas capacity to Europe.	8.40
2022-06-14	Nord Stream 1 export capacity is cut from 160 to 96 mcm/day due to delayed compressor-station repairs.	14.37
2022-08-22	Unplanned outages at Norway's Troll field and U.K. Continental Shelf maintenance.	10.79
Panel C: Euro Area — supply-security news events		
2006-09-29	Gazprom threatens Belarus with a fourfold gas price hike in a dispute over pipeline control and European transit.	4.42
2007-03-02	Gazprom and Fluxys advance a large Belgian gas-storage project, with drilling planned for the second half of 2007.	-3.74
2007-03-20	Azerbaijan seeks U.S.-EU backing for Caspian-to-Europe gas routes, strengthening future non-Russian Southern Corridor supply.	-1.55
2007-05-03	Pungla threatens BG's 8 bcm/year Brindisi LNG permit, putting future import capacity at risk.	2.68
2007-05-12	Russia, Turkmenistan, and Kazakhstan agree on Caspian gas pipeline via Russia, weakening trans-Caspian supply options to Europe.	6.74
2008-05-27	EU Parliament committee opposes Nord Stream and seeks a new environmental study, raising permitting risk for the Baltic pipeline.	2.10
2010-07-20	Eni advances talks to route 2-3 bcm/year of Turkmen gas via Azerbaijan to Turkey, supporting future Caspian supply diversification.	-3.47
2011-03-14	Japan's post-earthquake nuclear shutdowns raise the risk of LNG cargoes being diverted from Europe to Japan.	8.08
2011-08-29	Qatar gas announces autumn maintenance on multiple LNG trains, raising the risk of lower deliveries to Europe.	8.75
2014-04-07	Ukraine misses Gazprom debt deadline, raising the risk of Russian gas disruption via the Ukraine transit route.	5.47
2014-05-02	Russia threatens June gas cuts to Ukraine unless prepayment arrives by May 31, raising risks to Europe-bound transit.	2.11
2014-06-13	Ukraine prepares for Russian gas cuts after talks collapse, raising risks to EU transit supplies.	3.71
2014-12-15	Eustream tests higher Slovakia-Ukraine reverse-flow capacity, lowering risk of Ukraine transit disruption to Europe.	-2.75
2014-12-29	Hungary announces it will resume reverse-flow gas exports to Ukraine, easing supply-security concerns after months of suspension.	-4.12
2015-02-25	Putin threatens to halt gas deliveries to Ukraine over prepayment, threatening gas transit to Europe.	3.30
2015-05-10	Slovak prime minister says Russia plans to end Ukraine gas transit, raising Central European supply-security risk.	3.73
2016-12-03	Five binding offers are submitted for the 4.3 bcm Greece-Bulgaria gas link.	-4.19
2020-07-10	Denmark gives unexpectedly swift approval for Nord Stream 2 work to resume in Danish waters.	-2.29
2021-07-12	Gazprom says Nord Stream 2 could be completed by end-August, a month earlier than expected.	-5.12
2021-10-25	Algeria plans to shut the 13.5 bcm Maghreb-Europe pipeline via Morocco, raising risks over Spain's gas supply.	4.01
2021-10-27	Putin orders Gazprom to refill its depleted European storage sites from Nov. 8 after Russia's domestic injection campaign.	-3.15
2021-11-08	Promised Russian flows to refill Gazprom's European storage fail to appear, tightening supply expectations.	4.23
2022-03-21	Germany secures Qatar gas partnership as Berlin fast-tracks LNG terminals to cut Russian dependence.	-5.81
2022-03-23	Gazprom says "unfriendly" countries must pay for gas in rubles.	18.90
2022-05-06	Uniper advances Germany's first LNG terminal at Wilhelmshaven, while German utilities charter FSRUs to expand LNG imports.	-3.33
2022-05-11	Russia sanctions Gazprom Germania and EuRoPol GAZ, raising supply risks for Germany and the Yamal-Europe corridor.	1.87
2022-05-30	Gazprom is set to cut GasTerra over the ruble-payment dispute, while Ørsted faces similar risk after refusing the new terms.	2.51
2022-06-23	Germany triggers its gas "alarm level" after Nord Stream cuts, warning of winter shortages if Russian flows do not recover.	6.47
2022-06-29	Reports of National Grid emergency plans to shut U.K. gas links with Belgium and the Netherlands raise crisis supply-risk concerns.	4.78
2022-07-04	Gazprom seeks to extend the ruble-payment scheme from pipeline gas to LNG.	13.99
2022-09-28	Gazprom says its legal dispute with Naftogaz could halt remaining gas flows to Europe.	5.61
2022-11-22	Gazprom threatens to curb the last major Russian gas route to Europe via Ukraine over Moldova transit dispute.	6.34
2023-08-09	Australian LNG strike votes raise fears of tighter winter cargo competition for Europe and Asia.	11.39
2023-09-08	Strike action at two Australian LNG facilities, further raising concerns over stronger competition for Europe-bound LNG cargoes.	2.43
2023-11-25	Turkey and Russia agree a roadmap for a 2024 Turkish gas hub to price and route gas toward Southeast Europe.	-5.52

Table 1: *Natural gas supply events.*

Notes: Panel A reports U.S. flow events, Panel B Euro Area flow events, and Panel C Euro Area supply-security news events. The last column reports the corresponding instrument surprise, measured as the first principal component of event-day gas-price changes. Flow-event surprises use spot and front-month prices; news-event surprises use futures over the first year of the term structure, starting from the two-month contract.

shown in online Appendix Figure C32.

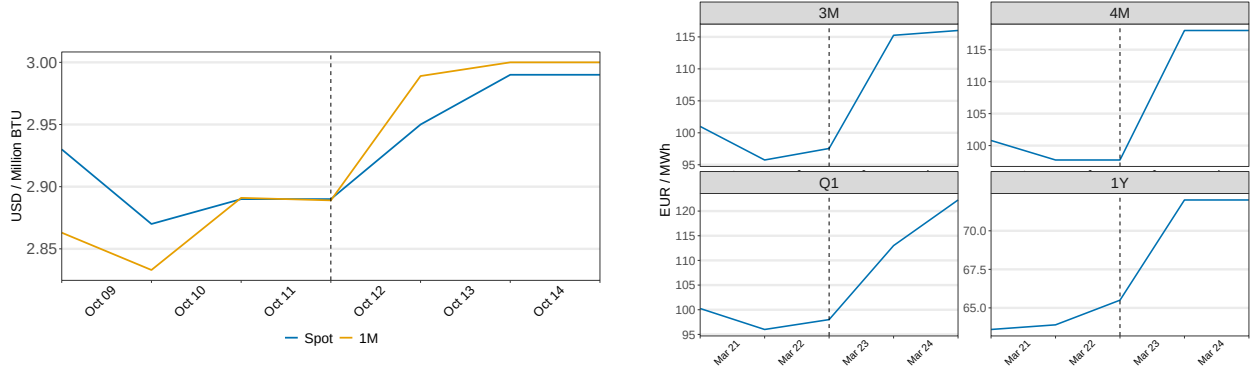


Figure 3: *Illustrative gas supply events in the United States and the Euro Area*

Notes: The left panel shows Henry Hub spot and front-month prices around the October 12, 2017 Texas Eastern Transmission force majeure, a U.S. flow-supply event. The right panel shows longer-dated TTF futures prices around the March 23, 2022 ruble-payment announcement, a Euro Area supply-security news event. The dashed vertical line marks the start of the daily event window; the event-window surprise is the price change from the previous close to the event-day close. The construction of the corresponding instruments is described below.

Construction of gas supply instruments. Using the retained gas-supply events, we construct event-day price surprises as the log change in natural gas prices from the previous trading-day close to the event-day close. For each event date d and contract maturity h , the surprise is

$$\Delta Gas_d^h = P_d^h - P_{d-1}^h, \quad (2)$$

where P_d^h denotes the log price of the gas contract with maturity h on date d , with $h = 0$ corresponding to the spot price.

Following Känzig (2021), we opt for a daily window. Unlike in other markets, natural gas lacks a centralized news source with regularly scheduled press releases that market participants closely follow. As a result, market participants may need more time to identify and process relevant information. An intraday window may fail to capture the full market reaction, while a multi-day window risks introducing unnecessary noise from unrelated market developments. A daily window thus balances the trade-off between capturing the entire response to the event and background noise.

We construct flow-supply instruments from spot and front-month prices, because these maturities are most sensitive to disruptions in current or near-term physical availability and align with the monthly frequency of the empirical analysis. For Euro Area future-supply

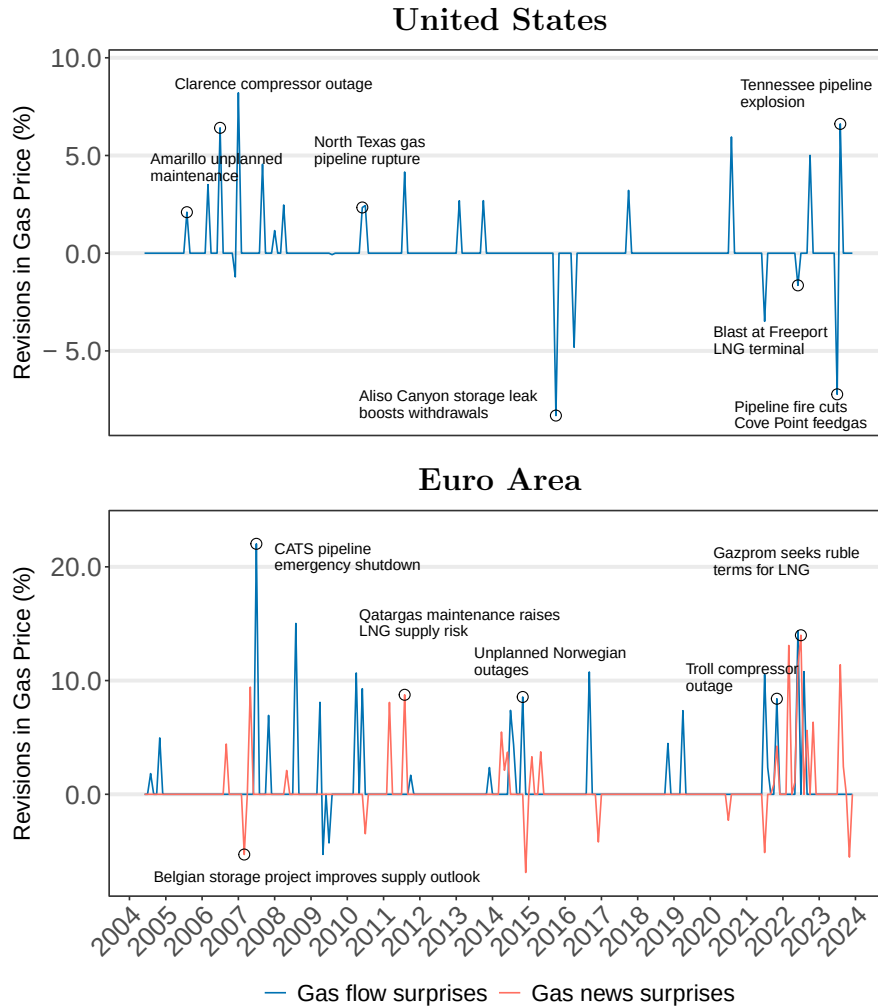


Figure 4: *Gas supply surprise series for the United States (top panel) and the Euro Area (bottom panel). In the Euro Area panel, the blue line denotes gas flow surprises, while the red line denotes gas news surprises.*

Notes: This figure shows the gas supply surprise series constructed as the first principal component of high-frequency changes in natural gas prices. The principal component is scaled to match the average volatility of the underlying price surprises so that the y-axis can be interpreted as percentage deviations in natural gas prices. Red circles highlight major gas supply events.

news, we instead use longer-dated TTF futures, up to one year ahead, since these events shift expectations about future gas availability rather than contemporaneous flows. For each instrument, we extract the common component across maturities using the first principal component of the corresponding event-day surprises. The first component explains 77% of the variation in the price changes used to construct the U.S. flow-supply instrument, 89% for the Euro Area flow-supply instrument, and 58% for the Euro Area future-supply-news instrument. These shares indicate that the retained events generate a broad repricing across the relevant maturities rather than isolated movements in individual contracts. Finally, we aggregate the resulting daily surprises to the monthly frequency by summing them within each calendar month. In months with no relevant events, the corresponding entry is set to zero. Figure 4 displays the resulting monthly series.¹¹

3.3 Diagnostics of the instrument series

In the following, we present diagnostic checks on the validity of the instrument series. These include a placebo exercise to assess the amount of noise in the supply series, correlation tests with other shocks, an informationally robust specification to address potential confounding factors, and standard tests for autocorrelation and forecastability.

Background noise. Even with a carefully chosen event window, concerns may remain that events unrelated to gas supply could affect gas prices within the same window. To evaluate the magnitude of such background noise in the surprise series, we compare daily changes in gas prices on gas-related event days with those observed on a set of control days. Control days are randomly selected from days with no gas supply events. As Figure 5 shows, price changes on event days are markedly larger than those in the control sample. The empirical CDFs of absolute price changes are shifted to the right on event days, indicating that gas-supply events are more likely to generate large price movements. Taken together, these results indicate that background noise is limited.

Informationally robust instruments. The placebo exercise above suggests that the retained event days contain a distinct gas-market signal. The key remaining concern is whether the associated price movements also convey information about other shocks that affect macroeconomic outcomes through channels unrelated to natural gas. This concern is

¹¹For the sectoral analysis in Section 5.4, we also construct a combined Euro Area supply instrument that pools the retained flow-supply and supply-news events. The event-specific maturities are the same as in the separate instruments: spot and front-month TTF prices for flow-supply events, and longer-dated TTF futures up to one year ahead for supply-news events.

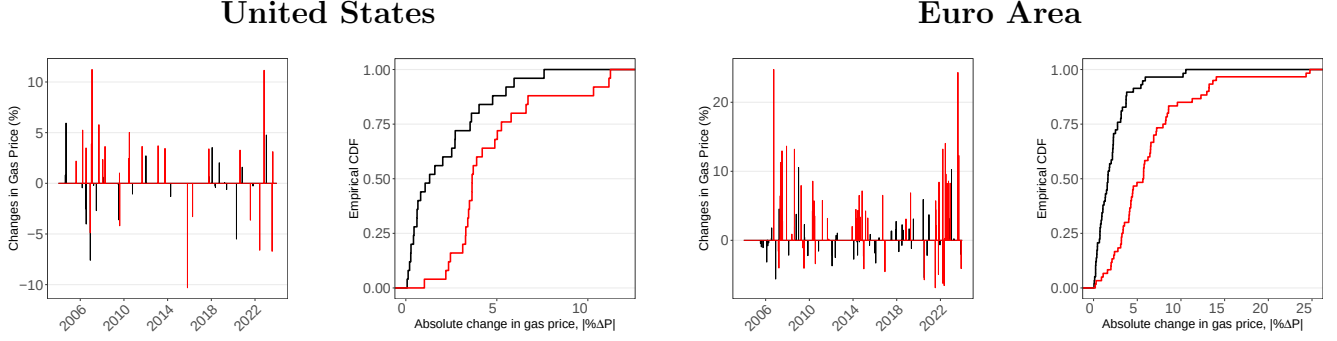


Figure 5: *Price changes on supply event days (red) and control days (black).*

Notes: Left panels within each region: daily changes in the one-month gas futures price on news and control days. Right panels within each region: empirical cumulative distribution functions of absolute percentage daily price changes. A one-sided Kolmogorov–Smirnov test rejects equality of the distributions in favor of larger news-day absolute price changes in both the Euro Area ($D = 0.69$, $p < 0.001$) and the United States ($D = 0.68$, $p < 0.001$). A permutation test based on mean absolute price changes with 10,000 draws also rejects the null that news-day price movements are no larger than control-day movements in both regions ($p < 0.001$).

analogous to the information-effects problem emphasized by Nakamura and Steinsson (2018) in the monetary policy context. We therefore first examine whether the instrument series are systematically related to existing measures of other structural shocks. As reported in Table C6 in the online Appendix, the instruments display little systematic comovement with a broad set of external shock measures.¹²

To go beyond simple correlations, we construct robust versions of the instruments by residualizing them with respect to observable confounders. For the supply-side instruments, the first concern is that event-window gas-price surprises may also reflect contemporaneous movements in related commodity markets. This is particularly relevant for food prices, which rose alongside natural gas prices during the Russia–Ukraine conflict (FAO, 2022; World Bank, 2022), and crude oil prices, which are closely linked to natural gas and have broader macroeconomic effects. More generally, we want to ensure that the gas-market events underlying the surprise series do not convey information about broader shocks that affect the economy through channels other than current or expected natural gas availability.

¹²The table reports correlations with oil supply, aggregate demand, and oil-specific demand shocks from Kilian (2009); oil supply and oil demand shocks from Baumeister and Hamilton (2019); oil supply expectations from Känzig (2021); the carbon-policy shock from Känzig (2023); monetary policy surprises for the Euro Area and the United States from Altavilla et al. (2019) and Bauer and Swanson (2023), respectively; financial uncertainty and credit-spread indicators from Bloom (2009) and Gilchrist and Zakrajšek (2012); and geopolitical risk from Caldara and Iacoviello (2022).

To this end, we adapt the standard approach from the monetary policy literature that isolates asset-price surprises from their informational component by controlling directly for observable confounders (e.g. Romer and Romer, 2004; Nakamura and Steinsson, 2018; Miranda-Agrippino and Ricco, 2021). We proceed in two steps. First, at the daily frequency, we construct food and oil surprise series on the same gas-event dates. For each commodity, we compute event-day futures-price surprises over the corresponding maturities and extract their first principal component, mirroring the construction of the gas-price surprise.¹³ We then residualize the daily gas instrument with respect to contemporaneous food and oil surprises:

$$Gas_d = \alpha_0 + \phi Food_d + \theta Oil_d + GasClean_d$$

where Gas_d denotes the daily gas surprise instrument, and $Food_d$ and Oil_d are the corresponding daily food and oil surprise series constructed around the same gas-related events. Second, after aggregating $GasClean_d$ to the monthly frequency, we residualize the instrument series with respect to its own lags and contemporaneous and lagged observable confounders. This step extends the static correlation checks above by allowing for dynamic confounding. The informationally robust instruments, IRS_t , are then defined as the residuals from the following monthly regression:

$$GasClean_t = \alpha_0 + \sum_{j=1}^l \phi_j GasClean_{t-j} + \sum_{j=0}^l \mathbf{x}_{t-j} \Gamma_j + IRS_t$$

where $\mathbf{x}_t$ collects monthly energy-market, macroeconomic, financial, monetary-policy, and power-market controls that may confound the interpretation of the instrument. We also include the other gas-market instruments as controls when residualizing a given instrument. The monthly residualization is applied to all instruments, including the demand instruments. The monthly control set includes the shock measures used in the correlation checks that are available over the same sample, augmented with weather-related controls for renewable electricity generation.¹⁴ The lag order is set to $l = 3$. As shown in online Appendix C.3.2, the impulse responses obtained using the informationally robust instruments are very similar to those based on the baseline instruments, suggesting that the main results are not materially driven by observable confounding factors.

¹³To construct food surprises, we use wheat prices as a proxy for overall food prices, since wheat was Russia’s main export (OECD, 2022) and is the most actively traded food commodity (CME, 2024). We use Matif wheat futures for the Euro Area and Hard Red Winter wheat futures for the United States.

¹⁴The additional controls include indicators for wind, precipitation, solar radiation, and drought conditions.

Other diagnostic checks. Online Appendix C.3 reports additional diagnostic tests on the instrument series, including tests for autocorrelation and Granger causality. We find no evidence of serial correlation in the instruments. We also find no evidence that gas-specific, macroeconomic, or financial variables help forecast the supply series, as none of the variables considered Granger-causes the instruments at conventional significance levels. Taken together, these results provide further support for the validity of the instruments.

4 Econometric framework

Consider the following structural VAR(p) model:

$$\mathbf{B}_0 \mathbf{y}_t = \mathbf{B}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{B}_p \mathbf{y}_{t-p} + \mathbf{w}_t, \quad (3)$$

where $\mathbf{y}_t$ is a $K \times 1$ vector of endogenous variables, assumed to have zero mean without loss of generality. The vector $\mathbf{w}_t$ is a $K \times 1$ vector of structural shocks. Following the standard definition of structural shocks as primitive exogenous disturbances, we assume that they are mutually uncorrelated, so that $\mathbb{E}(\mathbf{w}_t \mathbf{w}_t') = \mathbf{\Sigma}_w$ is diagonal (Ramey, 2016, Section 2.1). Since the matrices $\mathbf{B}_0$ and $\mathbf{w}_t$ are generally unobserved, we rely on the reduced-form representation to estimate the model:

$$\begin{aligned} \mathbf{y}_t &= \mathbf{B}_0^{-1} \mathbf{B}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{B}_0^{-1} \mathbf{B}_p \mathbf{y}_{t-p} + \mathbf{B}_0^{-1} \mathbf{w}_t \\ &= \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \end{aligned}$$

where $\mathbf{A}_i = \mathbf{B}_0^{-1} \mathbf{B}_i$ for $i = 1, \dots, p$, and $\mathbf{u}_t$ denotes the reduced-form innovation, with $\mathbf{\Sigma}_u = \mathbb{E}(\mathbf{u}_t \mathbf{u}_t')$. We estimate the reduced-form VAR parameters and residuals using Bayesian methods, implementing standard Minnesota, sum-of-coefficients, and dummy-initial-observations priors, with their hyperparameters selected using the hierarchical procedure of Giannone et al. (2015). Details are provided in online Appendix A.5. The key equation linking the reduced-form innovations to the structural shocks is:

$$\mathbf{u}_t = \mathbf{B}_0^{-1} \mathbf{w}_t. \quad (4)$$

Our objective is to identify the dynamic effects of the gas-market structural shocks associated with demand- and supply-side disturbances, which corresponds to recovering the relevant columns of the impact matrix $\mathbf{B}_0^{-1}$. To achieve this, we use the instruments constructed as described in Section 3.

Identification via external instruments. We are interested in identifying $r < K$ structural shocks. Without loss of generality, order these shocks as the first r entries of $\mathbf{w}_t$, and write $\mathbf{w}_t = (\mathbf{w}'_{1:r,t}, \mathbf{w}'_{r+1:K,t})'$, where $\mathbf{w}_{1:r,t}$ collects the target shocks and $\mathbf{w}_{r+1:K,t}$ collects the remaining shocks. Let $\mathbf{z}_t = (z_{1,t}, \dots, z_{r,t})'$ denote a vector of instruments. In our setting, each instrument is intended to identify one labelled target shock. Thus, for $j = 1, \dots, r$, the relevance and exogeneity conditions are

$$\mathbb{E}[z_{j,t}w_{j,t}] = \alpha_j \neq 0, \quad (5)$$

$$\mathbb{E}[z_{j,t}w_{\ell,t}] = 0 \quad \text{for all } \ell \neq j. \quad (6)$$

These conditions state that $z_{j,t}$ is relevant for its own target shock and orthogonal to all other structural shocks, thereby preserving the economic label supplied by the instrument. Let $\mathbf{b}_j$ denote the j -th column of $\mathbf{B}_0^{-1}$, i.e. the impact vector associated with $w_{j,t}$. Multiplying (4) by $z_{j,t}$ and taking expectations gives

$$\mathbb{E}[z_{j,t}\mathbf{u}_t] = \alpha_j \mathbf{b}_j, \quad j = 1, \dots, r. \quad (7)$$

Each instrument therefore identifies its labelled impact vector up to sign and scale.

Normalization. The scale can be set via any normalization subject to $\boldsymbol{\Sigma}_u = \mathbf{B}_0^{-1}\boldsymbol{\Sigma}_w(\mathbf{B}_0^{-1})'$. One common approach is to rescale the structural shocks to have unit variance, $\boldsymbol{\Sigma}_{\tilde{w}} = \mathbf{I}_K$. Under this normalization, a unit realization of $\tilde{w}_{j,t}$ corresponds to a one-standard-deviation structural shock, and its contemporaneous effects on the variables in $\mathbf{y}_t$ are given by the j -th column of $\tilde{\mathbf{B}}_0^{-1}$. This leads to the unit-variance representation of the model:

$$\tilde{\mathbf{w}}_t = \boldsymbol{\Sigma}_w^{-1/2}\mathbf{w}_t \quad (8)$$

$$\tilde{\mathbf{B}}_0^{-1} = \mathbf{B}_0^{-1}\boldsymbol{\Sigma}_w^{1/2} \quad (9)$$

$$\mathbf{y}_t = \mathbf{A}_1\mathbf{y}_{t-1} + \dots + \mathbf{A}_p\mathbf{y}_{t-p} + \tilde{\mathbf{B}}_0^{-1}\tilde{\mathbf{w}}_t \quad (10)$$

Alternatively, one can fix the scale of each identified impact vector by anchoring it to the contemporaneous response of a chosen variable. Let g denote the position of the real price of natural gas in $\mathbf{y}_t$, and partition $\mathbf{b}_j = (b_{g,j}, \mathbf{b}'_{-g,j})'$ and $\mathbf{u}_t = (u_{g,t}, \mathbf{u}'_{-g,t})'$. Then (7) implies

$$\mathbf{b}_{-g,j} = b_{g,j} \frac{\mathbb{E}[z_{j,t}\mathbf{u}_{-g,t}]}{\mathbb{E}[z_{j,t}u_{g,t}]}, \quad j = 1, \dots, r.$$

In the impulse responses reported in Section 5, we set $b_{g,j} = c$, with c chosen so that each

identified shock raises the real price of natural gas by 10% on impact. The empirical impact vectors are obtained by replacing the population moments in the expression above with their sample analogues.

Orthogonalization. The proxy moment conditions (5)–(6) identify the labelled impact vectors separately. Since these are population restrictions, the corresponding estimated vectors need not form an exactly orthogonal structural system in finite samples. In particular, if $\mathbf{S}$ collects the initially identified impact vectors, the sample analogue of $\mathbf{S}'\Sigma_u^{-1}\mathbf{S}$ need not be diagonal. This matters because recovering structural shocks, and constructing objects such as historical decompositions and forecast error variance decompositions, requires a mutually orthogonal system of shocks. We therefore apply a polar-whitening transformation to the initially identified impact vectors. Specifically, let $\mathbf{G} = \mathbf{S}'\Sigma_u^{-1}\mathbf{S}$ and set $\mathbf{S}^\perp = \mathbf{S}\mathbf{G}^{-1/2}$, so that $\mathbf{S}^{\perp'}\Sigma_u^{-1}\mathbf{S}^\perp = \mathbf{I}$. This transformation imposes exact orthogonality while minimizing the Σ_u^{-1} -weighted distance from the initially identified system.¹⁵ In Section 5.3, we show that the baseline results are robust to not imposing orthogonality, suggesting that the initially identified shocks are already close to orthogonal in sample. This also implies that a researcher interested in only one shock and using only its corresponding instrument would obtain very similar results to ours.

Invertibility. The external-instrument identification strategy described above assumes invertibility, in the sense that the relevant structural shocks are recoverable from the reduced-form VAR innovations. This assumption is important for our main object of interest, the structural elasticities discussed in the next section, because these are constructed from the contemporaneous structural impact vectors. It is also required for historical decompositions, which we report as part of our results and which require recovering the sequence of structural shocks from the reduced-form innovations. In Section 5.3, we formally test this assumption using the procedures in Plagborg-Møller and Wolf (2022) and Hamilton (forthcoming). We fail to reject invertibility in the baseline specification and also show that the main results are robust to an internal-instrument approach that does not rely on this assumption.

¹⁵Intuitively, if the instruments identify the correct shock directions and any lack of orthogonality is due to sampling variation, this transformation leaves the population target unchanged while enforcing exact orthogonality in the estimated system. We prefer this orthogonalization to alternatives that impose additional economic structure, such as recursive schemes based on an ordering of the identified shocks as in Mertens and Ravn (2013), because in our setting we do not see a compelling economic justification for such restrictions.

4.1 Estimating demand and supply elasticities

The physical accounting identity for the regional natural gas market is

$$DP_t + IM_t - EX_t = C_t + \Delta I_t, \quad (11)$$

where DP_t is domestic production, IM_t imports, EX_t exports, C_t current domestic gas use, and $\Delta I_t = I_t - I_{t-1}$ net injections into inventories.¹⁶ The left-hand side is total gas available to the regional market, while the right-hand side allocates this gas between current flow demand and storage demand. Together with the price of natural gas at the relevant wholesale hub, these balances summarize the aggregate gas market in each region.

We therefore define total available supply as $Q_t^s := DP_t + IM_t - EX_t$. In the Euro Area, this measure is largely driven by imports; in the United States, it mostly reflects domestic production, with net trade flows also contributing. When estimating the short-run supply elasticity, we use the flow-supply instrument, since the elasticity refers to the contemporaneous response of Q_t^s . Broader supply-security news shocks, which capture information about future supply conditions, are used below in the impulse-response analysis. On the demand side, we define current flow demand as $Q_t^d := C_t$, keeping inventory changes separate because they reflect intertemporal storage decisions rather than final contemporaneous gas use. This distinction is consistent with models of storable commodities, which separate current consumption from inventory dynamics (Kilian & Murphy, 2014; Baumeister & Hamilton, 2019). Our demand-side instrument, interpreted as an unexpected temperature innovation, provides exogenous variation in this flow-demand component.

To estimate the supply and demand elasticities we consider dynamic demand and supply equations of the form:

$$q_t^d = \alpha_{qp}^d p_t + \mathbf{x}' \boldsymbol{\beta}_d + w_t^d, \quad (12)$$

$$q_t^s = \alpha_{qp}^s p_t + \mathbf{x}' \boldsymbol{\beta}_s + w_t^s, \quad (13)$$

where p_t denotes the logarithm of the real price of natural gas, while q_t^d and q_t^s represent the logarithms of the quantity demanded and supplied, respectively. The parameters α_{qp}^d and α_{qp}^s denote the price elasticities of demand and supply. The terms w_t^d and w_t^s are the structural shocks to demand and supply, and $\mathbf{x}$ is a vector of control variables that includes a constant, contemporaneous values of other endogenous variables, as well as lags of all endogenous variables in the system. We are interested in estimating the parameters α_{qp}^d and

¹⁶In official energy balances, this identity may also include distribution losses and statistical discrepancies. We abstract from these residual components, which are small relative to the main flow variables.

α_{qp}^s .

These structural elasticities are recovered from the relevant rows of the contemporaneous structural matrix $\mathbf{B}_0$.¹⁷ Even when $\mathbf{B}_0^{-1}$ is only partially identified, the relevant rows of $\mathbf{B}_0$ can be recovered from the corresponding identified columns of $\mathbf{B}_0^{-1}$. To see this, consider the unit-variance representation in equations (8)–(10). Since $\boldsymbol{\Sigma}_u = \tilde{\mathbf{B}}_0^{-1}(\tilde{\mathbf{B}}_0^{-1})'$, it follows that

$$\tilde{\mathbf{B}}_0 = (\tilde{\mathbf{B}}_0^{-1})' \boldsymbol{\Sigma}_u^{-1}. \quad (14)$$

Thus, if the j -th column of $\tilde{\mathbf{B}}_0^{-1}$ is identified, the corresponding j -th row of $\tilde{\mathbf{B}}_0$ is $\tilde{\mathbf{b}}'_{0,j} = \left[(\tilde{\mathbf{B}}_0^{-1})_{:,j} \right]' \boldsymbol{\Sigma}_u^{-1}$. Since this row is identified only up to sign and scale, we normalize it so that the coefficient on the relevant quantity is equal to unity. Because the structural equations are written with price on the right-hand side, the elasticity α_{qp} is then given by the negative of the normalized coefficient on price, corresponding to the percentage change in quantity associated with a 1% increase in price, consistent with the standard convention.

4.2 Empirical specification

We estimate Bayesian VARs for the United States and the Euro Area. The data are monthly, and the estimation sample spans 20 years, from January 2004 to December 2023.¹⁸ Our baseline specification includes seven variables: the real price of natural gas, total available gas supply, gas consumption, gas inventories, the headline consumer price index, industrial production, and the real price of crude oil. This specification combines the key price and physical variables needed to characterize the natural gas market with aggregate macroeconomic indicators that allow us to assess the broader effects of gas price shocks. Including the real price of crude oil also helps control for developments in related energy markets that may otherwise confound the effects of natural gas shocks. To study additional transmission channels, we then augment the baseline VAR with additional variables or small blocks of variables tailored to each exercise, following a common practice in the VAR literature (Beaudry & Portier, 2014; Gertler & Karadi, 2015; Känzig, 2021; Miranda-Agrippino et al., 2026). When an additional series is not available over the full baseline period, we estimate the corresponding augmented specification over the largest common sample. We estimate

¹⁷The impact matrix $\mathbf{B}_0^{-1}$ describes the joint contemporaneous responses used for impulse response functions. Structural elasticities, instead, are *ceteris paribus* objects and correspond to rows of $\mathbf{B}_0$; see, e.g., Baumeister and Hamilton (2024a, Sec. 2.6).

¹⁸Our estimation sample begins in January 2004, the earliest date for which TTF data are available. To ensure comparability, we use the same sample period for the United States specification, although the limiting variable there is gas consumption, which begins in January 2001. As shown in the online Appendix, extending the U.S. sample does not materially affect the results.

the VARs in log levels, so impulse responses can be interpreted as percentage changes. A detailed description of the data and sources is provided in online Appendix B.1. We include six lags and only a constant term among the deterministic components; Section 5.3 shows that the results are robust to these choices.

5 Results

This section presents the empirical results for the Euro Area and the United States. We first estimate the short-run price elasticities of gas demand and supply and then use FEVDs and historical decompositions to assess how demand- and supply-side forces shape gas price fluctuations. We then examine the dynamic effects of gas price shocks, focusing on macroeconomic responses and on the transmission channels behind differences across shock types and regions. We also assess the robustness of the baseline results. Next, we extend the analysis for the Euro Area to study sectoral effects on prices and quantities and evaluate the extent to which gas price shocks can generate significant real effects. Finally, we quantify their contribution to the post-pandemic inflation surge.

5.1 Gas demand and supply elasticities and price volatility

Using the strategy described in Section 4.1, we estimate short-run price elasticities of natural gas demand and supply. For the Euro Area, the demand elasticity is -0.08 and the supply elasticity is 0.32 ; for the United States, the corresponding estimates are -0.04 and 0.23 . Table D9 in the online Appendix reports previous estimates based on autoregressive models, most of which adopt the identification strategy developed by Baumeister and Hamilton (2019) for the crude oil market. Our estimates exploit external exogenous variation and are broadly consistent with this literature, though our demand elasticities are lower in absolute value, while our supply elasticities lie toward the upper end of previous estimates. Figure 6 illustrates the implied demand and supply curves.

Gas demand is less price elastic than gas supply in both regions. This asymmetry implies that, for a given quantity disturbance, supply-side shocks generate larger price movements because adjustment occurs along a relatively inelastic demand curve. The forecast error variance decomposition in Table 2 is consistent with this implication: gas supply shocks account for a larger share of gas price fluctuations than demand shocks, especially at short horizons. Their contribution declines as the horizon lengthens, but supply shocks remain the larger source of variation. Across regions, supply is more elastic in the Euro Area, reflecting the faster adjustment of imports relative to domestic production in the United

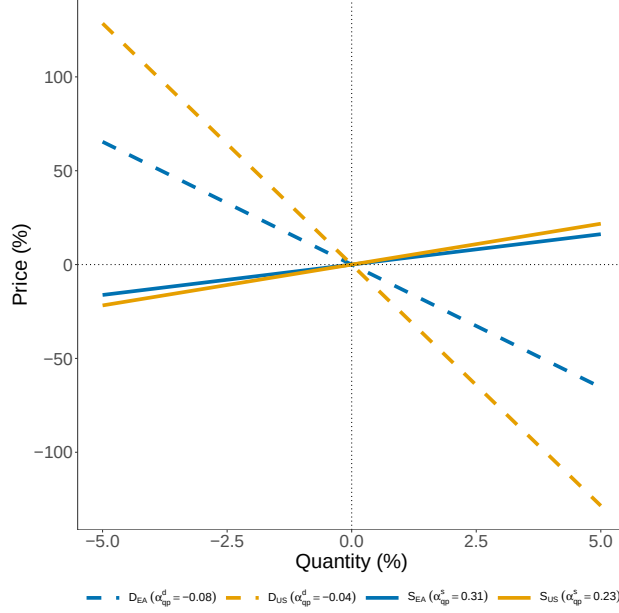


Figure 6: *Short-run gas demand and supply curves*

Notes: This figure depicts the implied short-run gas demand and supply curves for the Euro Area (blue) and the United States (orange), using the quantity definitions in Section 4.1. Dashed lines correspond to demand curves, while solid lines represent supply curves. Curve slopes reflect the inverse of the estimated elasticities. For illustrative purposes, supply and demand quantities are equated in the figure; this normalization does not affect the estimated slope coefficients.

States. Demand is relatively inelastic in both regions, with the U.S. estimate closer to zero; the FEVDs show that even small elasticity differences can matter for price dynamics when demand is inelastic (Caldara et al., 2019). These cross-region differences also reflect sectoral composition: residential gas use accounts for a larger share of consumption in the Euro Area (see online Appendix Figure B21), and existing estimates generally find household gas demand to be more price responsive than industrial gas use (Labandeira et al., 2017).

Region	Shock	$h = 1$	$h = 6$	$h = 12$	$h = 18$	$h = 24$
United States	Gas demand	11.9 [8.3, 15.7]	17.2 [11.6, 23.0]	17.3 [10.6, 24.0]	17.9 [11.4, 24.5]	19.0 [11.9, 25.9]
	Gas supply	73.6 [63.4, 83.8]	41.6 [33.5, 49.5]	34.5 [26.0, 42.8]	33.0 [24.9, 41.2]	31.6 [23.6, 39.4]
Euro Area	Gas demand	15.1 [10.9, 19.6]	16.2 [11.0, 21.3]	21.6 [13.2, 30.0]	23.5 [13.2, 33.8]	24.0 [13.2, 34.8]
	Gas supply	54.4 [42.5, 66.5]	35.2 [25.9, 44.6]	30.7 [20.0, 41.5]	29.6 [18.2, 40.7]	28.9 [17.4, 40.2]

Table 2: *FEVD of the gas price at horizons h . Entries report posterior mean contributions of gas demand and supply shocks, with 68% credible bands in brackets.*

Figure 7 presents the cumulative historical contributions of gas flow-supply and demand shocks to the real price of natural gas in the United States and the Euro Area, alongside the observed series. The identified shocks account for a substantial share of the observed price dynamics, with the combined contributions capturing several major episodes, especially during the 2021–2022 surge in European gas prices. Consistent with the FEVD results, gas flow-supply shocks account for a larger share of gas price fluctuations than demand shocks, although demand shocks contribute meaningfully in specific weather-related episodes.

5.2 Impulse-response analysis

We now present the dynamic effects of natural gas price shocks. Figure 8 reports the impulse responses from the baseline specification for the United States and the Euro Area; for the Euro Area, it also includes the supply-security news shock introduced in Section 3.2. All shocks are normalized to raise the real spot price of natural gas by 10% on impact.¹⁹ Median responses are shown by black lines; blue shaded credible bands refer to the Euro Area and orange shaded credible bands to the United States, at the 68%, 80%, and 90% levels.

¹⁹This normalization is close to the size of a typical one-standard-deviation shock. Under a one-standard-deviation shock normalization, the impact responses of the real gas price are 5.8% and 14.7% for U.S. demand and supply shocks, and 8.5%, 15.1%, and 8.0% for Euro Area demand, supply, and supply-news shocks, respectively.

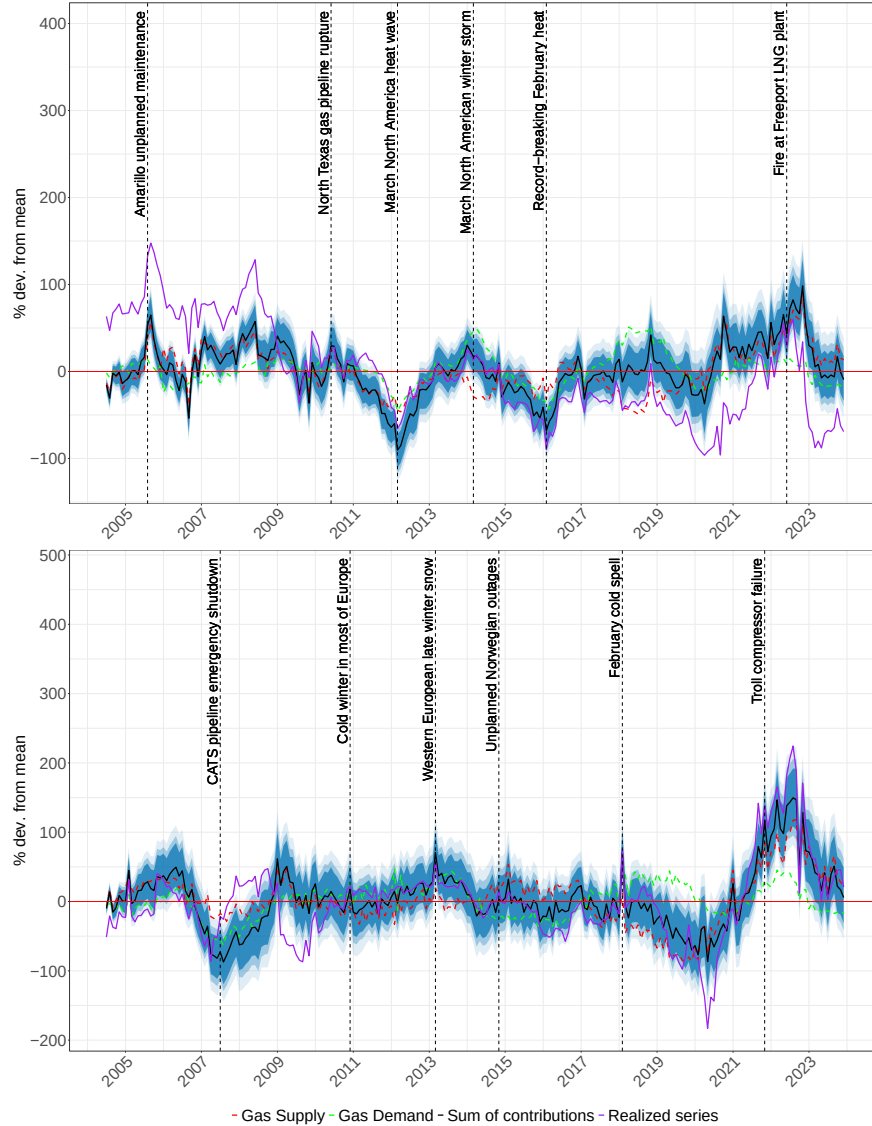


Figure 7: *Historical decomposition of the real price of natural gas in the United States (top panel) and the Euro Area (bottom panel).*

Notes: Estimated contributions of gas flow-supply shocks are shown by the red dashed line, gas demand shocks by the green dashed line, and their combined contribution by the solid black line with blue credible bands. All contributions are expressed as percentage deviations from the mean. The realized real gas price series is shown in purple. In the United States, vertical dashed lines mark unplanned maintenance on the Amarillo Mainline that reduced pipeline capacity in 2005M8; a rupture and explosion on Enterprise's North Texas gas pipeline in 2010M6; a heat wave across North America in 2012M3; a North American winter storm in 2014M3; record-breaking February warmth in 2016M2; and a blast at the Freeport LNG terminal that disrupted operations in 2022M6. In the Euro Area, vertical dashed lines mark the unexpected shutdown of the CATS pipeline due to condensate contamination in 2007M7; an abnormally cold winter in Europe in 2010M12; a late-winter snow event in Western Europe in 2013M3; unplanned Norwegian outages in 2014M11; a February cold spell in 2018M2; and a compressor failure at Norway's Troll field in 2021M11.

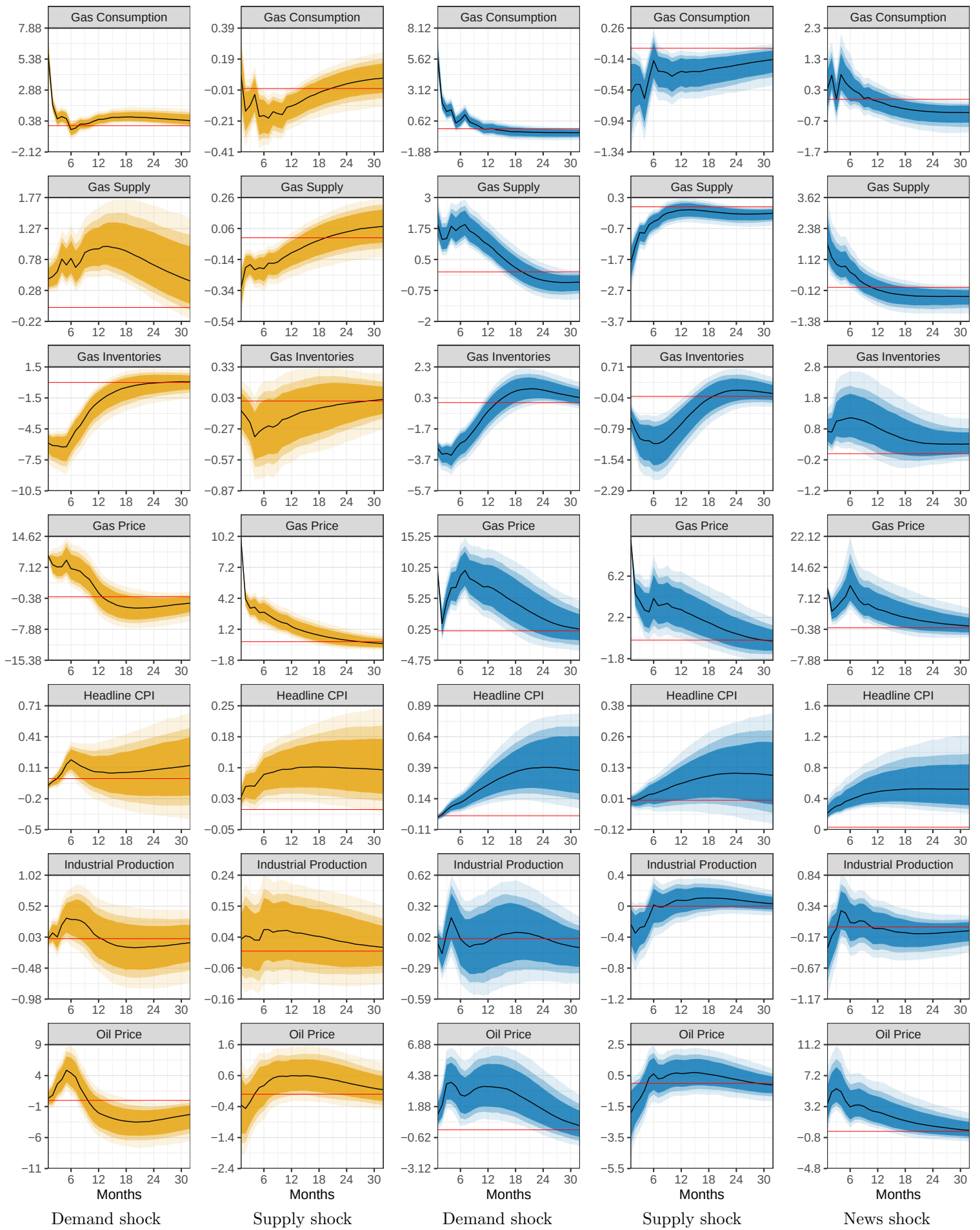


Figure 8: *Responses to natural gas shocks: United States (orange); Euro Area (blue)*

Responses of gas quantities. We first examine the quantity margins underlying the normalized gas-price responses. Figure 8 shows that demand shocks require substantially larger quantity adjustments than supply shocks to generate the same 10% increase in gas prices. In both regions, gas consumption rises by about 6% on impact following a demand shock. By contrast, flow-supply shocks generate the same price increase with much smaller movements in physical quantities, consistent with the short-run inelasticity of gas demand documented above. Following a demand shock, supply adjusts more gradually in the United States, where the response builds over subsequent months, than in the Euro Area, where imports allow for a larger immediate response. Consistent with this interpretation, Figure 9 shows that replacing total supply with domestic production for the United States and with net imports for the Euro Area yields similar dynamics. Following a flow-supply shock, gas consumption declines in both regions, but the adjustment is modest and unfolds gradually relative to the 10% increase in gas prices, consistent with the short-run inelasticity of demand. The decline is larger in the Euro Area, in line with the higher estimated demand elasticity reported above.

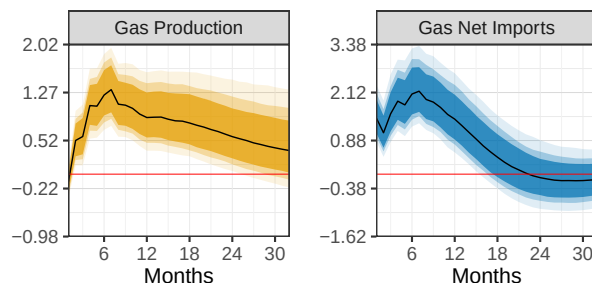


Figure 9: *Components of gas supply*

Notes: Impulse responses of U.S. gas production and Euro Area net gas imports following a gas demand shock. The responses are estimated from the baseline specification after replacing total gas supply with the corresponding supply component in each region.

These regional differences in gas-consumption responses are consistent with differences in sectoral composition. Existing estimates generally find residential gas demand to be more price responsive than industrial gas use, while manufacturing displays the lowest degree of price elasticity.²⁰ As discussed in Section 2, gas consumption in the Euro Area has a larger residential component, consistent with the larger absolute demand elasticity estimated above.

A second adjustment margin operates through the power sector. Because power generation accounts for a large share of gas use, fuel switching in electricity production can shape

²⁰See Asche et al. (2008), Serletis et al. (2010), Andersen et al. (2011), and Rubin and Auffhammer (2024) for estimates of gas demand elasticities across sectors and regions.

the short-run response of aggregate gas consumption to supply shocks, even if it represents only one component of total demand. To examine this margin, we estimate an alternative specification in which total gas consumption is replaced with coal- and oil-fired power generation. The impulse response functions presented in Figure 10 confirm significant interfuel substitution in the United States following a gas supply shock, with oil as the primary alternative. In contrast, the Euro Area exhibits a more limited degree of interfuel substitution, with coal serving as the main substitute.²¹

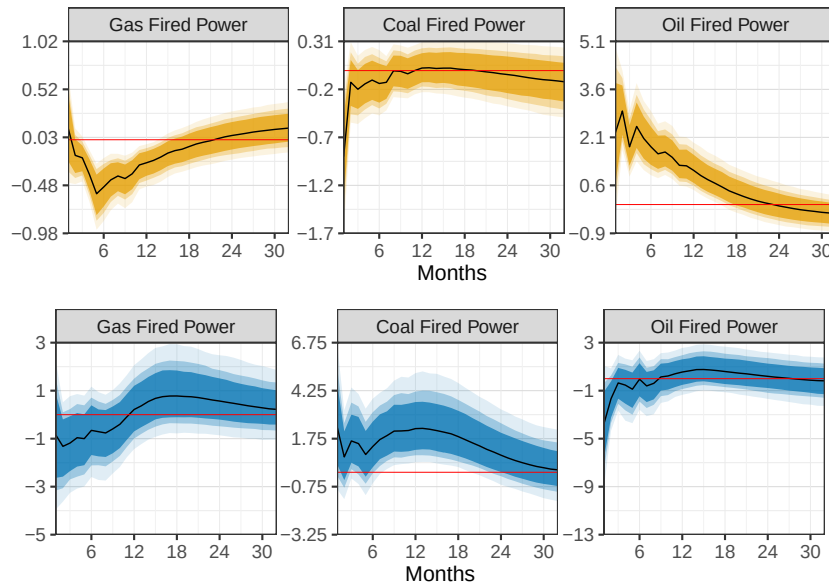


Figure 10: *Interfuel substitution in the power sector*

Notes: Impulse responses of gas-fired, oil-fired, and coal-fired power generation to a natural gas flow-supply shock. Estimates are based on the baseline specification augmented with the corresponding fuel-specific series. The Euro Area sample begins in 2007 due to data availability constraints.

The Euro Area supply-security news shock contrasts with the flow-supply shocks because it is not tied to an immediate physical shortfall. Its quantity response is more gradual, consistent with an initial repricing of expected future availability rather than an immediate contemporaneous disruption to physical flows.

²¹The power generation sector in the United States is comparatively more flexible, with approximately 20% of natural gas-fired combined-cycle capacity capable of switching to alternative fuels—primarily oil (EIA, 2022). In contrast, most gas turbines in Europe are not designed for dual-fuel operation, which significantly constrains short-term interfuel substitution. Coal, however, has remained a key substitute for gas: many European utilities operate mixed portfolios that include both gas- and coal-fired units, enabling them to adjust the fuel mix in response to relative price changes (Reuters, 2024).

Inventories, volatility, and uncertainty. The third row of Figure 8 shows that gas demand shocks lead to a large decline in inventories in both regions. This response is more pronounced in the United States in percentage terms, as shown in the IRFs, and because U.S. inventories are larger in levels, it reflects a larger absolute storage adjustment margin. Flow-supply shocks also generate a short-run inventory decline, but the response is smaller than after demand shocks and is more pronounced in the Euro Area, where storage is more directly used to buffer disruptions in external supply. The smaller inventory response to flow-supply shocks is consistent with the timing of the shocks. By construction, gas demand shocks primarily occur during winter, when inventories are more directly used to smooth seasonal fluctuations in consumption. Flow-supply shocks, instead, can occur at any point in the year, meaning that inventories are not necessarily positioned to absorb these disruptions effectively (European Council, 2022).

The Euro Area supply-security news shock, which captures news about expected gas availability rather than a contemporaneous disruption to physical gas flows, points to a precautionary storage and uncertainty channel. The underlying news events need not reduce current flows immediately; instead, when they raise the perceived risk of future scarcity, firms and storage operators may increase precautionary inventory demand, putting upward pressure on current gas prices. Consistent with this interpretation, the fifth column of Figure 8 shows that inventories increase after the shock. This mechanism is closely related to the evidence documented by Kilian and Murphy (2014) for the crude oil market.

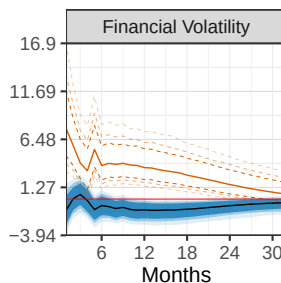


Figure 11: *Euro Area financial volatility responses to gas flow-supply and supply-security news shocks*

Notes: Responses of financial volatility are shown for Euro Area flow-supply shocks (black solid line with blue shaded credible bands) and Euro Area supply-security news shocks (red solid line with red dashed credible bands). Financial volatility is measured by the Euro Stoxx 50 Volatility Index (VSTOXX).

Figure 11 compares the response of Euro Area financial volatility to flow-supply and supply-security news shocks. Flow-supply shocks generate essentially no volatility response. By contrast, supply-security news shocks lead to a large and persistent increase in volatility.

This reinforces the interpretation that these shocks operate not only through expected future availability, but also through supply-security concerns and uncertainty.²² Similarly, Känzig (2021) documents an increase in crude oil inventories following a negative oil supply news shock and shows that expectations respond strongly, while financial uncertainty does not. This contrast is consistent with the different nature of the news: OPEC announcements provide information about future supply policy, whereas the Euro Area gas supply-news shocks studied here reflect supply-security concerns that heighten uncertainty about future availability.

Persistence of gas-price responses. The fourth row of Figure 8 illustrates the dynamics of the spot price following shocks normalized to raise the price of natural gas by 10% on impact. The results indicate that price responses to demand shocks are less persistent in the United States than in the Euro Area: the U.S. demand-driven response dissipates within about a year, whereas the Euro Area response declines more gradually. This difference is consistent with the adjustment margins discussed above: U.S. inventories respond more strongly to demand shocks, helping absorb the shock more quickly, while the larger Euro Area consumption response unfolds gradually and is not enough to prevent a more persistent price response. For flow-supply shocks, by contrast, price dynamics are broadly similar across the two regions. In both cases, the spot price declines steadily after impact and gradually converges back toward baseline over the horizon.

Euro Area supply-security news shocks display a distinct pattern. The spot price rises sharply over the first few months and then declines gradually, remaining above baseline for much of the horizon. This response is consistent with the expectation channel discussed above: inventories increase and financial volatility rises, pointing to precautionary inventory accumulation amid heightened uncertainty about future gas availability. Notably, gas prices can remain elevated even as the quantity responses of gas consumption, supply, and inventories move back toward baseline.

Macroeconomic propagation. The fifth and sixth rows of Figure 8 report the responses of headline CPI and industrial production. In the United States, gas demand shocks generate a modest increase in headline CPI, peaking at about 0.15–0.2%, and a temporary expansion in industrial production of roughly 0.3%. This implies a price-level pass-through of about 1.5–2% of the initial gas-price increase. U.S. flow-supply shocks also raise headline CPI, with a small but persistent pass-through close to 1%, while industrial production remains

²²Chîțu et al. (2024) document increased speculation on gas prices, particularly in Europe following the invasion of Ukraine.

essentially unchanged. This CPI response likely reflects the pass-through of domestic gas-supply disruptions into energy costs. The increase in industrial production after demand shocks is instead driven by higher activity in energy-related sectors, reflecting the rise in domestic gas production documented above.²³

In the Euro Area, demand shocks generate a persistent consumer-price response, with an implied pass-through of around 4% after roughly two years. Since these are gas-specific demand shocks rather than aggregate demand shocks, aggregate industrial production remains broadly unchanged. Flow-supply shocks have a different profile: their price-level pass-through is similar in magnitude to the U.S. response, around 1%, but is not meaningfully different from zero, whereas industrial production falls by about 0.4% after one month and then recovers within a few months. Thus, contemporaneous supply disruptions generate short-lived real effects rather than a persistent aggregate contraction.

The Euro Area supply-security news shock is instead strongly transmitted to consumer prices, with little evidence of aggregate real effects. The implied pass-through rises from about 2% on impact to roughly 5% over the medium run, while industrial production remains close to baseline. This contrast is central to the interpretation of the shock: news about future gas availability and supply security raises current prices and consumer-price pressures, but it does not require an immediate contraction in contemporaneous gas flows or aggregate production. Consistent with the precautionary storage and uncertainty channel discussed above, the supply-news shock therefore propagates mainly through prices rather than quantities, in line with evidence that natural gas price shocks can generate second-round pass-through via inflation expectations (Boeck & Zörner, 2025).

Taken together, the macro responses point to a clear regional asymmetry. Inflationary effects are larger and more persistent in the Euro Area, especially for demand and supply-security news shocks. Aggregate real effects remain limited in both regions, but they are not identical: U.S. flow-supply shocks leave industrial production essentially unchanged, whereas Euro Area flow-supply shocks generate a short-lived decline. This pattern is consistent with the structural differences discussed above: the U.S. market is anchored in domestic production and larger storage buffers, while the Euro Area is more exposed to import dependence and, for supply-news shocks, to precautionary and uncertainty channels related to future gas availability.

The next subsection examines whether these muted aggregate output responses conceal stronger effects in specific sectors and along the production chain. A central channel, shown in Figure 14, is the strong pass-through from gas prices to electricity prices through marginal

²³The energy sector is comparatively much larger in the United States, where it accounted for 5.2% of total employment in 2023 (U.S. Department of Energy, 2024).

pricing in European power markets (Baget et al., 2024). The sectoral price responses in Figure 15 show that these cost pressures are broader-based: producer prices rise first, cost increases are passed along the supply chain, although incompletely and with a delay, and consumer prices remain elevated even after producer-price pressures begin to ease. By contrast, output losses are concentrated in a few energy-intensive industries, while most sectors display only small responses. This pattern is consistent with firms passing higher input costs downstream while limiting output reductions where demand is relatively inelastic. Together, these results support the interpretation that Euro Area gas shocks transmit mainly through costs and prices, while aggregate real effects remain limited.

Gas and oil markets interrelation. The last row of Figure 8 shows how oil prices respond to the natural gas shocks identified in the baseline specification. Since gas and oil are imperfect substitutes, shocks to one commodity can affect the other, with transmission varying across shock types and regions.

Gas demand shocks raise oil prices in both regions, with peak responses of roughly 4–5% following a 10% increase in gas prices. By contrast, flow-supply shocks generate muted oil-price responses, consistent with the more localized nature of contemporaneous gas-supply disruptions. Euro Area supply-security news shocks also raise oil prices, suggesting that news about future gas availability can spill over to broader energy markets.

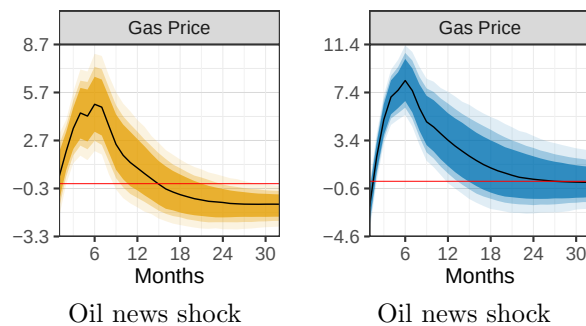


Figure 12: *Gas price responses to oil news shocks*

Notes: Responses of real gas prices to a 10% increase in real crude oil prices. Responses for the United States are shown in orange, while responses for the Euro Area are shown in blue. Oil supply news shocks are identified following Känzig (2021).

Figure 12 complements this evidence by showing the reverse direction of pass-through: the response of gas prices to oil news shocks estimated in the same baseline specification.²⁴ Oil news shocks raise gas prices in both regions, but the response is stronger and more persistent in the Euro Area than in the United States. The peak response is around one-half

²⁴The oil-price shock is identified using the instrument developed by Känzig (2021).

of the oil-price increase in the United States, compared with about 85% in the Euro Area, where the response also declines more gradually toward baseline.

This regional asymmetry is consistent with recent evidence that, as the LNG market has become more integrated, European and Asian gas prices have become more closely linked, whereas the North American market remains only partially integrated.²⁵ At the same time, Euro Area gas-market shocks that convey broader demand or supply-security information can also spill over to oil prices. Previous studies have explored the relationship between crude oil and natural gas markets in the U.S. context. While certain contributions find no significant long-run relationship between the prices of these commodities (Bachmeier & Griffin, 2006), others focus primarily on the influence of oil on the gas market (Jadidzadeh & Serletis, 2017). We add to this evidence by showing that oil-price shocks transmit to gas prices in the short run, while gas shocks also have economically meaningful effects on the oil market.

Environmental impact of gas price shocks. The power-sector results above show that gas-price shocks induce substitution away from natural gas toward more carbon-intensive fuels (see again Figure 10). To more directly evaluate the environmental consequences of gas price shocks, we draw on CO₂ emissions data from the EDGAR (Global Greenhouse Gas Emissions) database. Figure 13 shows the response of CO₂ emissions to natural gas price shocks.

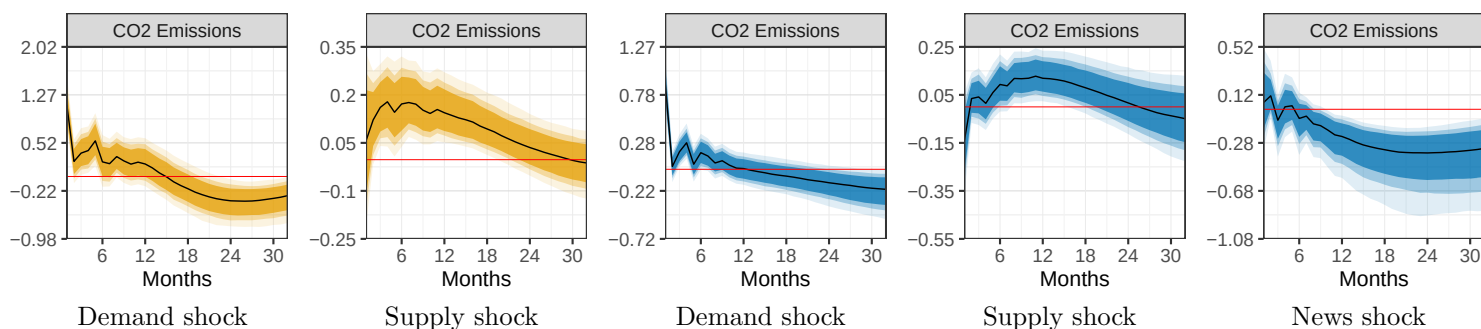


Figure 13: *Responses of CO₂ emissions to natural gas shocks*

Notes: Impulse responses of CO₂ emissions to natural gas demand, flow-supply, and supply-security news shocks, estimated using the baseline specification augmented with the CO₂ emissions series. The first two columns report responses for the United States, while the last three columns report responses for the Euro Area. The sample period ends in 2022M12 due to data availability constraints.

A gas demand shock that raises the price of natural gas by 10% leads to an immediate

²⁵Infrastructure constraints, particularly the limited capacity of LNG export terminals, restrict the volume of gas that can be shipped abroad (Albrizio et al., 2023).

increase in emissions in both the Euro Area and the United States, as more carbon-intensive fossil fuels are burned to satisfy demand. The increase is larger on impact in the United States, while in both regions the response fades over time and eventually falls below baseline. Following flow-supply shocks, emissions respond only modestly on impact but increase over the following months as higher gas prices induce substitution toward more carbon-intensive fuels; the response peaks at around 0.2% in the United States and around 0.1% in the Euro Area. The Euro Area supply-security news shock follows a different pattern: emissions do not show a clear positive impact response and then fall below baseline. Since these shocks convey information about future gas availability and can generate price increases perceived as more persistent, they are less likely to operate only through immediate interfuel substitution and may instead induce adjustment with a longer horizon.

These findings are in line with the distinction emphasized in the literature between short-run fuel substitution and longer-run adjustment. The responses to demand and flow-supply shocks mirror evidence that, in the short run, lower gas prices reduce emissions by encouraging coal-to-gas substitution (Knittel et al., 2015; Cullen & Mansur, 2017; Brehm, 2019): when gas prices rise, the same substitution margin works in the opposite direction and emissions increase. The supply-news response is instead consistent with the longer-run channel emphasized by Stock and Zaragoza-Watkins (2024), who show that sustained increases in gas prices can promote investment in cleaner energy sources and reduce emissions over time. Taken together, these dynamics underscore the intertemporal trade-offs associated with the role of natural gas in the energy transition. As Harstad and Holtsmark (2024) argue, expanding gas production can initially reduce emissions by displacing coal through price-driven substitution. Nonetheless, such policies may also deter long-term investment in renewables, a dilemma they describe as the “gas trap” (see also Acemoglu et al., 2023). This distinction also matters for policy: a gas-specific price increase is not equivalent to a broad carbon price. When short-run gas demand is inelastic, higher gas prices may redirect energy use toward coal or oil rather than reduce fossil-fuel use overall.

5.3 Robustness of the baseline results

We conclude the baseline analysis by assessing whether the main results depend on instrument construction, identifying assumptions, estimation method, lag length, sample period, or additional variables. Online Appendix E reports the corresponding figures and tables.

Instrument construction. We first consider alternative constructions of the instruments. Using the informationally robust instruments described in Section 3.3, which remove observable confounding information from the instruments before estimation, yields responses close

to the baseline. We then reconstruct the supply instruments using stricter event-selection thresholds, retaining only dates with absolute gas-price surprises above 3% and 4%, instead of the baseline 2% cutoff. The resulting estimates remain similar, indicating that the findings are not sensitive to the threshold choice. A leave-one-out jackknife likewise shows that no single event month, including the largest disruptions during the Russia–Ukraine crisis, drives the main dynamic patterns. For the temperature-based demand instrument, we check whether the results depend on how weather-driven gas demand is measured. The responses are similar when we include summer months, adjusting the sign of the index to account for cooling-related gas demand, or when we replace temperature deviations with heating degree days. Together, these exercises indicate that our findings are not an artifact of confounding information, a particular event sample, or the temperature measure.

Identifying assumptions. We next turn from instrument construction to the assumptions underlying identification. The external-instrument approach requires invertibility, meaning that the relevant structural gas-market shocks must be recoverable from the reduced-form VAR innovations. Following Plagborg-Møller and Wolf (2022) and Hamilton (forthcoming), we test whether lags of the instruments predict the VAR variables conditional on the baseline regressors; the joint tests provide no evidence against invertibility. As a complementary exercise, we estimate an internal-instrument specification (Ramey, 2011; Plagborg-Møller & Wolf, 2021), which orders the instrument first in a recursive VAR and does not require invertibility. The resulting responses remain close to the baseline. We also assess instrument relevance: the posterior distributions of the impact coefficients are well behaved and tightly concentrated (Caldara et al., 2019; Giacomini et al., 2022), and frequentist first-stage diagnostics provide additional support.²⁶ Finally, estimating the model without orthogonalizing the identified shocks produces similar impulse responses, indicating that the results are not driven by the finite-sample orthogonalization procedure.

Estimation and specification. Finally, we examine estimation and specification choices. Re-estimating the model by OLS and by the Bayesian local-projection approach of Ferreira et al. (2025), in which we adopt Minnesota-type priors, delivers similar responses, although some estimates are less precise. We also augment the VAR with electricity prices, electricity consumption, and renewable power generation to account for omitted power-market dynamics that could affect the estimated gas-demand responses. The results remain close to the

²⁶The heteroskedasticity-robust first-stage F-statistics from regressions of the gas-price reduced-form innovation on each instrument are 20.59, 16.88, and 16.22 for the Euro Area flow-supply, supply-security news, and demand instruments, and 10.01 and 25.99 for the U.S. supply and demand instruments, respectively; see Montiel Olea et al. (2021) on weak-instrument issues in SVARs identified with external instruments.

baseline. Lastly, we extend the lag length from six to twelve and, for the United States, expand the estimation sample back to 2001, the earliest date allowed by gas-consumption data. Neither modification materially alters the responses. The main results therefore do not hinge on the prior specification, the VAR representation of the dynamics, the lag structure, or the common post-2004 sample.

5.4 Propagation of Euro Area gas supply shocks

The aggregate results above show that Euro Area gas supply shocks generate strong price effects but more limited real effects. In this section, we use a combined Euro Area supply instrument, pooling flow-supply and supply-security news events, to study the sectoral margins behind this aggregate response: how supply-driven gas price increases propagate across energy prices, sectoral consumer and producer prices, and real activity. This approach is useful for tracing the overall economic incidence of gas supply disruptions, which is the relevant object for assessing the sectoral and policy implications of Europe’s exposure to gas-supply risk. The separate responses reported in online Appendix F clarify the role of the two supply components: supply-security news shocks account for most of the price pass-through, whereas flow-supply shocks generate short-lived real effects and drive the immediate output losses.

Transmission to broader energy prices. We first examine how gas supply shocks transmit to broader energy prices. To do so, we augment the baseline specification with electricity, gas, and fuel price series, as shown in Figure 14. Electricity spot prices respond quickly and strongly, peaking within the first year. This reflects the merit-order principle and marginal electricity pricing, which dictate that when natural gas is the marginal power source in use, it sets the price for electricity production (Baget et al., 2024; Segarra et al., 2024). Both households and businesses experience increases in gas and electricity prices; however, firms face larger price hikes because they are more directly exposed to market rates and benefit less from the protective measures available to households. Government interventions, primarily aimed at shielding consumers (Hemmerlé et al., 2023; Sgaravatti et al., 2023), mitigate the impact on household energy bills, while businesses absorb a larger share of the higher costs. Consumer prices for gas and electricity exhibit similar pass-through dynamics, with gas experiencing a slightly stronger response and peaking at approximately 1.5% after 12 months.

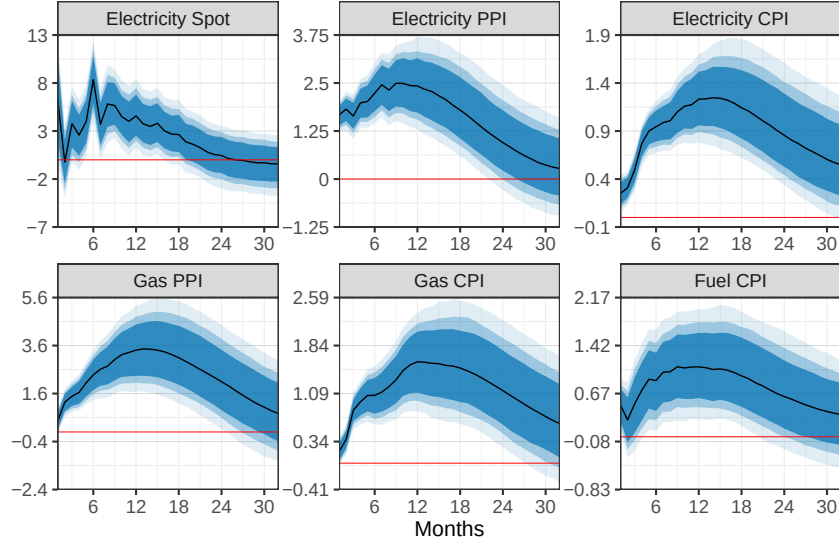


Figure 14: *Euro Area: Energy price responses to a gas supply shock*

Notes: Impulse responses of Euro Area electricity, gas, and fuel prices to a gas supply shock identified using the combined Euro Area supply instrument.

The inverted U-shape of the producer and consumer energy-price responses highlights the gradual pass-through of energy costs, shaped by contractual rigidities in producer and retail markets (Ason, 2022; Baget et al., 2024). These rigidities slow the adjustment of administered and contract-based prices, leading to delayed peaks and a gradual return toward baseline. This pattern is consistent with the broader literature on price stickiness, where mechanisms such as menu costs, information imperfections, and long-term agreements can slow price adjustment even after market-clearing prices have changed (e.g. Rotemberg, 1982; Mankiw, 1985; Borenstein and Shepard, 2002). Finally, fuel consumer prices rise by around 0.5% on impact and peak near 1%. This faster response reflects the closer link between retail fuel prices and wholesale energy markets, especially crude oil and refined petroleum products. This is consistent with Meyler (2009), who finds that oil price shocks in the Euro Area pass through rapidly to consumer fuel prices, with 90% of price changes transmitted within three to six weeks. By contrast, consumer gas and electricity prices adjust more gradually because of long-term contracting, regulatory frameworks, and weaker direct exposure to spot market fluctuations.

Sectoral effects on prices. Figure 15 reports aggregate consumer and producer price responses in Panel (a) and selected HICP consumer categories in Panel (b). Producer prices respond earlier and more strongly than consumer prices, peaking within the first year before gradually declining. Headline consumer prices rise more slowly and remain elevated,

while core inflation moves little initially but builds gradually over the horizon. This pattern indicates that gas supply shocks propagate beyond the direct energy component, with cost pressures passing along the supply chain before reaching consumer prices.

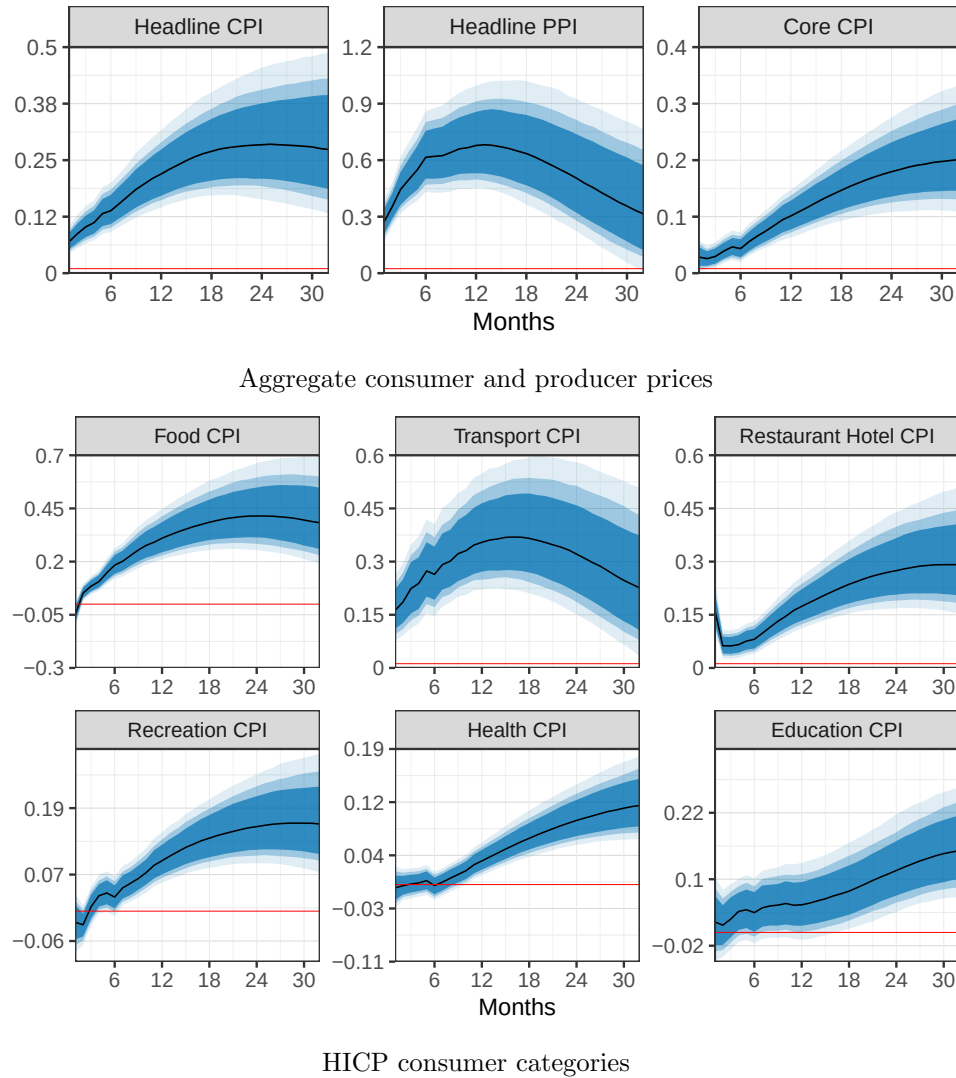


Figure 15: *Euro Area: Responses of aggregate prices and HICP consumer categories to a gas supply shock*

Notes: Impulse responses to a gas supply shock in the Euro Area. Panel (a) reports headline consumer prices, headline producer prices, and core consumer prices. Panel (b) reports six ECOICOP consumer-price categories. Numbers in parentheses denote their weights in the 2023 harmonized index, in per mille of the total basket: food (161), transport (150), restaurants and hotels (105), recreation and culture (89), health (49), and education (10); see Eurostat (2024b).

Inflationary pressures are visible across most consumption categories, but their timing and magnitude differ substantially. Goods prices respond faster and more strongly than

services. Transport reacts quickly, reflecting its direct exposure to fuel prices, while food displays the largest response overall, consistent with natural gas raising input costs through fertilizer production (IEA, 2021). Together, these two divisions account for close to a third of the consumption basket, so their response is central to the aggregate consumer-price effect. Services react later and by less: restaurants and hotels rise gradually, while recreation, health, and education respond more modestly and with a longer delay. Overall, the timing of the responses is consistent with faster direct effects in energy-intensive goods categories and more gradual indirect effects where input-cost and supply-chain channels are likely to matter.

Sectoral effects on quantities. Consistent with the limited aggregate real effect we documented, Figure 16 provides a more detailed analysis of how different subsectors of industrial production respond, ordered by value added. Machinery and vehicles display the clearest immediate declines in industrial production, whereas electricity, gas and steam, chemicals, and paper exhibit more delayed contractions.

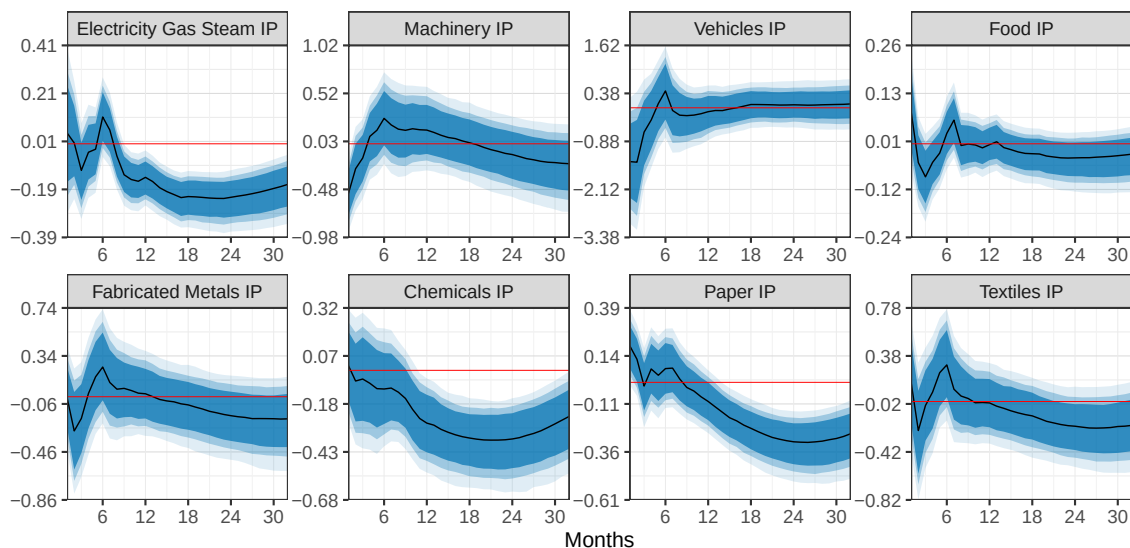


Figure 16: *Euro Area: Industrial-production responses to a gas supply shock*

Notes: Impulse responses to a gas supply shock in the Euro Area of selected sub-sectors, ordered by value added. Electricity, gas, steam and air conditioning supply represents 10.0% of total industrial production, measured by value added; Manufacture of machinery and equipment 9.8%; Manufacture of motor vehicles, trailers and semi-trailers 9.5%; Manufacture of food products 8.6%; Manufacture of fabricated metal products 8.0%; Manufacture of chemicals and chemical products 6.1%; Manufacture of paper and paper products 2.1%; Manufacture of textiles 1.0% (Eurostat, 2024a). The figures refer to 2017, the latest year for which the value added for Europe is available for all sectors.

The negative response in the Electricity, Gas, and Steam sector, the largest in terms of value added, reflects the central role of natural gas as a key input.²⁷ Within manufacturing, level 2 sub-sectors are affected unevenly. Chemicals and paper exhibit lagged but clear negative responses, consistent with their exposure to natural gas as a production input. In chemicals, gas is used as a feedstock for ammonia and other basic intermediates, while in paper manufacturing it is intensively used for process heat and steam generation, making substitution difficult in the short run. Other sectors, such as food and fabricated metals, show smaller or less persistent responses.

Considered alongside the sectoral HICP responses, these quantity responses are consistent with cost-push dynamics in energy-dependent sectors. Several exposed sectors display a U-shaped or inverted J-shaped output response, with an initial decline followed by gradual recovery, while prices tend to follow an inverted U-shaped pattern as higher production costs are passed on to consumers and then dissipate. The heterogeneity of these effects is a common feature of energy shocks (Davis & Haltiwanger, 2001). At the same time, the fact that clear output losses are concentrated in a limited set of sectors suggests muted cascading effects, helping to explain the limited aggregate response of industrial production.

Real effects of gas supply shocks, Euro Area and Germany. Having shown how Euro Area gas supply shocks propagate across sectors, we now examine their aggregate real effects. We compare the Euro Area with Germany, the economy at the center of the debate on the costs of a Russian gas cutoff (Deutsche Bundesbank, 2022; Albrizio et al., 2023; Bachmann et al., 2024; Di Bella et al., 2024). To this end, we augment the baseline specification with national expenditure variables, namely GNE, consumption, and investment.²⁸ We also estimate a Germany-specific version of the model, replacing the Euro Area macroeconomic and gas-quantity variables with their German counterparts. Figure 17 reports the corresponding impulse responses.

²⁷See Gunnella et al. (2022) for sectoral gas intensity statistics in the Euro Area.

²⁸As noted by Moll et al. (2023), Gross National Expenditure (GNE) is the welfare-relevant measure in this context, since GDP may not fully capture the terms-of-trade effects. However, the results for national expenditure should be interpreted with caution, as the data are originally available at a quarterly frequency and interpolated to monthly values for the analysis.

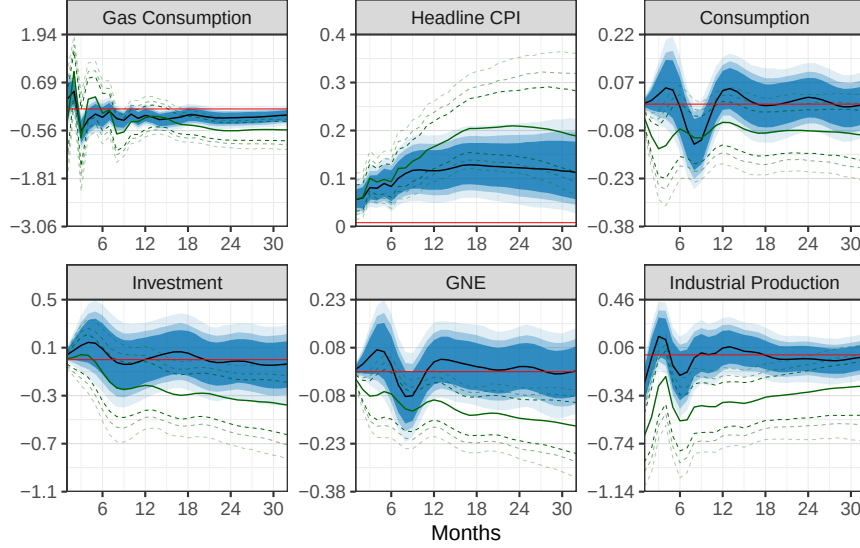


Figure 17: *Euro Area and Germany: Real effects of a gas supply shock*

Notes: Impulse responses to a gas supply shock. Black solid lines with blue shaded credible bands denote Euro Area responses; green solid and dashed lines denote Germany responses. Consumption, investment, and GNE are measured at quarterly frequency and interpolated to monthly frequency using cubic splines. The estimated elasticity of gas demand is -0.10 for Germany and -0.08 for the Euro Area.

Following a one-time increase in the real price of natural gas, gas consumption falls more clearly in Germany than in the Euro Area, especially after the initial months. Using the same identification strategy described in Section 4.1, we estimate a structural gas-demand elasticity of -0.10 for Germany, compared with -0.08 for the Euro Area. These elasticities are informative about the short-run response of gas use, but they do not by themselves identify the margin through which the shock is absorbed. Adjustment can occur through alternative gas sourcing and other relatively low-cost substitution margins, in which case aggregate losses are limited. It can also be costly: when substitution is limited, the shock operates as a real-income loss, either because agents pay substantially more for gas or because they forgo valuable uses of it, and this propagates to spending and production. The real supply-side benchmark of Moll et al. (2023) emphasizes the first type of adjustment: with small gas expenditure shares and scope for alternative sourcing or substitution, even sizable gas-import shortfalls imply limited aggregate losses and weak downstream propagation. Models with Keynesian demand amplification, such as Auclert et al. (2023) and Pieroni (2023), instead emphasize the propagation of the energy shock through lower real income, lower aggregate spending, and further reductions in output.

The German IRFs point to this amplification mechanism: lower gas use is accompanied by lower consumption, lower investment, and a persistent decline in industrial production.

This suggests that part of the adjustment reflects lower effective availability and reduced real income rather than smooth adjustment through alternative sourcing or substitution, consistent with the view that the gas-supply disruption contributed to Germany’s subsequent period of near-stagnation. By contrast, Euro Area real effects are more muted and short-lived, consistent with the sectoral evidence above and with lower exposure to imported natural gas: gas-import expenditure accounts for about 0.8% of nominal GNE in Germany, compared with 0.5% in the Euro Area.²⁹ Euro Area output losses are concentrated in a few exposed industries, whereas German industrial production remains below baseline throughout the reported horizon.

To gauge the magnitude of these responses, we compare our estimates with the quantitative predictions of these models. The comparison is only a benchmark, since those models study counterfactual crisis scenarios whereas our estimates are impulse responses to an identified gas-price shock. Applying the real supply-side approximation in Moll et al. (2023), calibrated with our gas-import expenditure shares and short-run elasticities, to the observed crisis-period import declines—30% for Germany and 12% for the Euro Area—implies small GNE losses, about 0.7% and 0.1%.³⁰ Rescaling our estimated impulse responses to match the same import declines gives trough GNE losses of about 2.6% and 2.0%. Since our estimated elasticities are close to the conservative 0.1 calibration used in this literature, the gap is unlikely to be explained by lower substitutability alone.³¹

The empirical magnitudes are instead closer to the demand-amplification benchmarks. Rescaling our 10% gas-price shock to the persistent energy-price doubling studied by Auclert et al. (2023) gives a German GNE decline of about 1.7%, close to their baseline contraction of about 2%; the corresponding Euro Area decline is about 0.8%. The German estimate is also within the range reported by Pieroni (2023), between 0.8% and 2% in the main scenarios and up to 3.4% in a fossil-gas-only scenario. Taken together, these comparisons suggest that the empirical losses are too large to be rationalized by the real supply-side benchmark alone, and point to amplification beyond the direct supply-side effect, through real-income and

²⁹Figures are pre-crisis values for 2019, based on own calculations using monthly bilateral gas-import volumes from the IEA, valued at the monthly European benchmark gas price, and nominal GNE from Eurostat.

³⁰The import declines are measured over July–March windows relative to pre-crisis gas-import volumes. The supply-side benchmark is computed using the second-order approximation in Moll et al. (2023), 2019 gas-import expenditure shares of nominal GNE, equal to 0.767% for Germany and 0.527% for the Euro Area, and the expenditure-share response implied by our estimated short-run elasticities. Since imports are not part of the baseline specification, the empirical rescaling uses an auxiliary specification in which gas imports and domestic production enter separately, and scales the responses so that the trough import response matches the observed import decline.

³¹Equivalently, matching our estimated losses within the real supply-side framework would require post-shock gas expenditure shares well above those observed during the crisis.

Keynesian aggregate-demand channels.

5.5 Contribution of gas price shocks to the inflation surge

In the introduction, we noted that the surge in natural gas and broader energy prices coincided with rising inflation globally (see again Figure 1). This raises two related questions: To what extent did gas price shocks contribute to the post-pandemic inflation surge? And how can we reconcile the fact that inflation followed a similar path in the United States and the Euro Area, despite stark differences in energy-price dynamics?

We provide a tentative answer to these questions through historical decompositions of headline inflation. Consistent with the impulse-response evidence discussed above, Figure 18 shows that gas price shocks, especially flow-supply and supply-security news shocks, were a key driver of the 2021–2023 inflation surge in the Euro Area. In contrast, supply shocks appear considerably less relevant in the United States. What, then, accounts for the inflationary pressures observed in the United States? A natural candidate, which can be evaluated within our baseline specification, is the role of oil price shocks, driven by global supply imbalances and heightened geopolitical uncertainty. Indeed, we find that oil shocks contributed more significantly to the inflation surge in the United States (see Figure G51 in the online Appendix). This is consistent with Kilian and Zhou (2023), who show that gasoline-price shocks were the largest energy-related contributor to the 2021–2022 U.S. inflation surge, while also emphasizing that non-energy shocks played an important role. The greater contribution of gas supply shocks in the Euro Area also helps explain the higher inflation peak observed there compared to the United States.

Our findings suggest that energy-related supply shocks were major contributors to the recent inflation surge, consistent with the conclusions of several recent studies (see, e.g., Bańbura et al., 2023; De Santis, 2024). However, the fact that gas and oil shocks account for a sizable part of the realized inflation dynamics does not rule out the influence of other, potentially offsetting, factors. Other contributors to the inflation surge likely include persistent supply-chain disruptions, such as component shortages and logistical bottlenecks (Benigno et al., 2022; Stiglitz & Regmi, 2023; De Santis, 2024), as well as the surge in aggregate demand during the reopening phase (Giannone & Primiceri, 2024). At the same time, the subsequent tightening of monetary policy may have partly offset these inflationary pressures, as central banks responded through sharp increases in interest rates.

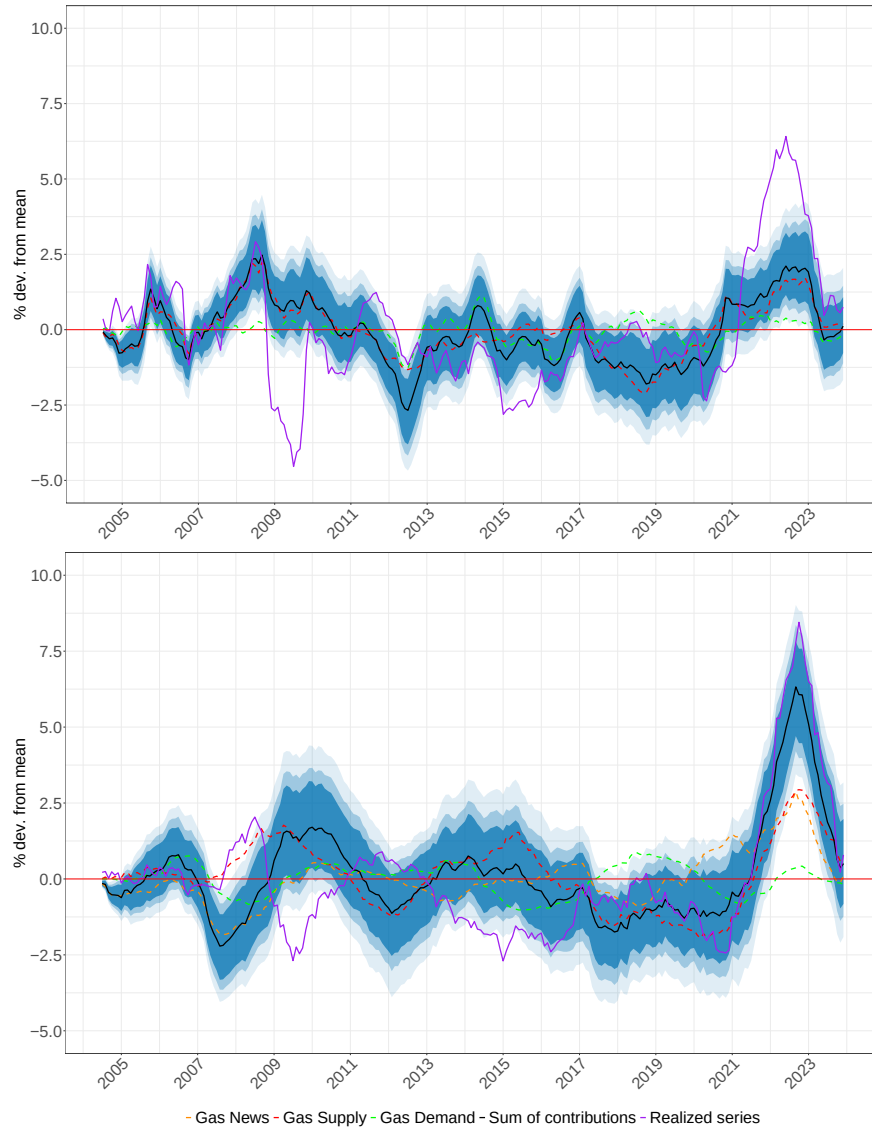


Figure 18: *Historical decomposition of headline inflation in the United States (top panel) and the Euro Area (bottom panel).*

Notes: Estimated contributions of gas flow-supply shocks are shown by the red dashed line, Euro Area supply-security news shocks by the orange dashed line, gas demand shocks by the green dashed line, and their combined contribution by the solid black line with blue credible bands. All contributions are expressed as percentage deviations from the mean. The realized headline inflation series is shown in purple.

6 Conclusions

In this paper, we propose a novel identification strategy to identify demand and supply shocks in natural gas markets. Using exogenous variation in temperatures, we identify gas demand shocks, while high-frequency movements in gas prices around gas market-relevant events allow us to distinguish between flow-supply shocks and supply-security news shocks. This approach enables us to estimate gas demand and supply elasticities for both the United States and the Euro Area and to present detailed evidence on the macroeconomic and sectoral impacts of natural gas shocks, offering new insights into their transmission mechanisms.

Our analysis reveals that natural gas shocks can have important macroeconomic effects, but their transmission differs sharply between the Euro Area and the United States, reflecting structural disparities in supply composition, demand elasticities, and market dynamics. In the United States, supply shocks have modest price effects and leave aggregate activity broadly unchanged. In the Euro Area, by contrast, supply shocks are primarily inflationary, especially when they capture news about future availability and supply security, while aggregate real effects remain limited. These news shocks operate through an expectation-driven mechanism linked to uncertainty, as reflected in increased inventory accumulation and heightened financial volatility. Their real effects, however, vary across sectors. Output losses are concentrated in industries where natural gas enters either as a difficult-to-substitute production input or as a source of process heat, while rising costs are passed downstream over time, sustaining inflationary pressures particularly in goods-producing sectors relative to services. In the case of Germany, however, our analysis shows that gas supply shocks have significant real effects, consistent with the interpretation that the country’s recent economic stagnation was, at least in part, a consequence of the disruption in Russian gas flows. Finally, we show that gas price shocks have been a key contributor to the recent inflation surge in the Euro Area, whereas their role has been considerably smaller in the United States.

Our findings highlight several important policy implications. First, the different transmission of flow-supply shocks and supply-security news shocks calls for different policy margins. Realized disruptions point to the value of supplier diversification, short-term procurement, and adequate storage capacity. The news shocks, by contrast, underscore the importance of credible and timely policy communication about future supply security, since expectations can affect current prices even before actual flows are disrupted. Second, the strong pass-through from gas prices to electricity prices in the Euro Area suggests that electricity-market design can help limit the amplification of gas shocks, especially where marginal gas-fired generation continues to shape wholesale and retail prices. Finally, our results speak to the green transition. Gas-specific price increases can redirect power generation toward more carbon-

intensive fuels in the short run, raising CO₂ emissions. This implies that policies aimed at reducing gas dependence need to be paired with measures that expand clean-energy substitutes or limit switching toward coal and oil; otherwise, lower gas use may come at the cost of higher emissions.

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Online Appendix

Gas Prices and the Macroeconomy

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A VAR econometrics

Consider the structural VAR(p) model:

$$\mathbf{B}_0 \mathbf{y}_t = \mathbf{B}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{B}_p \mathbf{y}_{t-p} + \mathbf{w}_t, \quad (15)$$

where $\mathbf{y}_t$ is a $K \times 1$ vector of endogenous variables, assumed to have zero mean without loss of generality. The vector $\mathbf{w}_t$ is a $K \times 1$ vector of structural shocks. Following the standard definition of structural shocks as primitive exogenous disturbances, we assume that they are mutually uncorrelated, so that $\mathbb{E}(\mathbf{w}_t \mathbf{w}_t') = \Sigma_w$ is diagonal (Ramey, 2016, Section 2.1). Since the matrices $\mathbf{B}_0$ and $\mathbf{w}_t$ are generally unobserved, we rely on the reduced-form representation to estimate the model:

$$\begin{aligned} \mathbf{y}_t &= \mathbf{B}_0^{-1} \mathbf{B}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{B}_0^{-1} \mathbf{B}_p \mathbf{y}_{t-p} + \mathbf{B}_0^{-1} \mathbf{w}_t \\ &= \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \end{aligned} \quad (16)$$

where $\mathbf{A}_i = \mathbf{B}_0^{-1} \mathbf{B}_i$ for $i = 1, \dots, p$, and $\mathbf{u}_t = \mathbf{B}_0^{-1} \mathbf{w}_t$. We estimate the reduced-form parameters $\mathbf{A}_1, \dots, \mathbf{A}_p$, $\Sigma_u = \mathbb{E}(\mathbf{u}_t \mathbf{u}_t')$, and the reduced-form residuals $\mathbf{u}_t$ using Bayesian methods (see below). The key equation linking the reduced-form innovations to the structural shocks is:

$$\mathbf{u}_t = \mathbf{B}_0^{-1} \mathbf{w}_t, \quad (17)$$

where the matrix $\mathbf{B}_0^{-1}$ has to be retrieved.

A.1 Impulse response functions

Impulse response functions (IRFs) describe the responses of each element of $\mathbf{y}_t = (y_{1t}, \dots, y_{Kt})'$ to a one-time impulse in each element of $\mathbf{w}_t = (w_{1t}, \dots, w_{Kt})'$:

$$\frac{\partial \mathbf{y}_{t+h}}{\partial \mathbf{w}_t'} = \Theta_h, \quad h = 0, 1, 2, \dots, H \quad (18)$$

This is a $(K \times K)$ matrix whose elements are given by

$$\theta_{jk,h} = \frac{\partial y_{j,t+h}}{\partial w_{kt}}.$$

In order to recover the IRFs, we first resort to the VAR(1) representation of the VAR(p) process:

$$\mathbf{Y}_t = \mathbf{A} \mathbf{Y}_{t-1} + \mathbf{U}_t, \quad (19)$$

with

$$\mathbf{Y}_t \equiv \begin{bmatrix} \mathbf{y}_t \\ \vdots \\ \mathbf{y}_{t-p+1} \end{bmatrix} \quad \mathbf{A} \equiv \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \dots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_K & \mathbf{0} & & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_K & & \mathbf{0} & \mathbf{0} \\ \vdots & & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I}_K & \mathbf{0} \end{bmatrix} \quad \mathbf{U}_t \equiv \begin{bmatrix} \mathbf{u}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}.$$

By recursive substitution, it can be shown that the response of variable $j = 1, \dots, K$ to a unit reduced-form innovation in component k , h periods in the past, for $k = 1, \dots, K$, is given by $\Phi_h = [\phi_{jk,h}] \equiv \mathbf{J} \mathbf{A}^h \mathbf{J}'$, where $\mathbf{J} \equiv \begin{bmatrix} \mathbf{I}_K & \mathbf{0}_{K \times K(p-1)} \end{bmatrix}$ is a selector matrix. These are sometimes called dynamic multipliers or reduced-form impulse responses.

Under covariance stationarity of $\mathbf{y}_t$, the system can be expressed as a weighted average of current and past shocks (multivariate MA(∞) representation), with weights Φ_h :

$$\mathbf{y}_t = \sum_{s=0}^{\infty} \Phi_s \mathbf{u}_{t-s} = \sum_{s=0}^{\infty} \Phi_s \mathbf{B}_0^{-1} \mathbf{w}_{t-s} = \sum_{s=0}^{\infty} \Theta_s \mathbf{w}_{t-s}, \quad (20)$$

where $\Theta_h \equiv \Phi_h \mathbf{B}_0^{-1}$. It follows that:

$$\frac{\partial \mathbf{y}_t}{\partial \mathbf{w}'_{t-h}} = \frac{\partial \mathbf{y}_{t+h}}{\partial \mathbf{w}'_t} = \Theta_h, \quad h = 0, 1, 2, \dots, H$$

These structural impulse responses can be obtained simply by post-multiplying Φ_h by $\mathbf{B}_0^{-1}$.

A.2 Forecast error variance decomposition

From equation (20), the h -step-ahead forecast error is given by

$$\mathbf{y}_{t+h} - \mathbb{E}_t[\mathbf{y}_{t+h}] = \sum_{s=0}^{h-1} \Theta_s \mathbf{w}_{t+h-s},$$

with forecast error variance:

$$\Sigma_h \equiv \mathbb{E}_t[(\mathbf{y}_{t+h} - \mathbb{E}_t[\mathbf{y}_{t+h}])(\mathbf{y}_{t+h} - \mathbb{E}_t[\mathbf{y}_{t+h}])'] = \sum_{s=0}^{h-1} \Theta_s \Theta_s'. \quad (21)$$

The j^{th} entry on the diagonal of Σ_h is therefore the variance of the forecast error of variable j due to all shocks $1, \dots, K$. This is:

$$[\Sigma_h]_{j,j} = \sum_{s=0}^{h-1} \sum_{\ell=1}^K ([\Theta_s]_{j,\ell})^2$$

while the contribution of a single shock k at horizon h is

$$\sum_{s=0}^{h-1} ([\Theta_s]_{j,k})^2$$

It follows that the percentage contribution of shock k to the total error variance of variable j at horizon h is:

$$\text{FEVD}_{j,k}(h) = \frac{\sum_{s=0}^{h-1} ([\Theta_s]_{j,k})^2}{\sum_{s=0}^{h-1} \sum_{\ell=1}^K ([\Theta_s]_{j,\ell})^2} \quad (22)$$

Partially identified case. When not all columns of Θ_h are identified, we cannot compute Σ_h directly using equation (21). Instead, leveraging the identity $\Theta_h \Theta_h' = \Phi_h \mathbf{B}_0^{-1} (\mathbf{B}_0^{-1})' \Phi_h' = \Phi_h \Sigma_u \Phi_h'$, we compute:

$$\text{FEVD}_{j,k}(h) = \frac{\sum_{s=0}^{h-1} ([\Theta_s]_{j,k})^2}{\sum_{s=0}^{h-1} [\Phi_s \Sigma_u \Phi_s']_{j,j}}. \quad (23)$$

Note that this is derived without loss of generality under the unit-variance normalization of the structural shocks: $\mathbb{E}[\mathbf{w}_t \mathbf{w}_t'] = \mathbf{I}_K$.

A.3 Structural shocks

In the partially identified case, where not all columns of $\mathbf{B}_0^{-1}$ are recovered, the matrix cannot be directly inverted to obtain the structural shocks via equation (17). Nevertheless, Stock and Watson (2018) demonstrated that the corresponding structural shock can still be recovered using the following expression:

$$\begin{aligned} \left[(\tilde{\mathbf{B}}_0^{-1})_{:,j} \right]' \Sigma_u^{-1} \mathbf{u}_t &= \tilde{\mathbf{b}}'_{0,j} \mathbf{u}_t = \mathbf{e}'_j \tilde{\mathbf{B}}_0 \mathbf{u}_t = \mathbf{e}'_j \tilde{\mathbf{B}}_0 \mathbf{B}_0^{-1} \mathbf{w}_t \\ &= \mathbf{e}'_j \tilde{\mathbf{B}}_0 \tilde{\mathbf{B}}_0^{-1} \Sigma_w^{-1/2} \Sigma_w^{1/2} \tilde{\mathbf{w}}_t = \mathbf{e}'_j \tilde{\mathbf{w}}_t = \tilde{w}_{j,t}, \end{aligned}$$

where $\mathbf{e}_j$ denotes the j -th standard basis vector and, without loss of generality, we adopted the unit-variance representation of the model in (16), given by:

$$\mathbf{y}_t = \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \tilde{\mathbf{B}}_0^{-1} \tilde{\mathbf{w}}_t$$

$$\begin{aligned}\tilde{\mathbf{w}}_t &= \Sigma_w^{-1/2} \mathbf{w}_t \\ \tilde{\mathbf{B}}_0^{-1} &= \mathbf{B}_0^{-1} \Sigma_w^{1/2}\end{aligned}$$

with $\Sigma_{\tilde{\mathbf{w}}} = \mathbf{I}_K$.

A.4 Historical decomposition

Structural impulse responses describe average movements in the data. However, we are often interested in quantifying how much a given identified structural shock contributes to the historically observed fluctuations of the variables in the VAR. For covariance stationary VAR models, such contributions—known as historical decompositions—can be computed by rewriting equation (20) as:

$$\mathbf{y}_t = \sum_{s=0}^{t-1} \Theta_s \mathbf{w}_{t-s} + \sum_{s=t}^{\infty} \Theta_s \mathbf{w}_{t-s}.$$

Since under covariance stationarity the MA coefficients decay over time, it holds that

$$\mathbf{y}_t \approx \sum_{s=0}^{t-1} \Theta_s \mathbf{w}_{t-s}. \quad (24)$$

This approximation can only be computed from $t = p + 1$ onward and is more accurate for periods toward the end of the sample.

A.5 Bayesian estimation

The Bayesian estimation procedure we implement employs Minnesota (Doan et al., 1984), sum-of-coefficients (see also Sims & Zha, 1998; Robertson & Tallman, 1999), and dummy-initial-observation (Sims, 1993) priors. These priors are incorporated through dummy observations, with the degree of prior informativeness selected based on the marginal likelihood of the observed data, following the approach of Bańbura et al. (2007). The optimal level of prior tightness is determined using the hierarchical procedure proposed by Giannone et al. (2015). Parameter estimation is conducted via a standard Gibbs sampling algorithm.

Prior distribution. Following Kadiyala and Karlsson (1997), we specify the prior distributions as

$$\text{vec}(\beta) \mid \Psi \sim \mathcal{N}(\text{vec}(\beta_0), \Psi \otimes \Omega_0) \quad \text{and} \quad \Psi \sim \mathcal{IW}(S_0, \alpha_0),$$

where $\beta = [\mathbf{A}_1, \dots, \mathbf{A}_p, \mathbf{c}]$, with $\mathbf{c}$ an intercept and Ψ the prior for the covariance matrix of reduced-form residuals Σ_u from (16). The prior parameters β_0, Ω_0, S_0 and α_0 are chosen so that prior expectations and variances coincide with those implied by Minnesota, sum-of-coefficients and dummy-initial observation priors (more below). We implement the prior by adding dummy observations: Bańbura et al. (2007) have shown that adding T_d dummy observations $\mathbf{y}_d$ and $\mathbf{x}_d$ to the VAR(p) induces the Normal-Inverted Wishart priors:

$$\begin{aligned}\beta_0 &= (\mathbf{x}_d' \mathbf{x}_d)^{-1} (\mathbf{x}_d' \mathbf{y}_d); \\ \Omega_0 &= (\mathbf{x}_d' \mathbf{x}_d)^{-1}; \\ S_0 &= (\mathbf{y}_d - \mathbf{x}_d \beta_0)' (\mathbf{y}_d - \mathbf{x}_d \beta_0); \\ \alpha_0 &= T_d - k,\end{aligned}$$

where S_0 is a $(k \times k)$ scale matrix and α_0 are the degrees of freedom of the Inverted Wishart distribution. Conveniently, this prior distribution is conjugate, implying that the posterior distribution has a known analytical form.

Minnesota prior. The Minnesota prior (Doan et al., 1984; Litterman, 1986) formalizes the belief that, ex ante, each variable in the system follows a random walk with drift, represented by the process $y_t = c + y_{t-1} + u_t$. This prior encodes the assumption that more recent lags contain greater predictive content than more distant ones, and that a variable's own history is more informative for forecasting its future values than the histories of other variables in the system. Accordingly, the prior shrinks the diagonal elements of $\mathbf{A}_1$ toward one, while shrinking the off-diagonal elements of $\mathbf{A}_1$ and all elements of $\mathbf{A}_2, \dots, \mathbf{A}_p$ toward zero. The mean of the prior distribution is given by:

$$\mathbb{E}[(\mathbf{A}_s)_{ij}] = \begin{cases} \delta_i & \text{if } i = j \text{ and } s = 1 \\ 0 & \text{otherwise} \end{cases}$$

Originally, $\delta_i = 1$ was imposed for all i , based on the notion that most variables exhibit strong persistence. However, this prior is not suitable for variables that display substantial mean reversion. In such cases, we adopt the belief that the variable behaves like white noise by setting $\delta_i = 0$. The prior variance of the parameters is

$$\mathbb{V}[(A_s)_{ij}] = \begin{cases} \frac{\lambda^2}{s^2} & \text{if } i = j \\ \vartheta \frac{\lambda^2}{s^2} \cdot \frac{\sigma_i^2}{\sigma_j^2} & \text{otherwise} \end{cases}$$

where the hyperparameter λ controls the relative importance of prior and data. When $\lambda \rightarrow 0$, no weight is given to the data and vice versa for $\lambda \rightarrow \infty$. The factor $1/s^2$ is the rate at which prior variance decreases with increasing lag length, and σ_i^2/σ_j^2 accounts for the different scale and variability of the data. The coefficient $\vartheta \in (0, 1)$ governs the degree of shrinkage applied to the lags of other variables relative to a variable's own lags.

The original formulation of the Minnesota prior assumes that the covariance matrix of the residuals is diagonal and known: $\Sigma_u = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$. However, for structural analysis, it is important to account for potential correlations among residuals across equations. To address this, we follow the approach of Kadiyala and Karlsson (1997) and Robertson and Tallman (1999), who propose a Normal-inverted Wishart prior that preserves the key features of the Minnesota prior. This specification is valid under the assumption $\vartheta = 1$, which we impose. The prior on the intercept c is assumed to be diffuse. These priors are implemented by adding the following dummy observations at the start of the sample:

$$\mathbf{y}_{(kp+k+1) \times k}^+ = \begin{bmatrix} \frac{\text{diag}(\delta_1 \sigma_1, \dots, \delta_k \sigma_k)}{\lambda} \\ \mathbf{0}_{k(p-1) \times k} \\ \hline \text{diag}(\sigma_1, \dots, \sigma_k) \\ \hline \mathbf{0}_{1 \times k} \end{bmatrix} \quad \mathbf{x}_{(kp+k+1) \times (kp+1)}^+ = \begin{bmatrix} J_p \otimes \frac{\text{diag}(\sigma_1, \dots, \sigma_k)}{\lambda} & \mathbf{0}_{kp \times 1} \\ \hline \mathbf{0}_{k \times kp} & \mathbf{0}_{k \times 1} \\ \hline \mathbf{0}_{1 \times kp} & \mathbf{1}_{k \times 1} \varepsilon \end{bmatrix}$$

where the first block of dummies imposes prior beliefs on the autoregressive coefficients, the second block implements the prior for the covariance matrix, the third block reflects the uninformative prior for the intercept (ε is a very small number).

Refinements of the Minnesota prior have been introduced to accommodate unit roots and cointegration, reflecting common practices in applied research. These refinements involve additional priors designed to diminish the influence of the deterministic component.

Sum-of-coefficients prior. This prior mitigates explosive roots by shrinking the sum of coefficients, so that posterior draws exhibit a unit root rather than explosive behavior. We can rewrite the VAR in its error correction form:

$$\Delta \mathbf{y}_t = \mathbf{c} - (\mathbf{I}_n - \mathbf{A}_1 - \dots - \mathbf{A}_p) \mathbf{y}_{t-1} + \tilde{\mathbf{A}}_1 \Delta \mathbf{y}_{t-1} + \dots + \tilde{\mathbf{A}}_{p-1} \Delta \mathbf{y}_{t-p+1} + \mathbf{u}_t.$$

The term $\tilde{\mathbf{A}}_1 \Delta \mathbf{y}_{t-1} + \dots + \tilde{\mathbf{A}}_{p-1} \Delta \mathbf{y}_{t-p+1}$ is made of stationary terms and hence does not impact the stationarity of the model. It follows that if $\Pi := \mathbf{I}_n - \mathbf{A}_1 - \dots - \mathbf{A}_p = \mathbf{0}$ (which is implied by a VAR in first differences) the model is stationary. The sum-of-coefficients prior imposes “inexact differencing” by shrinking the term towards 0 in the limit ($\Pi \rightarrow \mathbf{0}$). The

prior states that a no-change forecast for all variables is a good forecast at the beginning of the sample. Then, either all the variables are at their unconditional mean, which implies that the model is stationary despite the unit roots in the variables (implying cointegration), or the dynamic of the system is characterized by an unspecified number of unit roots, and the variables share a common stochastic trend. It is implemented by adding at the beginning of the sample dummy observations constructed in the following way:

$$\mathbf{y}_{k \times k}^{++} = \begin{bmatrix} \frac{\mu_1 \delta_1}{\tau} & 0 & \dots & 0 \\ 0 & \frac{\mu_2 \delta_2}{\tau} & \dots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \frac{\mu_k \delta_k}{\tau} \end{bmatrix} \quad \mathbf{x}_{k \times (kp+1)}^{++} = \left[\mathbf{y}^{++}, \dots, \mathbf{y}^{++}, \mathbf{0}_{k \times 1} \right],$$

where μ_i captures the unconditional mean of variable y_{it} , δ_i is the term previously defined, and the hyperparameter τ controls the variance of these prior beliefs: as $\tau \rightarrow \infty$, the prior becomes uninformative, while $\tau \rightarrow 0$ enforces the presence of a unit root in each equation and rules out cointegration entirely. This is because we can have cointegration (individual series display unit-roots but some linear combination of the series is stationary) only if Π has reduced rank but is not zero, otherwise the VAR becomes a purely differenced system.

Dummy-initial-observation. Since, in the limit, the sum-of-coefficients prior rules out cointegration, another prior can be implemented to push the variables toward cointegration. It is designed to correct the bias of the sum-of-coefficients prior against cointegration, while still addressing overfitting of the deterministic component. This is achieved by adding dummy observations at the beginning of the sample, constructed as follows:

$$\mathbf{y}_{k \times k}^{+++} = \begin{bmatrix} \frac{\mu_1}{\gamma} & 0 & \dots & 0 \\ 0 & \frac{\mu_2}{\gamma} & \dots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & \frac{\mu_k}{\gamma} \end{bmatrix} \quad \mathbf{x}_{k \times (kp+1)}^{+++} = \left[\mathbf{y}^{+++}, \dots, \mathbf{y}^{+++}, \mathbf{1}_{k \times 1} / \gamma \right]$$

The hyperparameter γ controls the tightness of the prior implied by this artificial observation. As $\gamma \rightarrow \infty$, the prior becomes uninformative. As $\gamma \rightarrow 0$, the model tends to a form in which either all variables are stationary with means equal to the sample averages of the initial conditions, or there are unit root components without drift terms.

Posterior distribution. When we augment the data with the dummy observations

$$\mathbf{y}_* = \begin{bmatrix} \mathbf{y} \\ \mathbf{y}^+ \\ \mathbf{y}^{++} \\ \mathbf{y}^{+++} \end{bmatrix}, \quad \mathbf{x}_* = \begin{bmatrix} \mathbf{x} \\ \mathbf{x}^+ \\ \mathbf{x}^{++} \\ \mathbf{x}^{+++} \end{bmatrix}$$

we can rewrite the $\text{VAR}(p)$ as

$$\mathbf{y}_* = \mathbf{x}_* \beta + \mathbf{u}_*$$

and it can be shown that the posterior distribution has the form

$$\text{vec}(\beta) \mid \Psi, \mathbf{y} \sim \mathcal{N}(\text{vec}(\tilde{\beta}), \Psi \otimes (\mathbf{x}_*' \mathbf{x}_*)^{-1}) \quad \text{and} \quad \Psi \mid \mathbf{y} \sim \mathcal{IW}(\tilde{\Sigma}, T_d + T + 2 - n),$$

where the posterior expectations of the coefficients correspond to the OLS estimates of the regression model based on the appended data, with:

$$\tilde{\beta} = (\mathbf{x}_*' \mathbf{x}_*)^{-1} (\mathbf{x}_*' \mathbf{y}_*) \quad \tilde{\Sigma}_u = (\mathbf{y}_* - \mathbf{x}_* \tilde{\beta})' (\mathbf{y}_* - \mathbf{x}_* \tilde{\beta})$$

Choosing the informativeness of the priors. As explained, the prior distribution of our Bayesian VAR is governed by a set of hyperparameters (or hyperpriors), which control the strength and structure of the imposed beliefs. δ_i is an indicator that takes the value 1 or 0, reflecting whether the i^{th} variable is assumed to follow a highly persistent process (random walk, $\delta_i = 1$) or a stationary process (white noise, $\delta_i = 0$). σ_i is the standard deviation of the residuals for variable i under the prior, used to scale the variance and account for differences in variable volatility. λ controls the overall tightness of the prior distribution. When $\lambda \rightarrow 0$, the prior dominates and the posterior shrinks heavily toward the prior mean. When $\lambda \rightarrow \infty$, the data dominate and the posterior approaches the OLS estimate. τ governs the strength of the sum-of-coefficients prior. γ controls the tightness of the dummy initial observation prior. The objective is to choose these hyperparameters to ensure that the prior structure is neither too tight nor too loose relative to the data. When well-calibrated, they help regularize the estimation in high-dimensional settings, improve forecast performance, and mitigate overfitting.

Following the hierarchical approach of Giannone et al. (2015), we select the hyperparameters that maximize the marginal likelihood of the data. From a Bayesian standpoint, selecting the informativeness of the prior distribution is conceptually analogous to conducting inference on any other unknown parameter of the model. Consider, for example, a model defined by a likelihood function $p(y \mid \theta)$ and a prior distribution $p_\gamma(\theta)$, where θ denotes the

vector of $\text{VAR}(p)$ parameters and γ represents the set of hyperparameters (or hyperpriors) that govern the specification of the prior. Although these hyperparameters do not directly influence the likelihood, they shape the prior distribution. Consequently, it is natural to treat them within a hierarchical modeling framework by substituting $p_\gamma(\theta)$ with $p(\theta \mid \gamma)$ and performing inference on γ , evaluating their posterior.

B Data and descriptive statistics

This appendix documents the data and provides descriptive evidence on the natural gas markets studied in the paper. It first reports the sources, transformations, and coverage of the main empirical variables, then compares the energy market structures of the Euro Area and the United States, and finally documents the role of the TTF as the benchmark price for European natural gas.

B.1 Data sources

Table [B3](#) reports the sources, sample coverage, and transformations for the main data used in the empirical analysis.

Variable	Description	Source	Time Coverage	Units
EA Instruments				
TRNLTTFE-h (RIC)	TTF natural gas futures contracts (settlement price); monthly (1M-1M), quarterly (1Q-1Q), seasonal (1S-2S), and calendar-year (1Y)	Datstream	05/01/2004-01/01/2024	100Δ log
LCOe-h (RIC)	Brent crude oil futures h-month contracts from 1M to 12M (settlement price)	Datstream	18/03/1980-01/01/2024	100Δ log
ERA5 surface temperature	2m daily mean temperature on a 0.25° grid, aggregated to country level (GADM0) and weighted by lagged national gas consumption	ERA5, EUROSTAT	01/01/1990-01/01/2024	100Δ log
Heating and cooling degree days	Monthly HDD and CDD, aggregated to country level and weighted by lagged national gas consumption	EUROSTAT	1975M1-2024M1	None
EA Variables (baseline)				
Gas Consumption	EA gas consumption (GDINCTROAD), SA	IEA	1984M1-2024M1	100 * log
Gas Supply	EA dry marketable production (INDPROD) plus gas imports from non-EA countries (IMPORTS) minus gas exports to non-EA countries (EXPORTS), SA	IEA	1984M1-2024M1	100 * log
Gas Inventories	EA closing stock levels (CSNAFTER), SA	IEA	1984M1-2024M1	100 * log
Gas Price	Dutch TTF spot price at close (TRNLTFED1) in Euro per MWh, deflated by the EA HICP all-items index	Datstream	2004M1-2024M1	100 * log
Headline CPI	HICP all-items index (000000), SA	EUROSTAT	2000M12-2024M1	100 * log
Industrial Production	Total industry excluding construction (B-D), SA	EUROSTAT	1995M1-2024M1	100 * log
Oil Price	Brent spot price at close (DCOILBRETEU) in Euro per barrel, deflated by the EA HICP all-items index	FRED	1990M1-2024M1	100 * log
US Instruments				
NGe-h (RIC)	NYMEX HH natural gas futures h-month contracts from 1M to 12M (settlement price)	Datstream	03/04/1990-01/01/2024	100Δ log
CLe-h (RIC)	WTI crude oil futures h-month contract from 1M to 12M (settlement price)	Datstream	18/03/1980-01/01/2024	100Δ log
ERA5 surface temperature	2m daily mean temperature on a 0.25° grid, aggregated to state level (GADM1) and weighted by lagged state gas consumption	ERA5, EIA	01/01/1940-01/01/2024	None
Heating and cooling degree days	Monthly HDD and CDD by state, aggregated to the national level and weighted by lagged state gas consumption	NOAA, EIA	1985M1-2024M1	None
US Variables (baseline)				
Gas Consumption	U.S. Gas total consumption (N9140CIS1), SA	EIA	2001M1-2024M1	100 * log
Gas Supply	U.S. Dry gas production (N9070US1) plus gas imports (N9100US2) minus gas exports (N9130US2), SA	EIA	1984M1-2024M1	100 * log
Gas Inventories	U.S. Total Natural Gas in Underground Storage Working Gas (N5020US2), SA	EIA	1975M9-2024M1	100 * log
Gas Price	NYMEX HH spot price at close (NG-W-HH-SM) in Dollar per Million BTU, deflated by the US CPI all-items index	FRED	1991M8-2024M1	100 * log
Headline CPI	Headline CPI index (CPIAU CSL), SA	FRED	1960M1-2024M1	100 * log
Industrial Production	Total index (INDPRO), SA	FRED	1974M1-2024M1	100 * log
Oil Price	WTI spot price at close (DCOILWTCO) in Dollar per barrel, deflated by the US CPI all-items index	FRED	1986M1-2024M1	100 * log
Additional Variables				
Gas Production (EA)	EA Dry marketable production (INDPROD), SA	IEA	1984M1-2024M1	100 * log
Gas Net Imports (EA)	EA Gas imports from non-EA countries (IMPORTS) minus EA Gas exports to non-EA countries (EXPORTS), SA	IEA	1984M1-2024M1	100 * log
Gas, Oil, Coal, and Renewable Power (EA)	Monthly electricity generation by source in GWh, SA	ENTSO-E	2007M1-2024M1	100 * log
Financial Volatility (EA)	Euro Stoxx 50 Volatility Index (VSTOXX)	STOXX	1990M1-2024M1	100 * log
Core CPI (EA)	HICP index excluding energy and food seasonally adjusted (TOTLXNHCFOOD)	EUROSTAT	2000M12-2024M1	100 * log
Headline PPI (EA)	Total industry producer price index, SA (PRONNS020)	EUROSTAT	2000M1-2024M1	100 * log
Nominal Exchange Rate (EA)	Nominal Broad Effective Exchange Rate for EA (NBXMBIS)	FRED	1994M1-2024M1	100 * log
IP C10-C38 (EA)	NACE2 2-digit level Manufacturing Industrial Production indexes, SA	EUROSTAT	1991M1-2024M1	100 * log
PPI D35-10/20 (EA)	NACE2 4-digit level Electricity and Gas Producer Price indexes, SA	EUROSTAT	2000M1-2024M1	100 * log
HICP CP01-CP12 (EA)	ECOIPOP 1-digit level indexes, SA	EUROSTAT	2000M1-2024M1	100 * log
Electricity Price (EA)	Consumption-weighted average of euro-area day-ahead spot prices, deflated by the EA HICP	Refinitiv, EUROSTAT	2004M1-2024M1	100 * log
Electricity Consumption (EA)	Monthly electricity demand, EA19 aggregate, SA	ENTSO-E	2007M1-2024M1	100 * log
German macro variables (DE)	Headline HICP (CP90.DE) and industrial production (B-D), defined and transformed as the corresponding euro-area series	EUROSTAT	2000M12-2024M1	100 * log
German gas variables (DE)	German gas consumption, supply and inventories, defined as the corresponding euro-area series	IEA	1984M1-2024M1	100 * log
Consumption, Investment, GNE (EA, DE)	National accounts aggregates, chain-linked volumes, SA; quarterly, interpolated to monthly by cubic spline	EUROSTAT	1995Q1-2023Q4	100 * log
Gas Production (US)	U.S. Dry Natural Gas Production (N9070US1) in Bcf, SA	EIA	1973M1-2024M1	100 * log
Financial Volatility (US)	CBOE Volatility Index (VIXCLS)	FRED	1990M1-2024M1	100 * log
Gas, Oil, and Coal Power (US)	Fossil fuel consumption by the electric power sector, SA	EIA	2004M1-2024M1	100 * log
Nominal Exchange Rate (US)	Nominal Broad Effective Exchange Rate for US (NBUSBIS)	FRED	1994M1-2024M1	100 * log
Electricity Price (US)	On-peak wholesale electricity spot price, deflated by the US CPI	EIA, ICE	2001M1-2024M1	100 * log
Electricity Consumption (US)	Electricity retail sales, SA	EIA	2001M1-2024M1	100 * log
Renewable Power (US)	Renewable power generation in thousand MWh, SA	EIA	2001M1-2024M1	100 * log
CO2 Emissions (EA, US)	Total CO ₂ emissions from fossil fuel combustion and industrial processes, measured in gigagrams	EDGAR CO2	1970M1-2022M12	100 * log
Weather controls (EA, US)	Maximum wind gust, precipitation, SPEI drought index and solar radiation, aggregated to the region with fixed 2015 technology-specific generation weights, seasonally adjusted and standardised	ERA5-Land, OWD	1901M1-2024M1	Standardized

Table B3: *Data description, sources, and coverage*

B.2 Energy statistics in the United States and Euro Area

Figures B19 and B20 place natural gas within the broader energy mix. Natural gas accounts for a substantial share of energy supply in both regions, but the composition of that supply and the exposure to external producers differ markedly across regions.

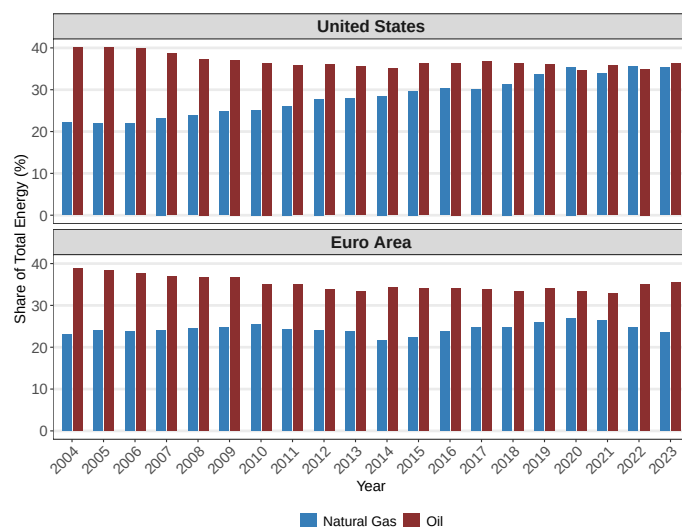


Figure B19: *Natural gas and crude oil and petroleum products as a share of total energy supply*

Notes: Share of natural gas and crude oil and petroleum products in total energy supply (2004–2023); United States in the top panel, Euro Area in the bottom panel. Energy supply is defined as production plus imports minus exports minus stock changes. Source: IEA.



Figure B20: *Energy supply by source*

Notes: Composition of total energy supply by source, averaged over 2004–2023; United States in the left panel, Euro Area in the right panel. Energy supply is defined as domestic production plus imports minus exports minus stock changes. Source: IEA.

Figure B21 shows that gas is used across residential, industrial, and power-generation activities, with residential use accounting for a larger share in the Euro Area. Figure B22 highlights the sharper difference on the supply side: the Euro Area relies heavily on imported gas, whereas the United States has become increasingly self-sufficient.

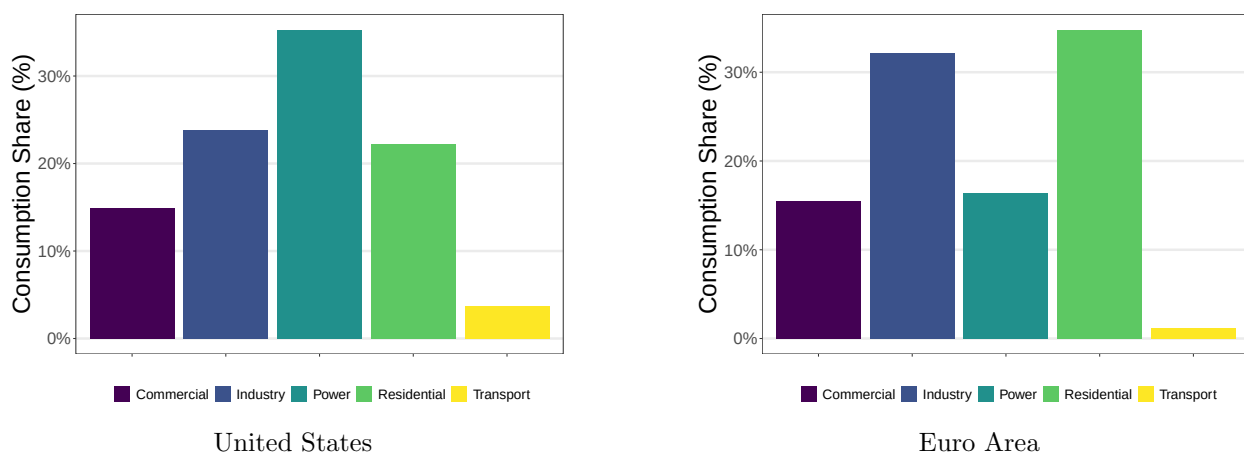


Figure B21: *Natural gas consumption share by sector*

Notes: Sectoral shares of natural gas consumption, averaged over 2004–2022; United States in the left panel, Euro Area in the right panel. Total consumption encompasses deliveries to the power generation, industrial, commercial, residential, and transportation sectors. Source: IEA.

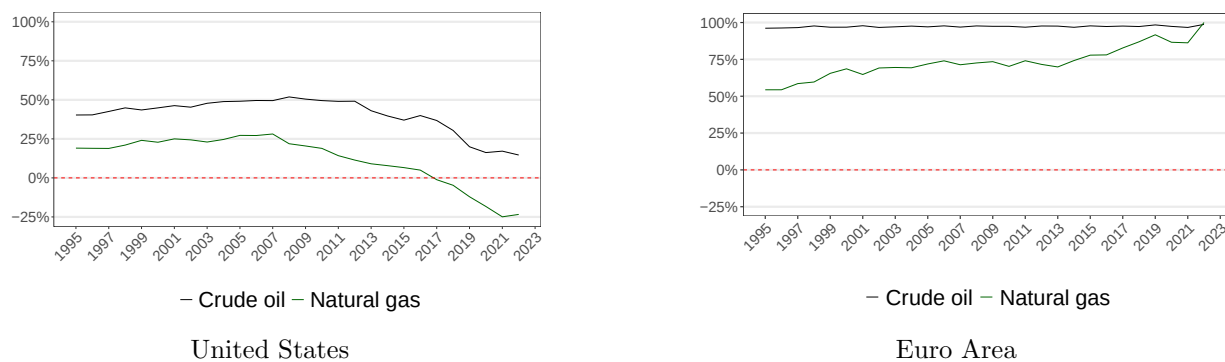
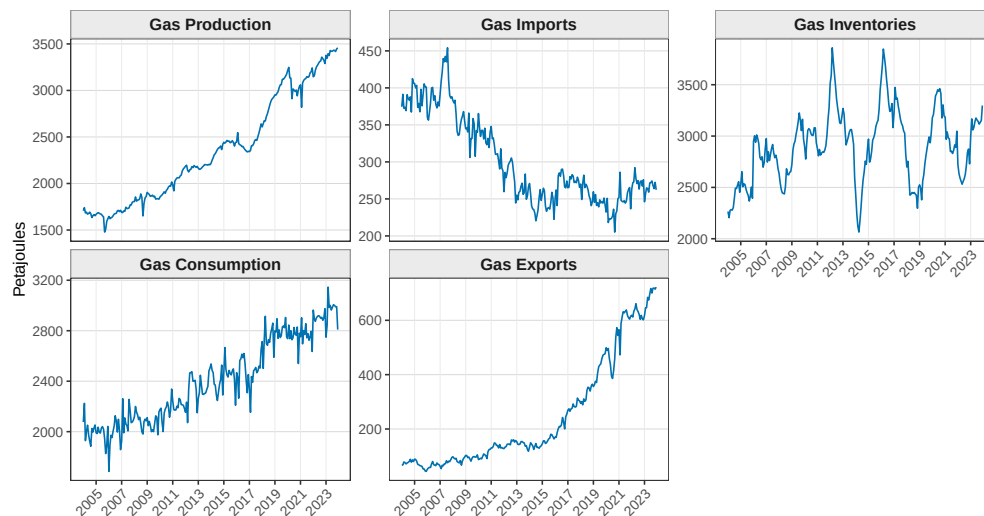


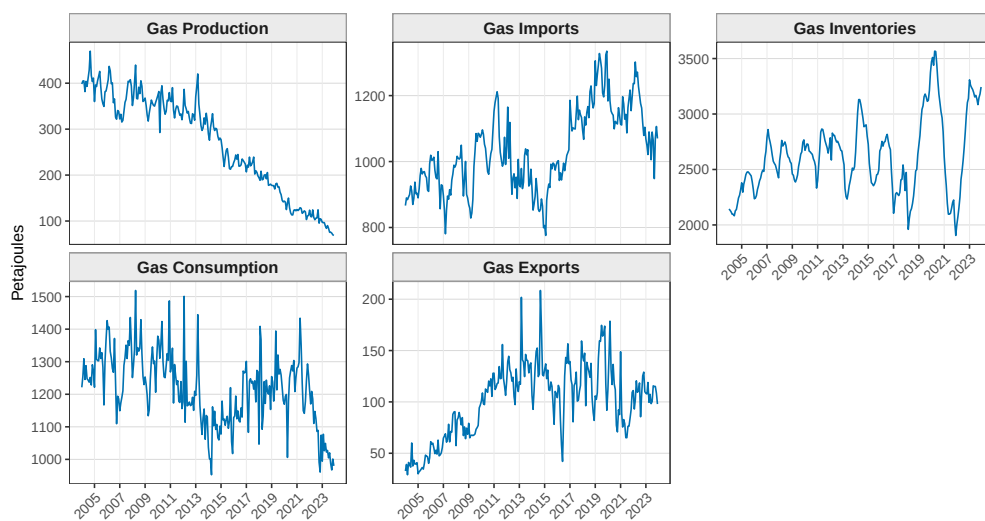
Figure B22: *Energy import dependency*

Notes: Import dependency on natural gas and oil, 1990–2022, measured as the share of net imports in total consumption of each fuel; United States in the left panel, Euro Area in the right panel. Sources: Eurostat, EIA, Energy Institute.

Figure B23 and Table B4 report the gas-market quantities behind these supply-side differences. They show that the Euro Area combines declining domestic production with continued import dependence, whereas U.S. production has expanded steadily.



United States



Euro Area

Figure B23: *Gas market variables for the United States and the Euro Area*

Notes: All series are expressed in petajoules and seasonally adjusted. Sources: IEA and EIA.

Region	Production	Imports	Exports	Consumption	Inventories	Supply
United States	2,354	301	246	2,399	2,897	2,409
Euro Area	271	1,033	100	1,216	2,616	1,204

Table B4: *Gas market variables for the United States and the Euro Area*

Notes: Monthly averages in petajoules over 2004–2023. Supply is defined as production plus imports minus exports. Sources: IEA and EIA.

Figures B24 and B25 provide a longer-run comparison of external energy trade. The Euro Area remains a large net importer of both fuels, while rising U.S. production has reversed the gas trade balance and substantially narrowed net crude-oil imports.



Figure B24: *Natural gas imports and exports*

Notes: Natural gas imports and exports in exajoules; United States in the left panel (1990–2024), European Union in the right panel (1990–2022). Sources: Eurostat and EIA.



Figure B25: *Crude oil imports and exports*

Notes: Crude oil imports and exports in exajoules; United States in the left panel (1990–2024), European Union in the right panel (1990–2022). Sources: Eurostat and EIA.

B.3 The TTF as the European benchmark gas price

This subsection documents the role of the Dutch TTF as the benchmark price for European natural gas markets. Historically, natural gas pricing in Europe has been predominantly linked to oil products. However, over the last two decades, gas prices have moved away from oil indexation to spot pricing through a series of regulatory reforms aimed at liberalizing the European gas market. The European Union began its liberalization process in 1992 with the EU energy market regulatory framework. However, meaningful developments only started in the late 1990s, leading to the issuance of three European Directives designed to foster competition and create a single market for natural gas. This reform process culminated in the “Gas Regulation” of 2009, which further strengthened market integration efforts.¹ These regulatory reforms led to the development of trading hubs across Europe, with 11 main active hubs as of 2021, though varying significantly in liquidity and infrastructure (Heather, 2021).²

While full liberalization of the European gas market remains incomplete (Cardinale, 2019; ACER, 2022), regional integration has nonetheless advanced significantly and has been a defining feature of the market for over two decades. The Dutch Title Transfer Facility (TTF) gas hub, recognized as the most liquid trading hub, has emerged as the benchmark for European gas prices. The TTF, listed on the ICE ENDEX futures exchange in Amsterdam, was established in 2003, whereas the first gas hub in the region, the National Balancing Point (NBP), was created in the United Kingdom in 1996.

Additionally, the share of hub-indexed imported gas relative to fixed contracted prices has grown significantly over time, representing approximately 80% of total gas imports in the European Union in 2021 (IEA, 2021). As part of this trend, oil-indexed contracts, which constituted over 90% of European gas imports in 2005, declined sharply to only 25% by 2019 (IEA, 2020). These developments support our use of the TTF as the European benchmark gas price. TTF-linked futures are the most liquid natural-gas contracts in Europe, with trading volume reaching 5.7 million contracts in May 2023 (ICE, 2023). Indeed, most studies that examine the role of gas prices in Europe focus on the TTF price (e.g. Adolfsen et al., 2024; Boeck and Zörner, 2025; López et al., 2026). Jotanovic and D’Ecclesia (2021) provide detailed evidence of a high level of integration among the European trading hubs, with the TTF playing the role of the reference trading hub.

This integration is reflected in Figure B26 and Table B5, which show that price dynamics across European hubs are highly correlated. Figure B26 shows the natural gas spot prices at selected European trading hubs: the Italian European Gas Network (EGN), the British

¹Regulation (EC) No 715/2009 (the “Gas Regulation”).

²While there are approximately 30 gas trading hubs in Europe, not all of them are actively operational.

Greater Buchan Area (GBA), the Spanish *Mercado Ibérico del GAS* (MIBGAS), the British National Balancing Point (NBP), the German NetConnect Germany (NCG), the French *Point d'échange de Gaz* (PEG), the Italian *Punto di Scambio Virtuale* (PSV), the Austrian Virtual Trading Point (VTP), and the Belgian Zeebrugge Trade Point (ZTP). These prices closely followed the TTF not only in the period before the pandemic but also amidst the subsequent market disruptions. Table B5 quantifies the comovement between TTF and these gas prices. The correlations are very high, ranging from 0.93 for the British NBP to 1.00 for NCG.

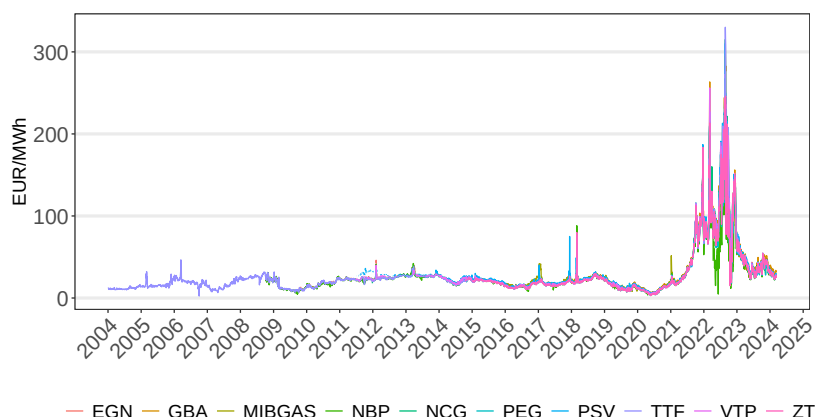


Figure B26: *TTF and other European gas prices*

Notes: Daily Dutch TTF spot price alongside spot prices at other European trading hubs. Source: Datastream.

Hub price	TTF
NCG	1.00
VTP	1.00
PSV	1.00
ZTP	0.97
EGN	0.98
NBP	0.93
GBA	1.00
PEG	0.97
MIBGAS	0.97

Table B5: *Correlation between TTF and other European gas prices*

Notes: Correlations between the Dutch TTF spot price and natural gas spot prices at other European trading hubs. Source: Datastream.

Crucially, while LNG has become increasingly important in the European market, its expansion has not resulted in greater price divergence within the region. On the contrary, the integration between European and global LNG markets has strengthened significantly,

driven by the substantial growth of seaborne gas trade (Albrizio et al., 2023). Historically, LNG prices did not closely track the TTF, reflecting its limited role in the European market at the time. However, as LNG has gained prominence, its price has become more closely aligned with the TTF, reducing divergence. This convergence is documented in Figures B27 and B28.

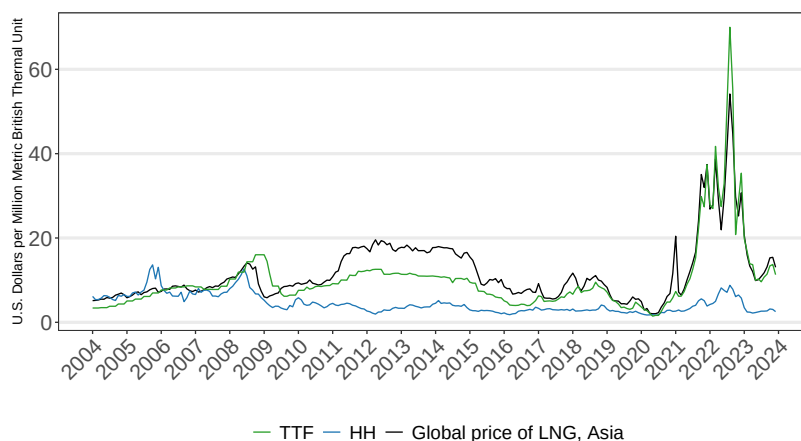


Figure B27: *TTF, HH, and Global LNG gas prices comparison*

Notes: The figure plots the monthly TTF spot price, converted to U.S. dollars for illustrative purposes, alongside the Henry Hub and the JKM LNG price. Source: Datastream.

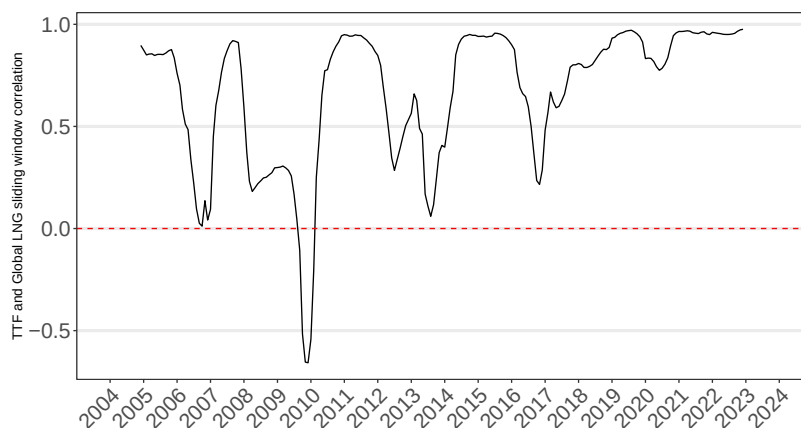


Figure B28: *TTF and Global LNG gas prices correlation*

Notes: Rolling correlation between the TTF and global LNG spot prices, computed over a 24-month window. Source: Datastream.

C Additional details on natural gas instruments

This appendix provides additional information on the construction and validation of the instruments. It first describes the gas demand instrument and the evidence supporting the heating-demand channel. It then illustrates the selection and audit of gas-supply events and reports further diagnostics for all instrument series.

C.1 Gas demand instrument

C.1.1 Construction of gas demand instrument

ERA5 surface temperature data. The daily temperature data are taken from ERA5’s single-levels dataset, the fifth-generation atmospheric reanalysis produced by the European Centre for Medium-Range Weather Forecasts. Weather data from ERA5 (Hersbach et al., 2020) at a regular latitude-longitude grid of $0.25^\circ \times 0.25^\circ$ are taken from the *reanalysis era5 single levels* dataset. Average daily temperature corresponds to the *2m temperature* (daily mean) variable. The series are aggregated from daily to monthly by taking temporal averages. To aggregate the grid-level data to the country level, we employ the Database of Global Administrative Areas (GADM), using the country-level layer, GADM0.³⁴

Natural gas consumption data. Annual natural gas consumption data are used to construct the aggregation weights for the regional temperature series. For Europe, the data are taken from Eurostat’s *Supply, transformation and consumption of gas* dataset, from which we use inland consumption of natural gas at the country-year level. For the United States, we use state-level annual natural gas consumption data from the U.S. Energy Information Administration’s *Natural Gas Consumption* database. In both cases, these data are used only to define predetermined aggregation weights rather than to measure contemporaneous demand conditions directly.

C.1.2 Validation of the demand channel

Alternative computations. We have explored several constructions of the temperature index, all of which yield very similar series and leave the main results unchanged. These robustness checks are available upon request. In particular, we consider detrending the temperature series prior to seasonal adjustment. As an alternative, seasonality can be removed by subtracting the mean temperature for each calendar day, computed across all years in

³<https://gadm.org/>.

⁴When using U.S. temperature data we average across all U.S. states and aggregate using the first sub-national level, GADM1.

the sample, rather than the corresponding calendar-month mean, although this tends to introduce more noise. The index can also be constructed using daily maximum or minimum temperatures instead of daily averages. Another weighting scheme incorporates demographic or geographic factors, such as night lights (Gortan et al., 2024), to refine the temperature series. Finally, we consider a rolling seasonal adjustment, in which the reference means are computed over a moving window of preceding years rather than over the full sample. We also verify that the results are robust to using fixed gas-consumption weights computed over the full sample and to omitting cross-sectional weights altogether.

Anticipation effects. Figure C29 demonstrates that the anticipation effects of temperature fluctuations on natural gas prices are likely minimal. The top-left panel illustrates the correlation between daily temperatures and daily natural gas prices, revealing that the strongest (negative) correlation occurs contemporaneously. This indicates that price movements are primarily driven by actual temperature realizations rather than expectations about future temperatures. Additionally, the temporal correlation of daily temperatures can explain the presence of some correlations in lead terms without necessarily implying anticipation effects. In practice, correctly forecasting average temperatures over a period may be more common than predicting daily temperatures with precision. However, the other panels show that even when considering the average temperature over several subsequent days, the strongest correlations remain contemporaneous. Overall, this analysis suggests that anticipation effects on natural gas prices, if present, are quite limited.

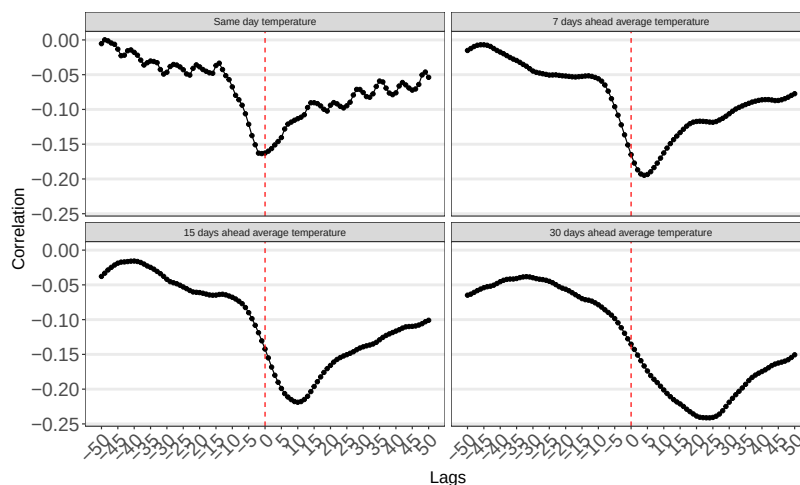


Figure C29: *Temperatures and gas price correlations*

Notes: The figure plots the correlations between the TTF spot price of natural gas and average temperatures at several leads and lags. The different panels use temperature averages computed over different time spans.

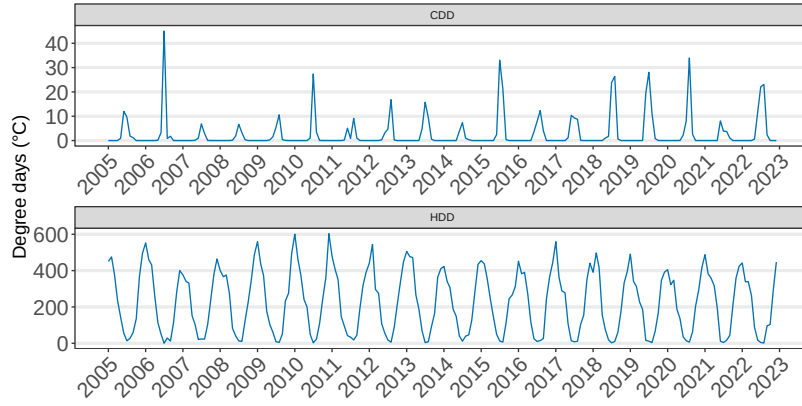


Figure C30: *Cooling degree days and heating degree days, average across selected European countries*

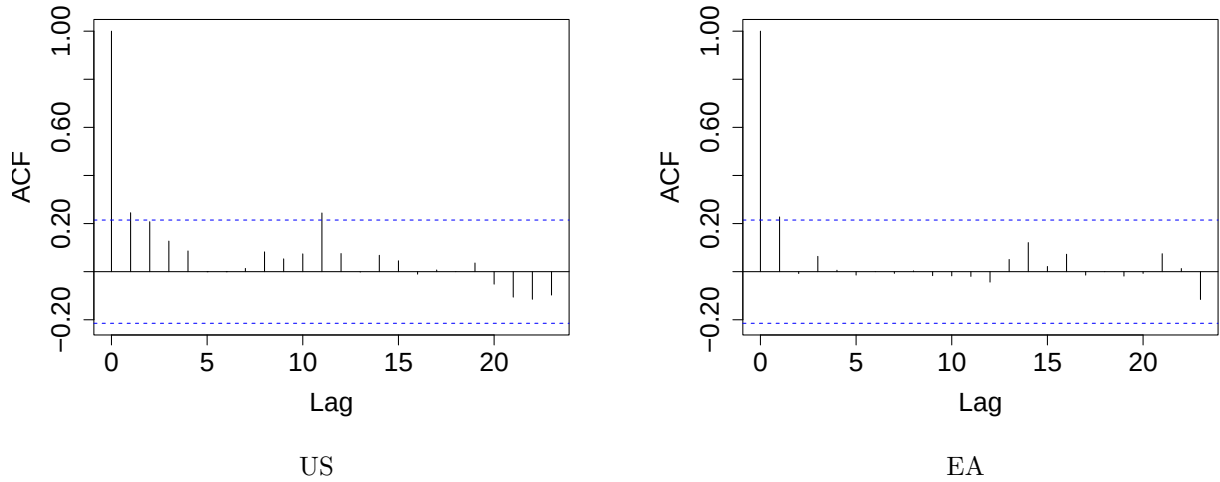


Figure C31: *Sample autocorrelation function of the temperature index*

C.2 Gas supply instruments

This subsection provides additional details on the gas-supply event-selection procedure. We first discuss illustrative examples of retained events and of excluded candidates. We then describe the structured large-language-model audit used as an additional check on the manual classification.

C.2.1 Illustrative retained and excluded supply events

Figure C32 illustrates a Euro Area flow-supply event. On July 2, 2007, BP unexpectedly shut the U.K.’s Central Area Transmission System (CATS) pipeline after damage to the pipeline casing, preventing gas from the J-Block North Sea fields from reaching Britain and continental Europe. The disruption reduced short-run physical gas availability and was followed by a clear increase in TTF spot and front-month prices over the daily event window.

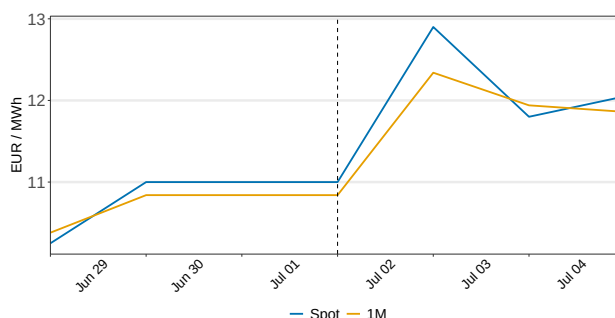


Figure C32: *Illustrative Euro Area flow-supply event*

Notes: TTF spot and front-month prices around the July 2, 2007 shutdown of the U.K.’s Central Area Transmission System (CATS) pipeline. The dashed vertical line marks the start of the daily event window; the event-window surprise is the price change from the previous close to the event-day close, equal to 15.9% for spot and 13.0% for front-month prices.

Illustrative excluded candidates. Several news items clearly related to natural gas markets are nonetheless excluded for failing one or more of the criteria detailed in Section 3.2. For the United States, an instructive case is Hurricane Katrina. Coverage on August 29, 2005, the day of landfall on the northern Gulf Coast, reported that the storm had already shut in more than 15 Bcf of Gulf production, with the extent of damage to offshore platforms, pipelines, and treatment plants still unknown. Despite its evident relevance for gas supply, the episode is not a clean, isolated supply event: the same coverage emphasized extreme summer demand, tightening storage, oil and gasoline disruptions, refinery damage, and potentially broad macroeconomic effects. We therefore exclude it.

Other flow-supply candidates are excluded on different grounds. Chesapeake’s announcement of voluntary output curtailments on April 17, 2009 is discarded as endogenous: it implied lower production, but the response was driven by prevailing low prices rather than an exogenous disruption. For Europe, the reduced Langede nominations of April 18, 2019 are excluded because the supply signal is confounded, as the article attributed the price move

chiefly to warm weather, weak residential demand, and high wind generation, and noted that the lower nominations might reflect commercial decisions rather than an unplanned outage. The Kollsnes shutdown of May 25, 2009 is excluded as anticipated: it was planned valve work already reflected in production guidance, and so revealed no new supply information.

For the future-supply-news instrument, the binding criterion is specificity to expected European gas availability, so broad geopolitical or commodity-market news is discarded even when gas-related. One such case is the announcement, on February 28, 2022, of an earlier-than-planned opening of the Poland-Lithuania Gas Interconnector, which would improve Baltic access to alternative gas supplies. Although the announcement carried favorable implications for future regional gas availability, markets were simultaneously processing new restrictions on the Russian Central Bank, the announced removal of selected Russian banks from SWIFT, inconclusive Russia-Ukraine negotiations, and intensified fighting around Kharkiv. European gas prices rose despite the favorable interconnector news, indicating that the price movement over the event window reflected these war-related developments rather than the announcement itself. The EU sanctions package of July 25, 2014 is excluded because it concerned capital markets, dual-use goods, and oil technology without imposing a concrete restriction on gas deliveries. Lastly, Gazprom’s reduction in planned Power of Siberia capacity, announced on November 12, 2018, is excluded because that route serves Asian markets, with no direct channel established to European supply.

C.2.2 Large-language-model validation audit

As an additional check on the manual classification, we audited the retained event set using a large language model (ChatGPT 5.4, classifications obtained at temperature zero). The audit was not used to discover new events or to draw on outside information. Instead, for each retained event it received only the retrieved Factiva articles and a prompt reproducing the selection criteria. The prompt, summarized in Box C.2.2, asked the model to verify that each event was supported by the source material, assigned to the correct instrument category, dated according to the price-relevant market-information date, and consistent with the sign of the constructed surprise. Cases flagged as invalid, uncertain, or incorrectly dated were manually re-examined against the underlying articles, supplementing the record with additional sources where coverage was incomplete. Events were retained only when this re-examination confirmed that the news was discrete, newly revealed, gas-supply-specific, and consistent with the timing of the price surprise.

Two examples illustrate the role of this audit. The first is an August 28, 2009 report on the UK gas market. The item initially appeared to describe a clean North Sea supply disruption: prompt prices rose as St. Fergus receipts fell, and market participants pointed

to a technical problem at the Britannia field. The audit flagged that the article did not in fact verify an outage. St. Fergus flows had already recovered, ConocoPhillips stated that Britannia was in operation, and the company did not confirm any earlier trip. The candidate was therefore excluded as resting on unverified market talk rather than on a confirmed, discrete supply event. Second, the audit rejected a September 25, 2009 U.S. candidate based on Pennsylvania’s order requiring Cabot Oil and Gas to cease hydraulic fracturing operations in Susquehanna County. The headline suggested a regulatory supply restriction, but the article concerned safety and environmental requirements after spills at a local well site. It did not document a contemporaneous production shut-in, pipeline disruption, or measurable short-run restriction in gas available to the U.S. market. The candidate was therefore excluded despite being natural-gas-related.

Condensed representation of the LLM audit prompt

You are conducting a final audit of natural-gas supply-event instruments. Use only the provided Factiva articles; do not use outside information.

Each event belongs to one of three categories:

- **US flow-supply:** a discrete, plausibly exogenous operational development changing short-run gas available to the U.S. market (e.g., pipeline ruptures, force majeure declarations, compressor or processing-plant failures, Gulf production shut-ins, LNG export outages, meaningful restorations). Exclude demand-side and weather-driven demand events, price commentary, broad production forecasts, stale follow-ups, and voluntary price-driven production responses.
- **EA flow-supply:** a realized, unexpected disruption or easing in gas availability relevant for Europe, already affecting contemporaneous or very near-term physical availability (e.g., pipeline or transit interruptions, unplanned field, processing, or terminal outages, unexpected changes in Norwegian or Russian flows). Exclude planned maintenance without an unexpected component and forward-looking news without a realized near-term physical effect.
- **EA news:** unexpected credible news about future gas availability or supply security relevant for Europe (e.g., payment-rule changes, gas-specific sanctions, transit disputes, strike risks, import-infrastructure decisions) that can move prices before physical flows change. Exclude generic geopolitical or macroeconomic news without a concrete gas-supply channel, and non-European developments unless the articles explicitly link them to European gas availability or prices.

For each event, verify that the cited articles support the event; that the category is correct; that the recorded date is the price-relevant market-information date; and that the event

direction is consistent with the sign of the price surprise.

Mark weak, stale, speculative, or insufficiently supported cases as uncertain or invalid; do not retain an event merely because gas prices moved, and do not rescue weak cases with background context from unrelated articles. Return a structured row-level audit with the category and date, validity, and price-sign checks, a corrected date where needed, a brief reason, and the supporting article.

C.3 Additional instrument diagnostics

This subsection reports three complementary validity checks. We examine whether the instruments comove with other structural shocks, reconstruct them after purging observable confounding information, and test whether the supply surprises are serially correlated or predictable from lagged macroeconomic and financial variables.

C.3.1 Correlations with other shocks

Table C6 compares the instruments with established measures of oil, monetary-policy, uncertainty, financial, and geopolitical shocks. Across the 60 correlations the largest in absolute value is 0.18, and only two differ from zero at the 5% level against three expected by chance. Moreover, no external shock correlates with the same instrument in both regions, and no instrument correlates with more than one external shock.

Source	Indicator	US demand		US flow-supply		EA demand		EA flow-supply		EA news		n
		ρ	p-value	ρ	p-value	ρ	p-value	ρ	p-value	ρ	p-value	
Kilian (2009)**	Oil supply	0.02	0.76	0.01	0.84	0.03	0.68	-0.04	0.58	-0.03	0.64	237
Kilian (2009)**	Aggregate demand	-0.08	0.25	0.06	0.34	0.02	0.72	-0.06	0.35	-0.04	0.57	237
Kilian (2009)**	Oil-specific demand	-0.10	0.11	0.00	0.97	-0.10	0.14	-0.05	0.43	0.07	0.29	237
Baumeister and Hamilton (2019)*	Oil supply	0.06	0.39	-0.03	0.67	0.06	0.36	0.01	0.94	-0.07	0.29	237
Baumeister and Hamilton (2019)*	Oil demand	-0.01	0.84	0.02	0.73	-0.05	0.43	0.01	0.89	-0.05	0.44	237
Känzig (2021)**	Oil supply expectations	-0.03	0.68	-0.01	0.89	-0.04	0.58	-0.05	0.41	0.02	0.75	237
Känzig (2023)	Carbon-policy shock	0.17	0.01	0.05	0.46	0.02	0.71	0.01	0.86	0.05	0.45	237
Altavilla et al. (2019)*	Target monetary policy (EA)	0.10	0.13	0.00	0.98	0.08	0.20	0.01	0.93	0.07	0.31	237
Bauer and Swanson (2023)*	Monetary policy surprise (US)	-0.09	0.16	-0.11	0.10	-0.05	0.42	0.04	0.54	0.18	0.00	237
Bloom (2009)**	VXO-VIX	0.10	0.14	0.07	0.29	0.08	0.19	0.00	0.95	0.06	0.38	237
Gilchrist and Zakrajšek (2012)*	Corporate credit spread index	-0.06	0.35	-0.02	0.77	-0.10	0.13	-0.01	0.89	-0.07	0.31	237
Caldara and Iacoviello (2022)	Geopolitical risk index	0.11	0.10	0.00	0.98	0.11	0.13	0.07	0.31	0.08	0.28	201

Table C6: *Correlation of instruments with other indicators*

Notes: This table presents Pearson correlation coefficients (ρ) and p-values for the US demand and flow-supply instruments and for the EA demand, flow-supply, and supply-security news instruments in relation to a variety of economic shocks and indicators from the literature. The p-values correspond to two-sided tests for the null hypothesis of no correlation. The sample size (n) varies across shocks, as some are unavailable or not easily extendable to our full sample (2004-2023).

*Extended by the original authors beyond the original sample used in the published paper.

**Extended by the authors of this study.

C.3.2 Informationally robust instruments

Figure C33 compares the baseline instruments with the informationally robust series, obtained by purging monthly energy, macro-financial, weather-related information from all instruments, and daily oil and food-price surprises from the supply instruments. The refined series preserve the timing and, to a large extent, the magnitude of the baseline series. As shown in Figure E35, they also produce impulse responses close to the baseline, suggesting that informational confounding is not material for the results.

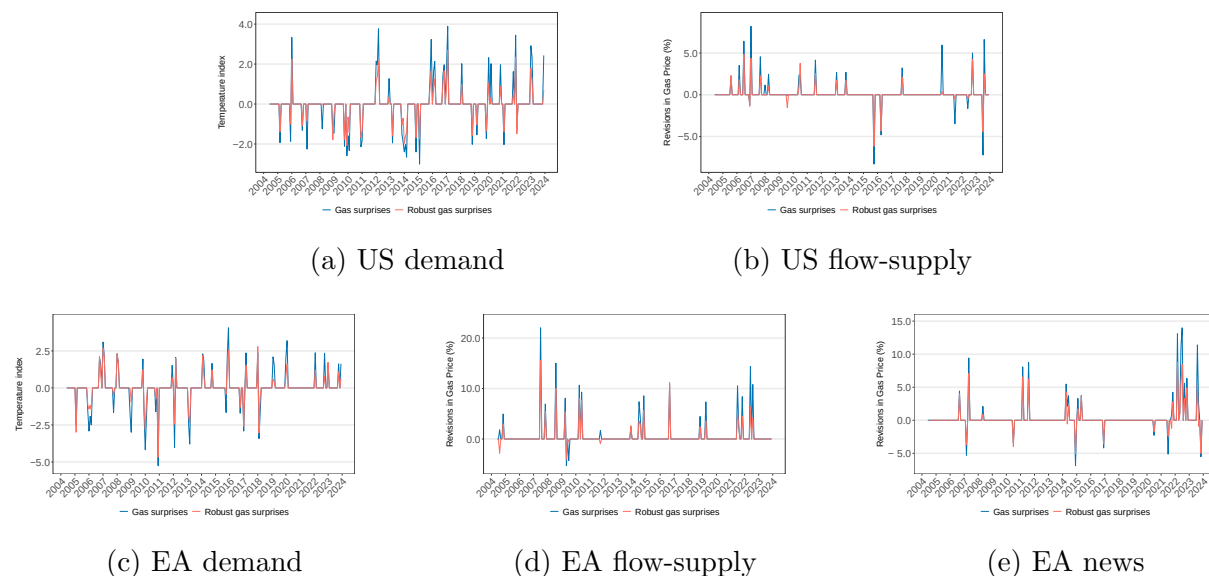


Figure C33: *Informationally robust gas surprise series*

Notes: This figure shows the baseline gas surprise series (blue line) alongside the informationally robust gas surprise series (red line). Panels (a)–(b) report the US demand and flow-supply instruments. Panels (c)–(e) report the Euro Area demand, flow-supply, and supply-security news instruments.

C.3.3 Serial correlation and forecastability of supply surprises

Figure C34 shows little autocorrelation in the surprise series, with only mild short-run serial dependence in the Euro Area supply-security news series. We then assess predictability using Granger's causality tests. Table C7 shows that none of the instruments is forecastable from its own lags or from the lagged gas-market, macroeconomic, or financial variables considered, either individually or jointly.

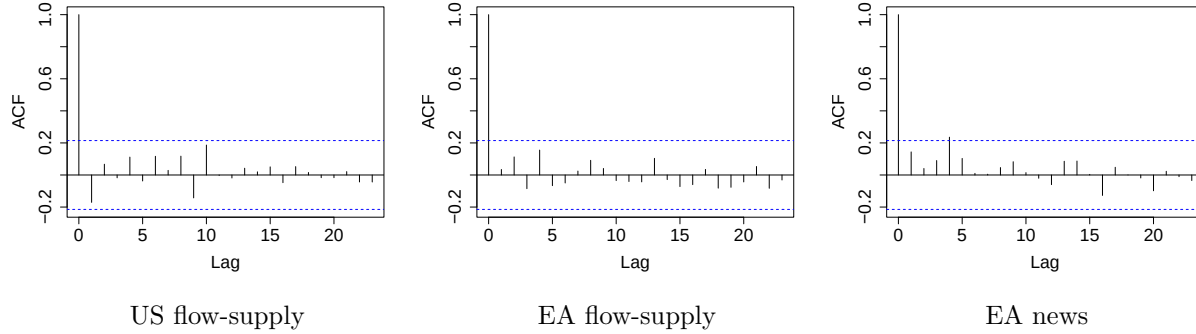


Figure C34: *Sample autocorrelation functions of the US flow-supply (left), Euro Area flow-supply (center), and Euro Area supply-security news (right) surprise series.*

Variable	US flow-supply	EA flow-supply	EA news
Instrument Lags	0.13	0.81	0.67
Gas price	0.79	0.75	0.96
Oil price	0.81	0.95	0.71
Gas consumption	0.87	0.76	0.77
Gas total available supply	0.97	0.88	0.95
Gas inventories	0.95	0.81	0.66
Headline CPI	0.11	0.75	0.10
Industrial production	0.40	0.87	0.53
Interest rate	0.61	0.35	0.99
Nominal exchange rate	0.40	0.17	0.52
Stock market (SP500/STOXX50E)	0.36	0.59	0.64
Financial volatility	0.53	0.67	0.23
Real economic activity	0.53	0.96	0.94
Joint Test	0.85	0.99	0.54

Table C7: *Granger causality tests*

Notes: The table reports p-values from Granger’s causality tests of the gas flow-supply and supply-security news surprise series using the set of variables included in our baseline specification, expanded with financial and real-activity variables. Series are rendered stationary by taking first or second differences as required. The tests include 12 lags and a constant term. The interest rate is EURIBOR for the Euro Area and the effective FFR for the United States; the stock market indices are the EURO STOXX 50 and the S&P 500; real economic activity is the index of Kilian (2009).

D Short-run gas market elasticities

This appendix reports the estimated structural price elasticities of gas demand and supply and compares them with previous estimates.

Estimated gas demand and supply elasticities. Table D8 reports the posterior medians and 68% credible bands. Short-run gas demand is relatively inelastic in both regions,

with estimates of -0.039 for the United States and -0.076 for the Euro Area. Gas supply is more responsive, with elasticities of 0.230 and 0.309 , respectively.

Region	Elasticity	Estimate
United States	Demand	-0.039 [-0.073, -0.008]
United States	Supply	0.230 [0.150, 0.403]
Euro Area	Demand	-0.076 [-0.113, -0.046]
Euro Area	Supply	0.309 [0.205, 0.516]

Table D8: *Estimated price elasticities of gas demand and supply. Point estimates are posterior medians; brackets report 68% credible bands.*

Comparison with previous estimates. Table D9 summarizes selected estimates from the literature. Our demand elasticities lie toward the less elastic end of the reported range, and smaller in absolute value than the short-run elasticity of -0.18 reported in the meta-analysis of Labandeira et al. (2017). On the supply side, our estimates are close to those of Casoli et al. (2024) and Wiggins and Etienne (2017), and substantially larger than the near-zero elasticities reported by Rubaszek et al. (2021) and Farag (2024). These comparisons should nevertheless be interpreted cautiously, as the studies differ in their measures of gas quantities, sample periods, data frequencies, and identification strategies.

Authors (year)	Region	Demand		Supply		Identification	Sample period
		Quantity	Estimate	Quantity	Estimate		
Wiggins and Etienne (2017)	United States	Production	[-0.18; -0.08]**	Production	[0.15; 0.3]**	Sign restrictions	1976Q1–2015Q2
Rubaszek et al. (2021)	United States	Consumption	-0.42	Production	0.01	Priors and sign restrictions*	1978Q1–2020Q3
Farag (2024)	United States	Consumption	-0.177	Production + Imports	0.019	Priors and sign restrictions*	1992M1–2023M10
Casoli et al. (2024)	Europe	Consumption	-0.47	Production	0.34	Priors and sign restrictions*	2010M1–2022M7

Table D9: *Previous estimates of short-run gas market elasticities*

Notes: “Quantity” refers to the specific aggregate used in each study to evaluate the percentage response to a change in the gas price.

*Baumeister and Hamilton (2015, 2019) identification strategy.

**Wiggins and Etienne (2017) report a range from a time-varying parameter model.

E Robustness of the baseline results

This appendix reports sensitivity exercises for the baseline results. We first consider alternative constructions of the instruments, including informationally robust refinements,

alternative event-selection thresholds, leave-one-out exercises, and alternative temperature-based demand instruments. We then assess the identifying assumptions by testing invertibility, estimating an internal-instrument specification, examining instrument relevance, and relaxing the orthogonality restriction across shocks. Finally, we consider alternative estimators, lag lengths, specifications, and samples, which leave the baseline shock construction unchanged but alter the empirical system or estimation approach.

Unless stated otherwise, each robustness figure compares an alternative exercise with the baseline model, uses the same sample and variables as the baseline, and normalizes each shock to raise the real price of natural gas by 10% on impact. The baseline median response is drawn as a black solid line, with credible bands shaded orange for the United States and blue for the Euro Area. Alternative estimates are shown in red or green.

E.1 Alternative instrument construction

Informationally robust instruments. The baseline instruments could contain information about observable confounders that affect the outcome variables through channels unrelated to natural gas. We therefore apply the informationally robust refinement described in Section 3.3, which residualizes the instrument series with respect to daily oil and food-price surprises, where applicable, and monthly energy, macro-financial, weather, and power-market controls. Figure C33 compares the baseline and refined instruments, while Figure E35 reports the associated impulse responses. The refined instruments remain close to the baseline series and yield similar responses.

Event selection and influential observations. The baseline supply instruments retain events associated with absolute gas-price surprises above 2%. Figure E36 reports responses obtained after reconstructing the instruments using stricter 3% and 4% thresholds. Figure E37 reports a leave-one-out jackknife, in which one nonzero event month is set to zero at a time. Across both exercises, the responses remain close to the baseline, indicating that neither the screening cutoff nor any single event month drives the results.

Alternative gas demand instruments. The baseline demand instrument excludes summer months to isolate heating-related variation. Figure E38, panel (a), reports responses obtained when summer observations are retained, with the sign of the temperature index adjusted in those months to account for cooling-related gas demand. Panel (b) replaces temperature deviations with heating degree days. Both alternatives yield responses similar to the baseline.

Informationally robust instruments

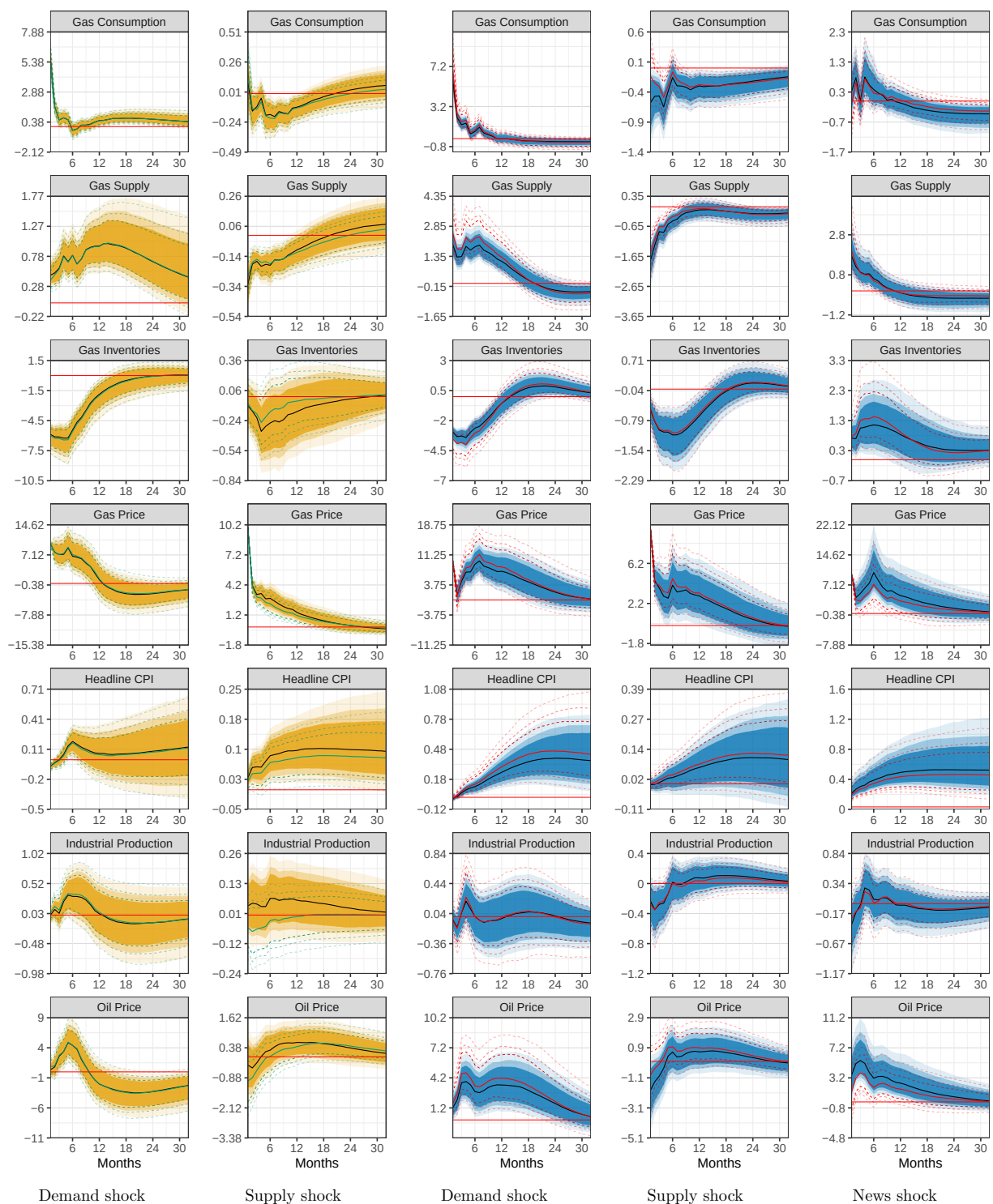


Figure E35: *Comparison with baseline responses: informationally robust instruments.*

Alternative event-selection thresholds

(a) 3% surprise threshold

(b) 4% surprise threshold

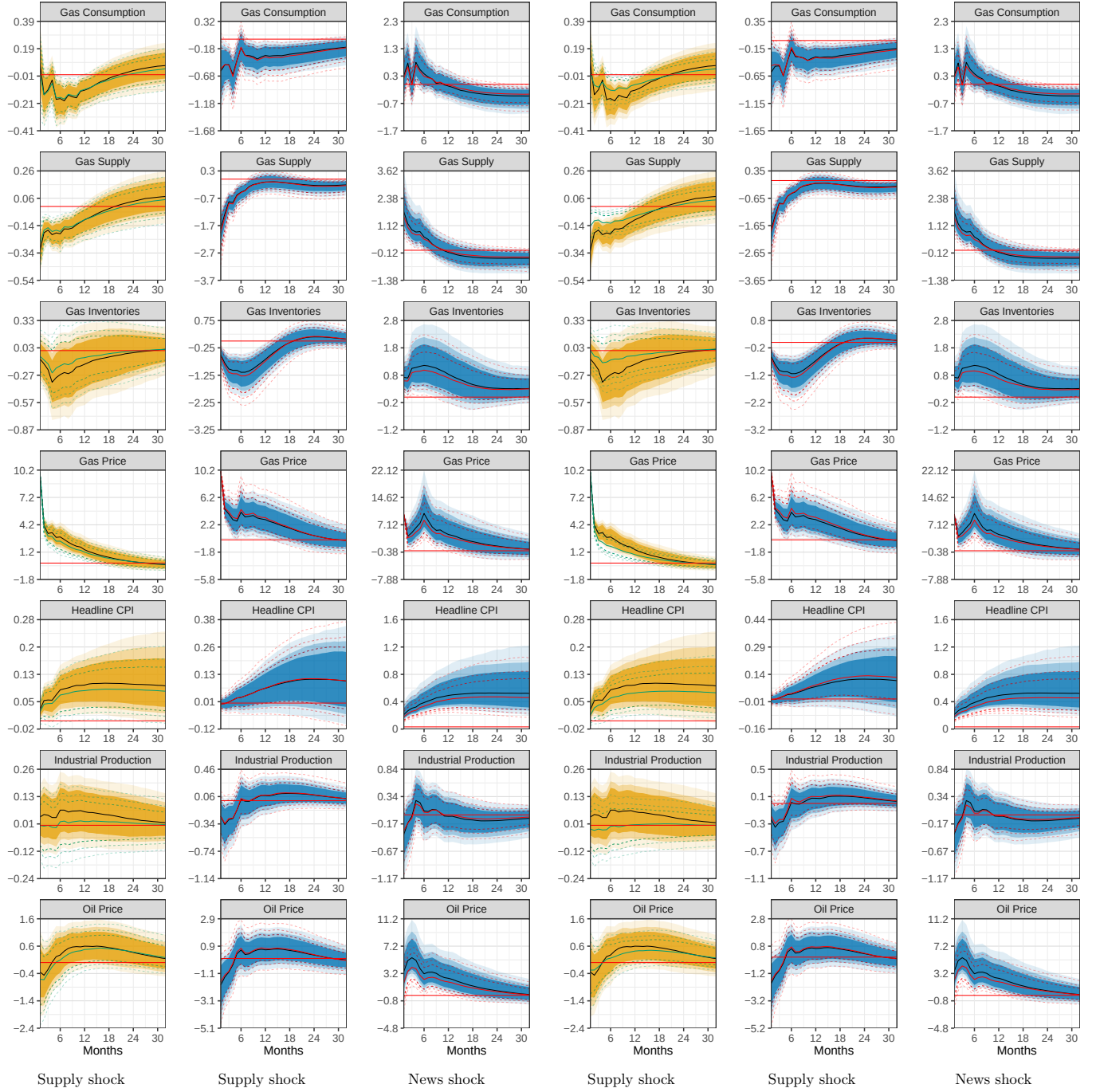


Figure E36: Comparison with baseline responses: supply instruments constructed using 3% and 4% event-selection thresholds.

Jackknife exercise

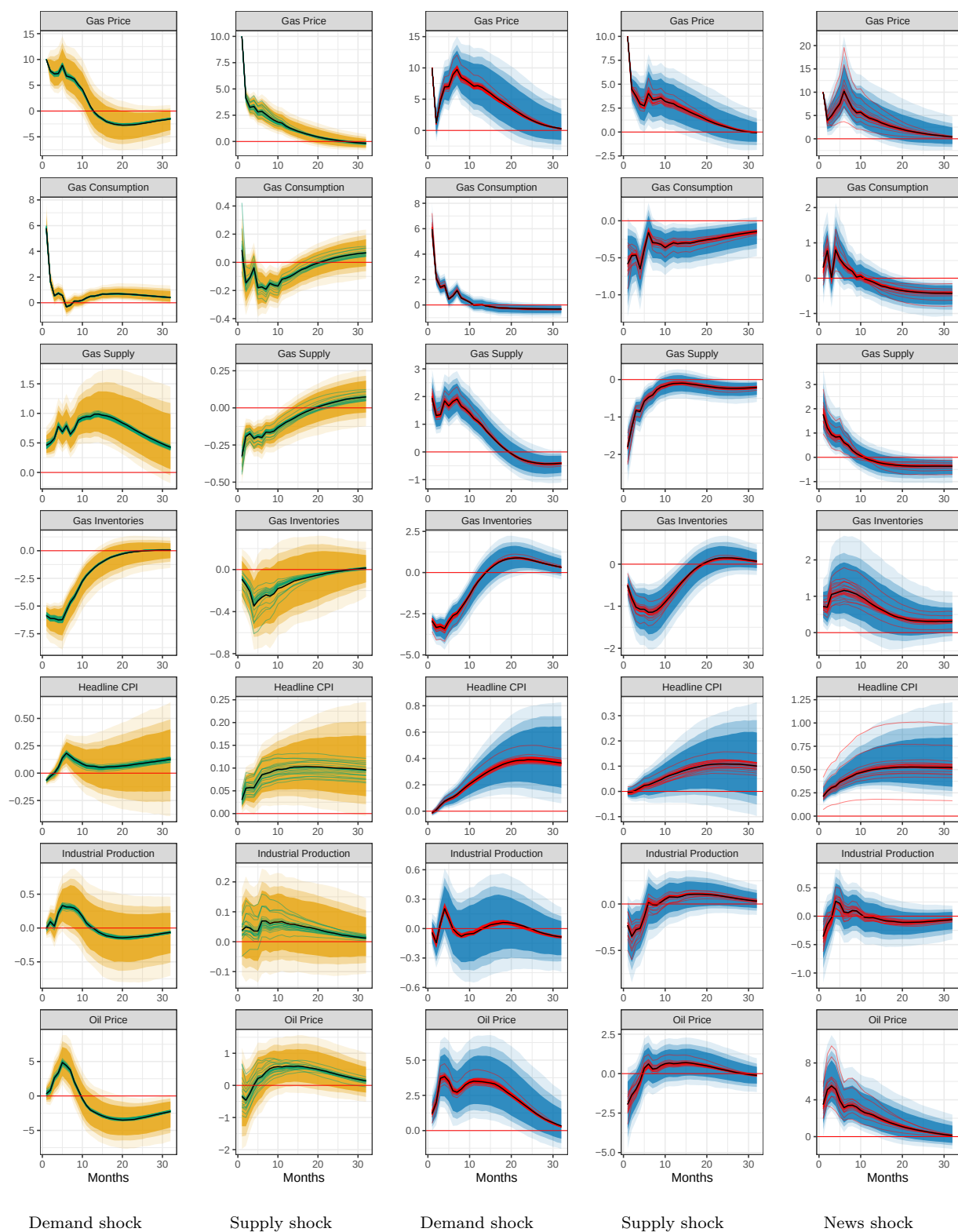


Figure E37: *Comparison with baseline responses: leave-one-out jackknife over nonzero instrument observations.*

Alternative demand instruments: summer months and heating degree days

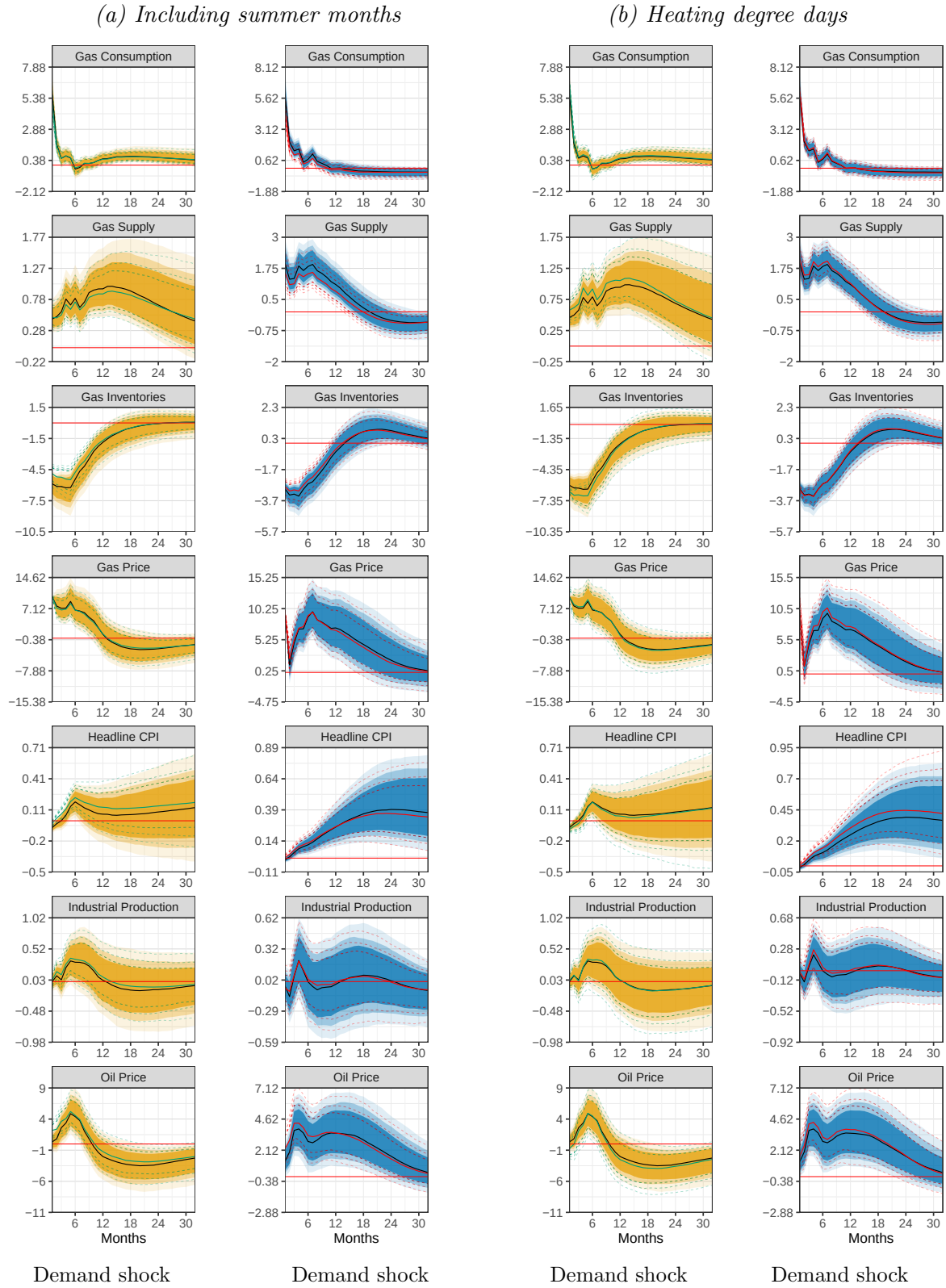


Figure E38: Comparison with baseline responses: alternative gas demand instruments. Panel (a) includes summer months; panel (b) uses heating degree days.

E.2 Identifying assumptions

Invertibility. Table E10 reports the invertibility tests of Plagborg-Møller and Wolf (2022) and Hamilton (forthcoming). We test whether lags of each instrument predict the VAR variables after conditioning on the baseline regressors, using specifications with six to twelve lags of the instrument and reporting both equation-by-equation and joint p-values. The joint tests provide no evidence against invertibility for any instrument. Figure E39 reports a complementary internal-instrument specification (Ramey, 2011; Plagborg-Møller & Wolf, 2021). The resulting responses remain close to the baseline.

Instrument relevance. Figure E40 reports the posterior distributions of the identified impact coefficients (Caldara et al., 2019; Giacomini et al., 2022), while Figure E41 plots the relationship between each instrument and the relevant reduced-form innovation. The posterior distributions are well behaved, and the instruments display clear comovement with the reduced-form innovations used for identification. The heteroskedasticity-robust first-stage F-statistics from regressions of the gas-price reduced-form innovation on each instrument are 20.59, 16.88, and 16.22 for the Euro Area flow-supply, supply-security news, and demand instruments, and 10.01 and 25.99 for the U.S. supply and demand instruments, respectively. Together, these diagnostics support instrument relevance.

Finite-sample orthogonalization. The baseline estimates impose exact orthogonality among the identified shocks in finite samples. Figure E42 reports responses obtained after relaxing this restriction. The responses are broadly unchanged, indicating that the finite-sample orthogonalization step is not driving the baseline results.

Invertibility tests

US demand								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	Joint
6	0.430	0.033	0.900	0.345	0.292	0.190	0.529	0.353
7	0.472	0.056	0.831	0.447	0.358	0.214	0.589	0.502
8	0.582	0.047	0.882	0.393	0.293	0.292	0.649	0.524
9	0.476	0.048	0.930	0.498	0.187	0.388	0.651	0.334
10	0.550	0.083	0.899	0.578	0.242	0.474	0.581	0.502
11	0.457	0.088	0.926	0.562	0.244	0.417	0.666	0.523
12	0.384	0.120	0.874	0.602	0.197	0.459	0.798	0.534
US flow-supply								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	Joint
6	0.798	0.299	0.633	0.831	0.303	0.310	0.925	0.640
7	0.794	0.238	0.301	0.890	0.280	0.244	0.817	0.526
8	0.839	0.148	0.337	0.836	0.099	0.328	0.641	0.434
9	0.827	0.213	0.290	0.870	0.121	0.429	0.657	0.542
10	0.805	0.290	0.270	0.920	0.113	0.507	0.736	0.532
11	0.726	0.334	0.326	0.953	0.135	0.591	0.756	0.511
12	0.789	0.414	0.217	0.958	0.156	0.470	0.842	0.553
EA demand								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	Joint
6	0.145	0.353	0.998	0.579	0.422	0.698	0.542	0.630
7	0.162	0.369	0.999	0.680	0.427	0.766	0.642	0.721
8	0.213	0.417	1.000	0.759	0.238	0.842	0.558	0.785
9	0.196	0.523	0.998	0.823	0.306	0.852	0.641	0.852
10	0.230	0.585	0.858	0.877	0.365	0.677	0.500	0.684
11	0.134	0.596	0.848	0.904	0.364	0.749	0.567	0.566
12	0.178	0.063	0.890	0.165	0.265	0.797	0.646	0.249
EA flow-supply								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	Joint
6	0.181	0.858	0.868	0.591	0.548	0.312	0.646	0.203
7	0.259	0.825	0.922	0.544	0.644	0.411	0.708	0.377
8	0.355	0.764	0.735	0.635	0.531	0.328	0.782	0.352
9	0.455	0.767	0.821	0.714	0.615	0.261	0.848	0.511
10	0.421	0.832	0.497	0.719	0.521	0.290	0.822	0.453
11	0.283	0.869	0.142	0.462	0.587	0.246	0.653	0.223
12	0.246	0.893	0.188	0.559	0.640	0.243	0.773	0.329
EA news								
Lags	Gas price	Gas consumption	Gas supply	Gas inventories	Headline CPI	Industrial production	Oil price	Joint
6	0.217	0.294	0.075	0.405	0.564	0.976	0.877	0.593
7	0.147	0.390	0.108	0.501	0.333	0.987	0.794	0.493
8	0.182	0.470	0.150	0.613	0.221	0.978	0.837	0.580
9	0.230	0.474	0.176	0.700	0.135	0.990	0.889	0.667
10	0.294	0.583	0.189	0.748	0.148	0.994	0.936	0.796
11	0.339	0.327	0.143	0.760	0.200	0.978	0.961	0.784
12	0.299	0.238	0.194	0.756	0.177	0.958	0.745	0.605

Table E10: *Testing the invertibility assumption*

Note: The table reports p-values from F-tests that the coefficients $\alpha_1, \dots, \alpha_m$ are zero in the regression $y_{it} = \pi_i'x_{t-1} + \alpha_1 z_{t-1} + \dots + \alpha_m z_{t-m} + \eta_{it}$. The test is conducted for each series included in the baseline VAR model and for different numbers of instrument lags, reported in the first column. The VAR includes seven lags, one more than in the baseline specification; using six lags gives similar results. Only a constant is included as deterministic term. The last column reports the p-value of the joint system test that lags of z do not enter any equation.

Estimation by internal instrument

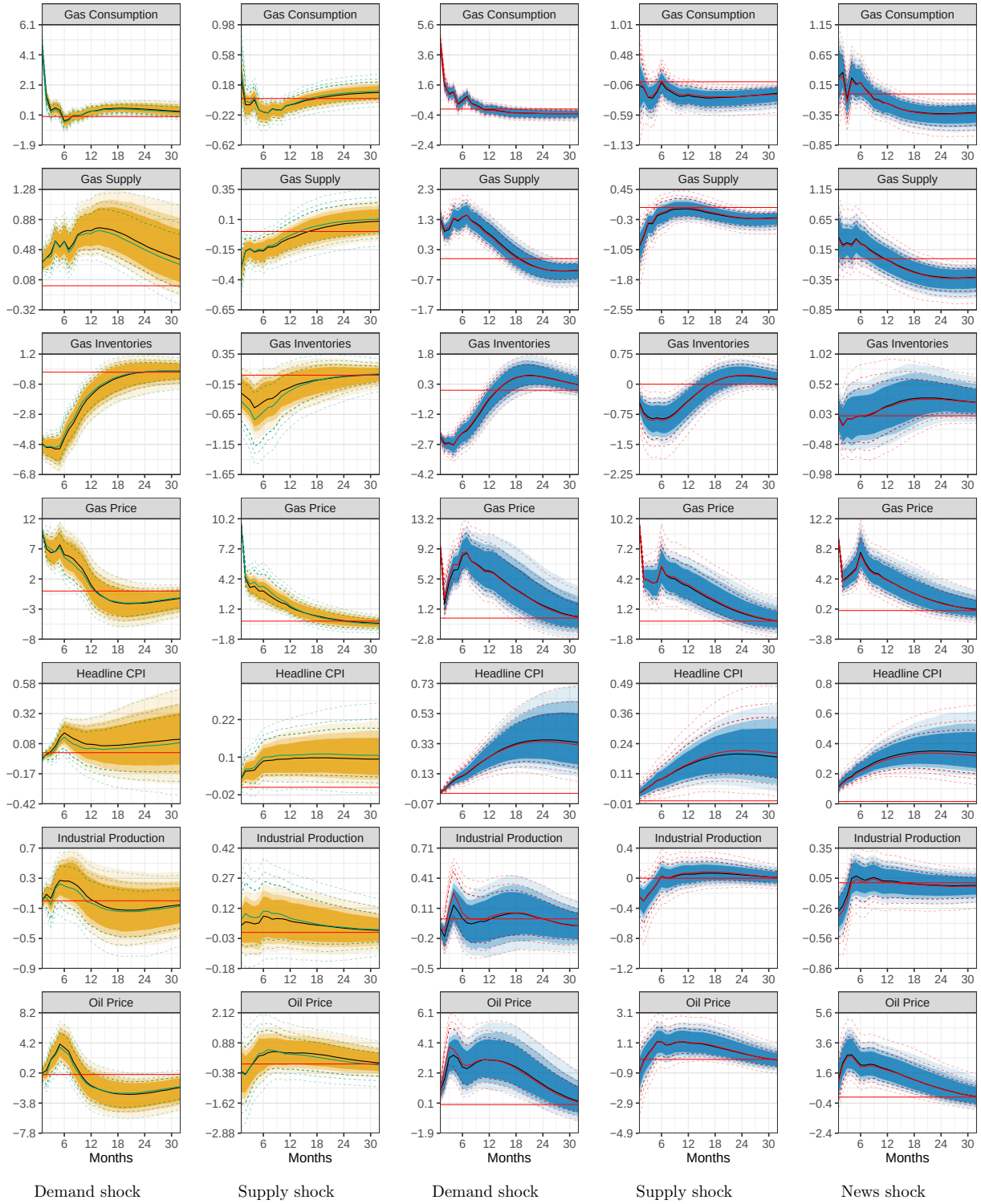
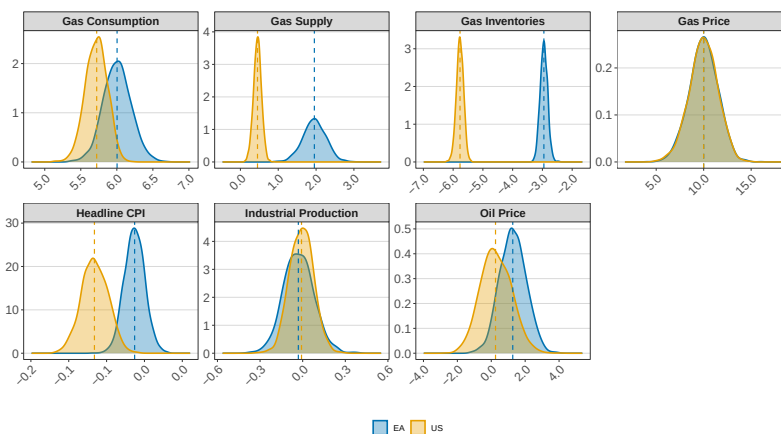
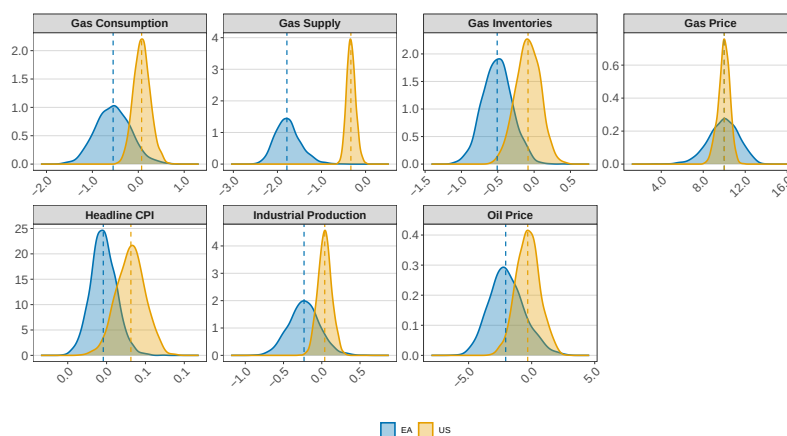


Figure E39: *Comparison with baseline responses: internal-instrument specification.*

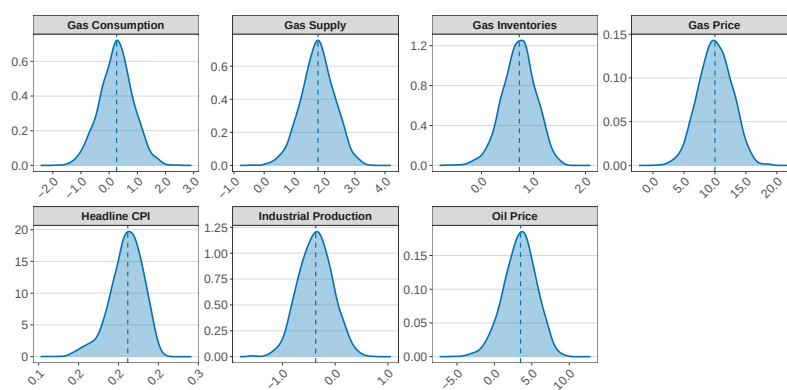
Instrument relevance



Demand shock



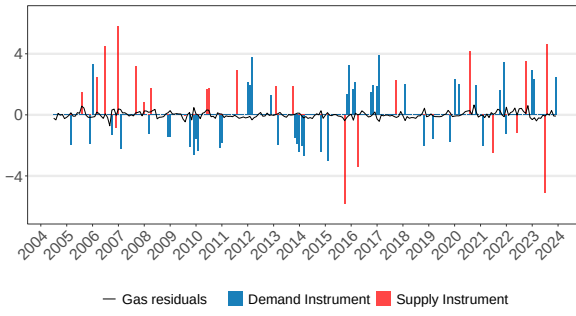
Supply shock



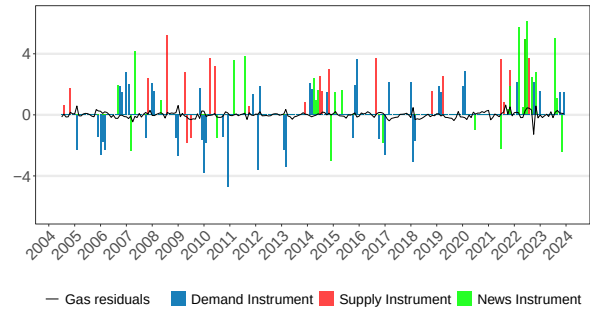
News shock

Figure E40: *Posterior distributions of impact coefficients*

Notes: This figure shows the posterior distributions of the impact coefficients identified for gas demand, flow-supply, and supply-security news shocks. Dashed vertical lines indicate the median of each distribution.



United States



Euro Area

Figure E41: *Instrument strength*

Notes: The figure shows how the gas-market instruments are related to the relevant reduced-form residuals of our baseline specification. All series are rescaled to have unit variance for comparability. Panel (a) reports results for the United States, and panel (b) reports results for the Euro Area.

Baseline without orthogonalization

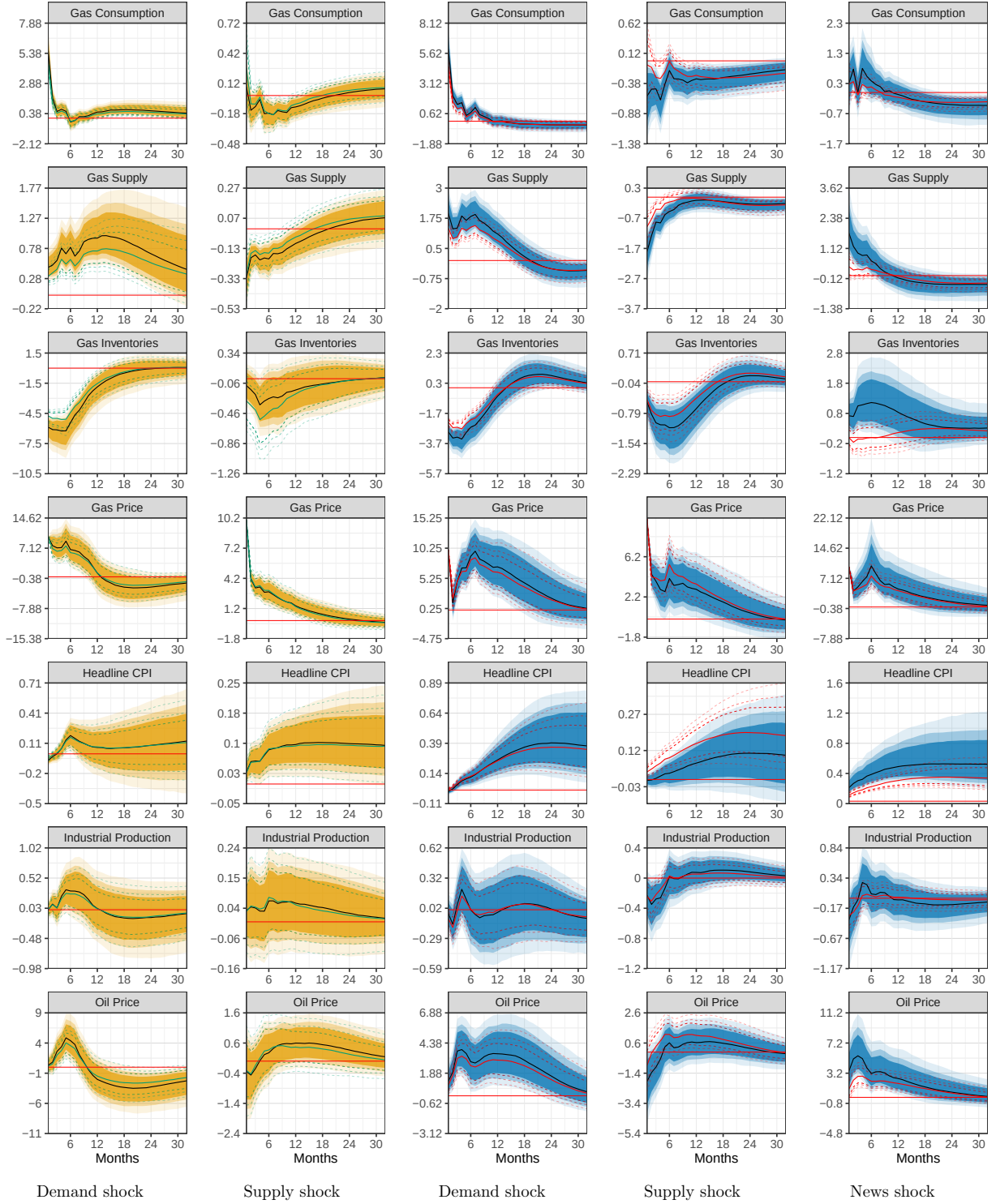


Figure E42: *Comparison with baseline responses: relaxing the orthogonality restriction across shocks.*

E.3 Estimation, specification, lag length, and sample

Alternative estimators. To assess the role of Bayesian regularization and the VAR representation, we re-estimate the model by OLS and with the Bayesian local-projection approach of Ferreira et al. (2025), using Minnesota-type priors. Figures E43 and E44 show similar dynamic patterns, although the alternative estimates are sometimes less precise.

Additional variables. We also augment the VAR with electricity prices, electricity consumption, and renewable power generation to account for omitted power-market dynamics. Figure E45 shows that the responses remain close to the baseline.

Lag length. The baseline VAR contains six monthly lags. Figure E46 reports estimates from an otherwise identical specification with twelve lags; the responses are similar.

Extended U.S. sample. The common baseline sample begins in 2004, the earliest date for which TTF prices are available. For the United States, the data permit estimation from 2001. Figure E47 shows that extending the U.S. sample to 2001–2023 does not materially alter the demand- or supply-shock responses.

Estimation by VAR-OLS

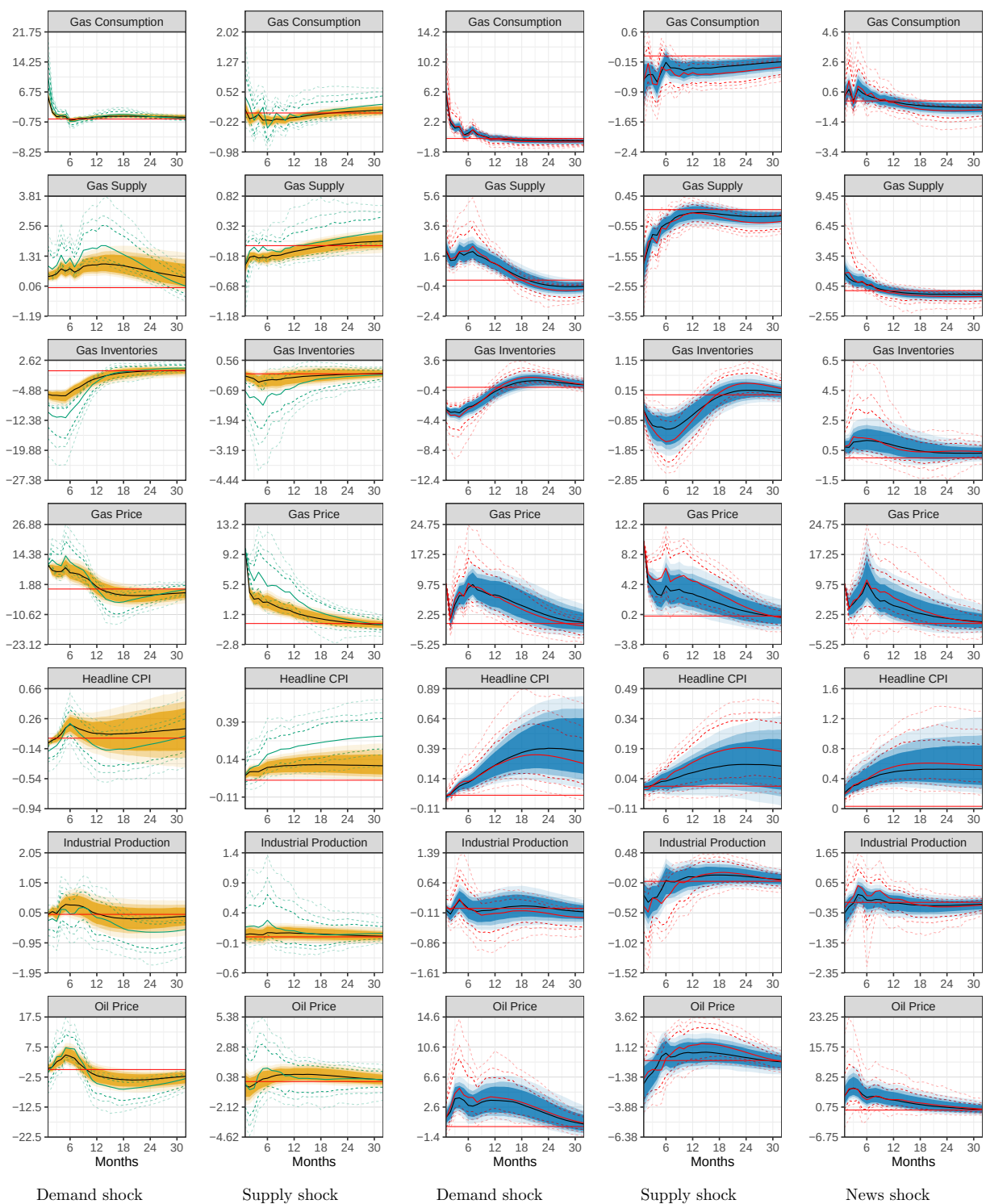


Figure E43: Comparison with baseline responses: VAR estimated by OLS.

Estimation by Bayesian Local Projections

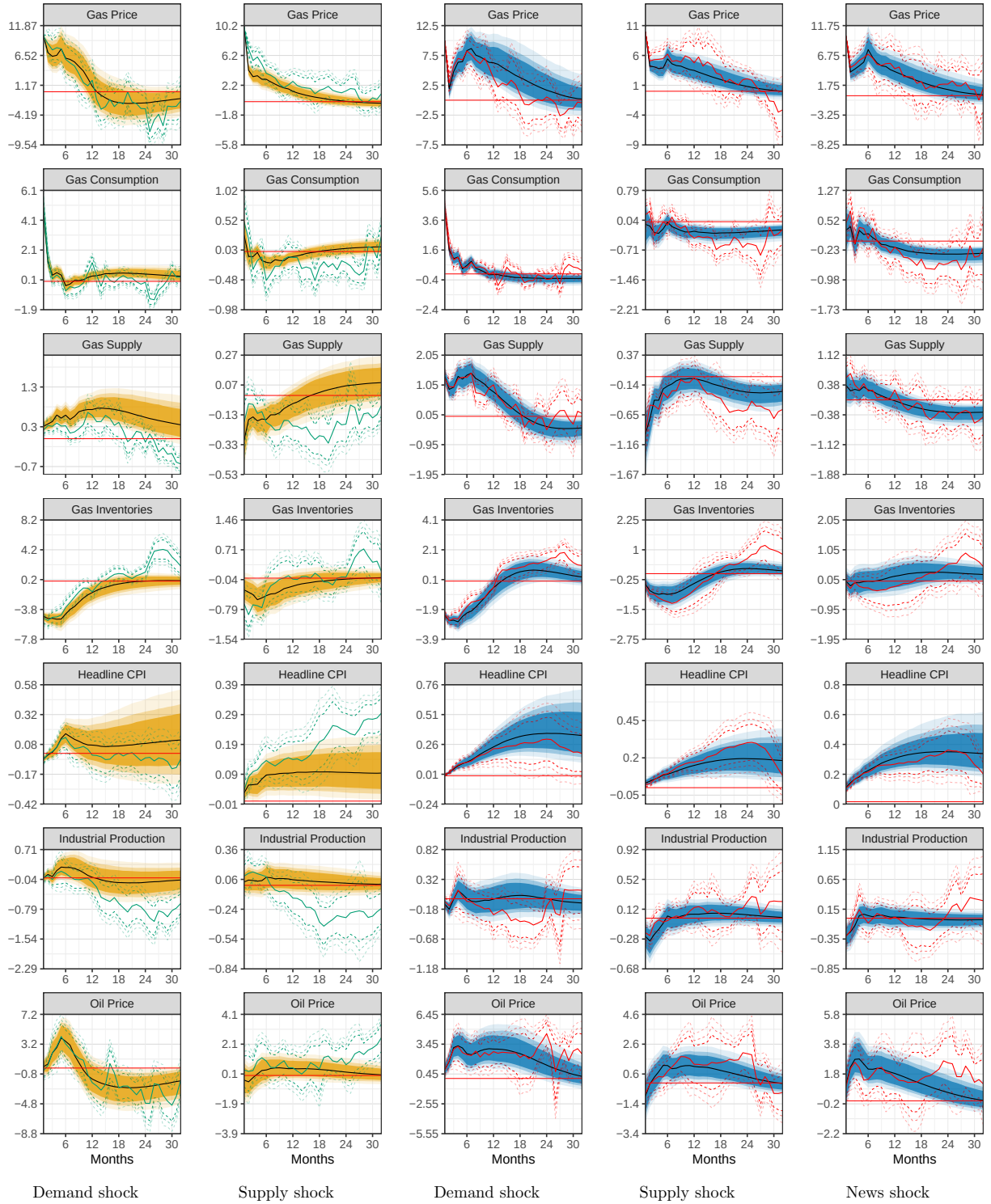


Figure E44: *Comparison with baseline responses: Bayesian local projections.*

Specification with power and renewable-energy variables

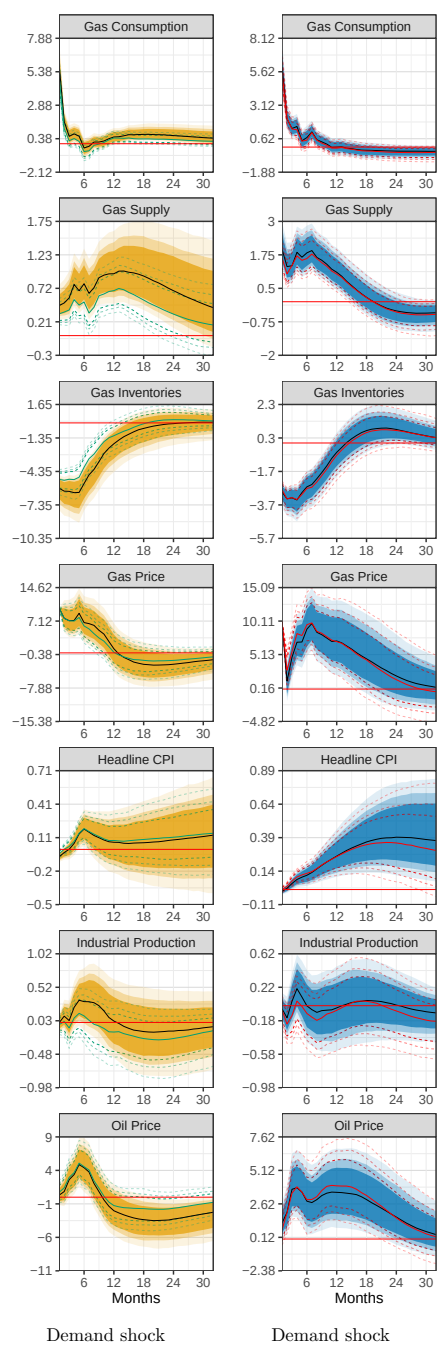


Figure E45: *Comparison with baseline responses to gas demand shocks: VAR augmented with electricity prices, electricity consumption, and renewable power generation.*

Specification with 12 lags

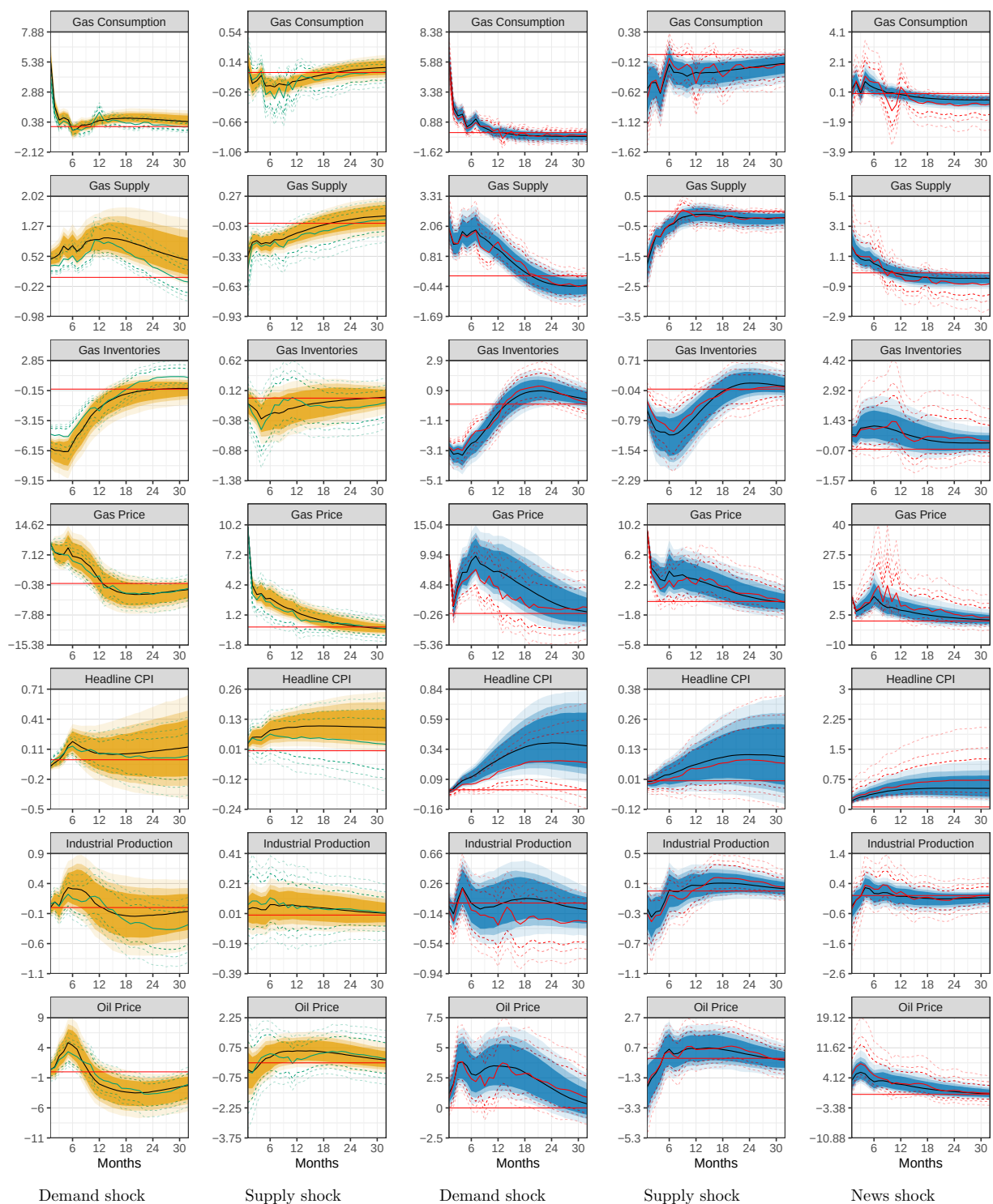


Figure E46: *Comparison with baseline responses: VAR with twelve lags.*

Extended estimation sample for the U.S. (2001-2023)

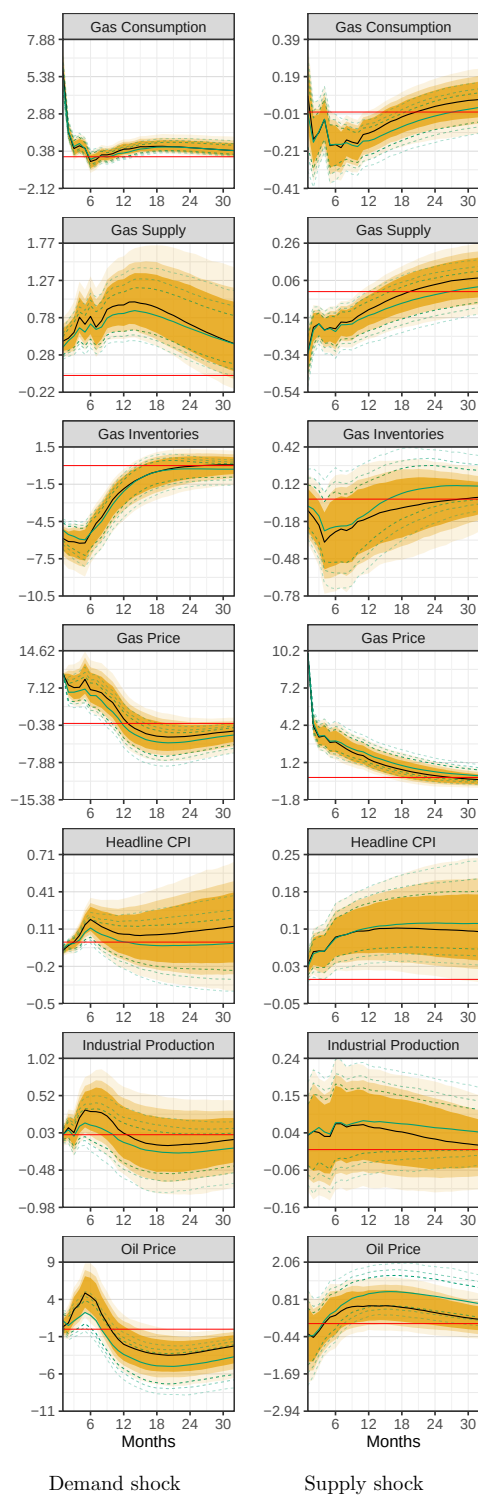


Figure E47: *Comparison with baseline responses: extended U.S. sample, 2001–2023.*

F Sectoral effects of Euro Area gas supply shocks

This appendix reports separate sectoral responses to Euro Area flow-supply and supply-security news shocks. Both shocks are normalized to raise the real price of natural gas by 10% on impact. In all figures, black solid lines with blue shaded credible bands denote responses to flow-supply shocks, while red solid lines with red dashed credible bands denote responses to supply-security news shocks. The comparison clarifies the composition of the main sectoral results: supply-security news shocks account for most of the price pass-through, whereas flow-supply shocks drive the immediate output losses.

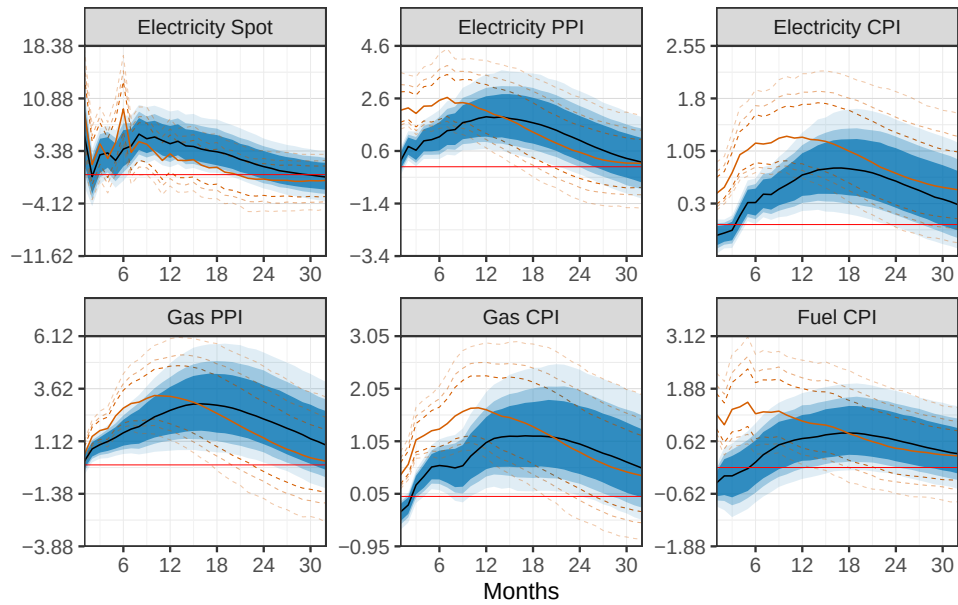


Figure F48: *Energy-price responses to Euro Area flow-supply and supply-security news shocks*

Figure F48 shows that both shocks pass through to downstream energy prices, but unevenly along the pricing chain. Electricity spot prices respond immediately and most strongly, producer prices for gas and electricity build up over the first year, and consumer prices for gas, electricity, and fuels adjust more gradually, consistent with retail price stickiness. Across most price measures, supply-security news shocks generate larger near-term responses than flow-supply shocks, consistent with expected future scarcity being priced before physical flows adjust.

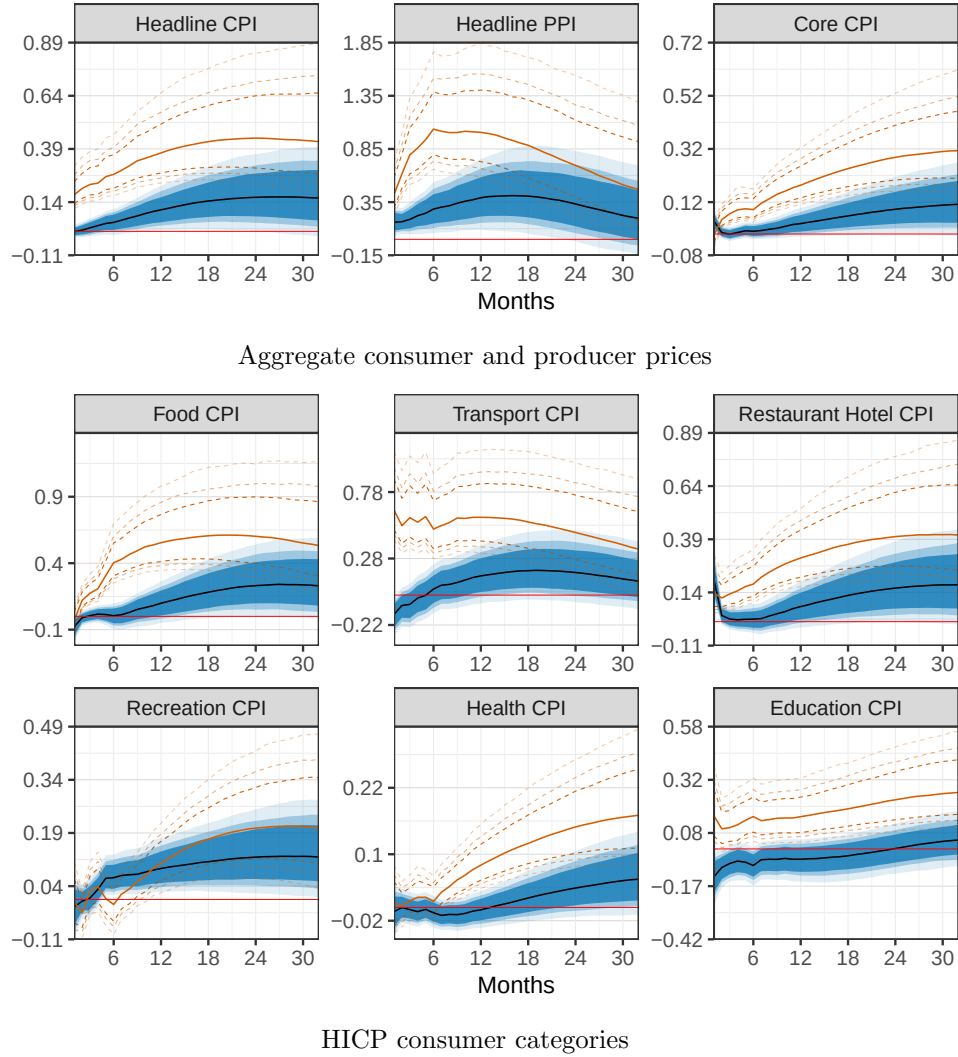


Figure F49: *Price responses to Euro Area flow-supply and supply-security news shocks*

Figure F49, Panels (a) and (b), shows that the price effects documented in the combined specification are driven mainly by supply-security news shocks. In Panel (a), supply-security news shocks generate substantially larger responses of headline consumer prices, producer prices, and core consumer prices than flow-supply shocks; the difference is especially pronounced for headline and core consumer prices over medium horizons. Panel (b) shows a similar pattern across consumer categories. Flow-supply shocks generate comparatively modest price responses, while supply-security news shocks raise food and transport prices more strongly and generate larger increases in services over time, especially in restaurants and hotels, health, and education. This pattern is consistent with news about future supply security raising expected input costs, whereas part of the price effect of contemporaneous

flow disruptions unwinds as physical supply recovers.

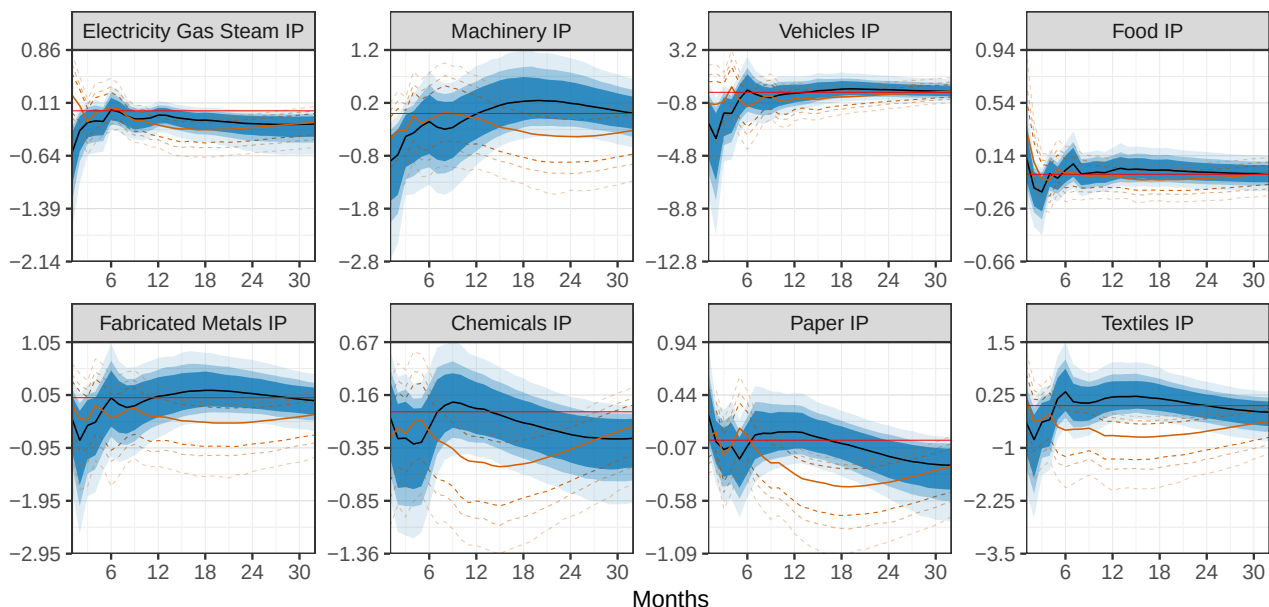


Figure F50: *Sectoral industrial-production responses to Euro Area flow-supply and supply-security news shocks*

Finally, Figure F50 considers sectoral output. Real effects are concentrated in energy supply and in energy-intensive manufacturing, including textiles, paper, chemicals, fabricated metals, and vehicles, while several other sectors respond only modestly. The two shocks differ primarily in the timing of the contraction. Flow-supply shocks depress activity on impact in the exposed sectors, and these declines largely unwind within a few months. Supply-security news shocks instead generate contractions that build more gradually, particularly in chemicals and paper, consistent with the delayed price pass-through documented above.

G Contribution of oil shocks to the historical decomposition of inflation

This appendix reports the contribution of oil supply shocks, identified following Känzig (2021), to the historical decomposition of year-on-year headline inflation. Figure G51 shows that these shocks contributed to the post-pandemic inflation surge in both regions, but played a more prominent role in the United States. This contrasts with gas supply shocks, whose contribution was substantially larger in the Euro Area.

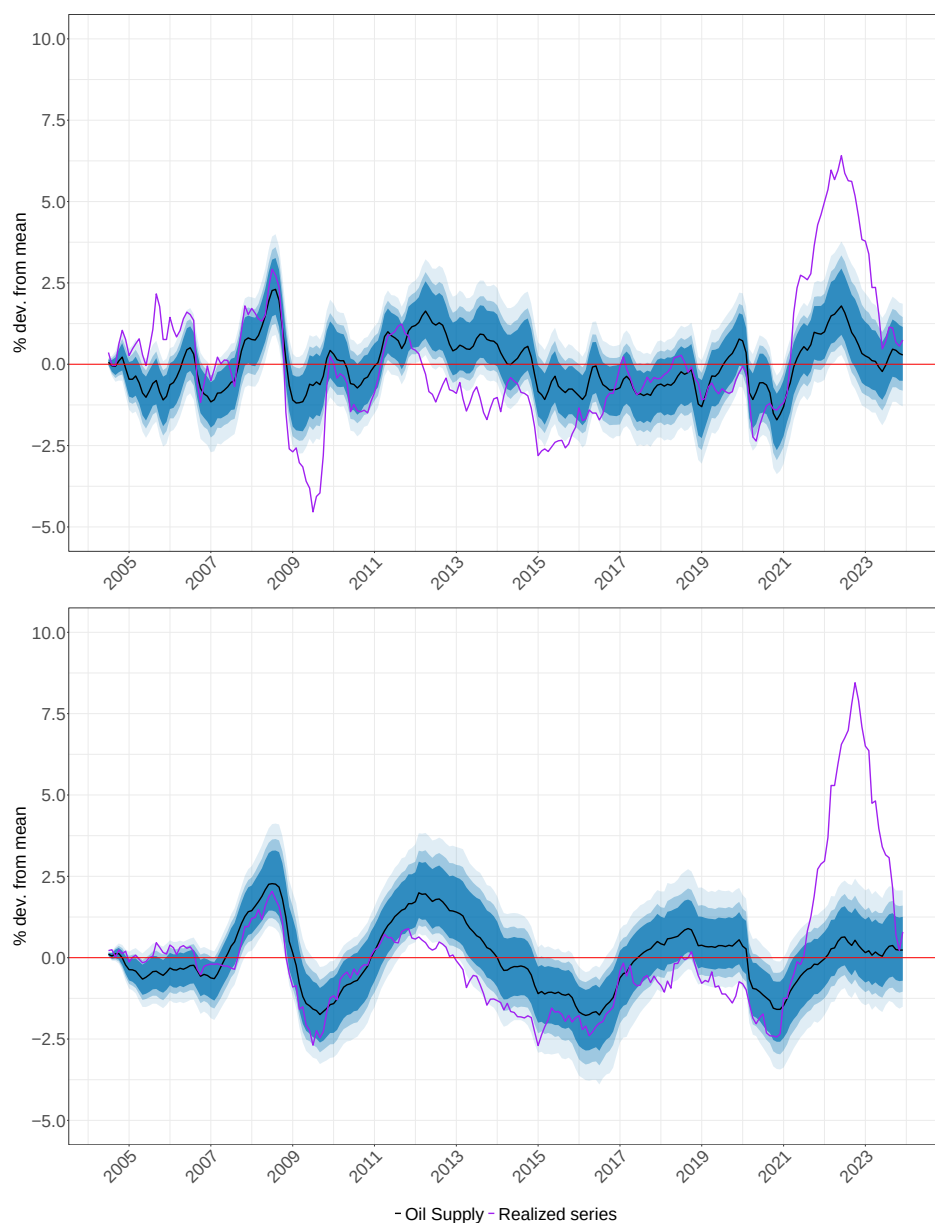


Figure G51: *Historical decomposition of headline inflation: contribution of oil supply news shocks*

Notes: Estimated contribution of oil supply news shocks identified following Känzig (2021), shown by the solid black line; United States in the top panel, Euro Area in the bottom panel. Contributions are expressed as percentage deviations from the mean. Realized headline inflation is shown in purple.

H Comparison with an alternative gas supply instrument for Europe

Alessandri and Gazzani (2025) recently proposed an instrument for identifying gas supply shocks in Europe, employing a high-frequency approach similar to the one used in this paper. In this Appendix, we discuss both the technical and substantive differences between their instrument and ours.

A first distinction lies in the data source: their instrument is based on the ENDEX futures series, whereas we rely on TRPC continuation values. Although the two series are highly correlated, the TRPC data covers a longer time span, enabling us to construct an instrument over an extended sample. Specifically, their instrument spans the period from 2010M1 to 2022M11, while their estimation sample extends from 2000M1 to 2022M11. By contrast, our instrument covers our full estimation sample, from 2004M1 to 2023M12.

Another important difference concerns the treatment of futures contracts. Alessandri and Gazzani (2025) focus exclusively on surprises in the front-month contract. In contrast, we follow the standard approach in the literature (see e.g. Känzig, 2021) and extract the first principal component from surprises across multiple maturities, up to one year. While Alessandri and Gazzani (2025) include a robustness check in which they incorporate 1-, 2-, 3-, and 12-month futures, their baseline specification does not. We argue that including longer-dated contracts is important, as gas supply shocks are also transmitted via expectations about future supply conditions, as we have shown. Moreover, as discussed in Section 3.2, using the principal component allows us to isolate a cleaner signal, especially in cases where futures with different maturities respond heterogeneously to market-relevant news. Figure H52 compares the two gas supply shock measures. Despite some overlap, the correlation between the two instruments is relatively modest, at 0.30.

Perhaps the most substantial difference between the two instruments emerges when comparing the results they produce. When we use the instrument proposed by Alessandri and Gazzani (2025) in our baseline specification, we obtain the impulse response shown in Figure H53. Unlike the responses generated using our instrument, the results on quantities indicate a positive impact on gas demand and no decrease in gas supply up to at least a year. This finding is somewhat puzzling, as the instrument is intended to capture purely supply-side disruptions. In our specification, the first-stage F statistic associated with their instrument is also large, but falls below the conventional threshold of 10 when estimated robust to heteroskedasticity.

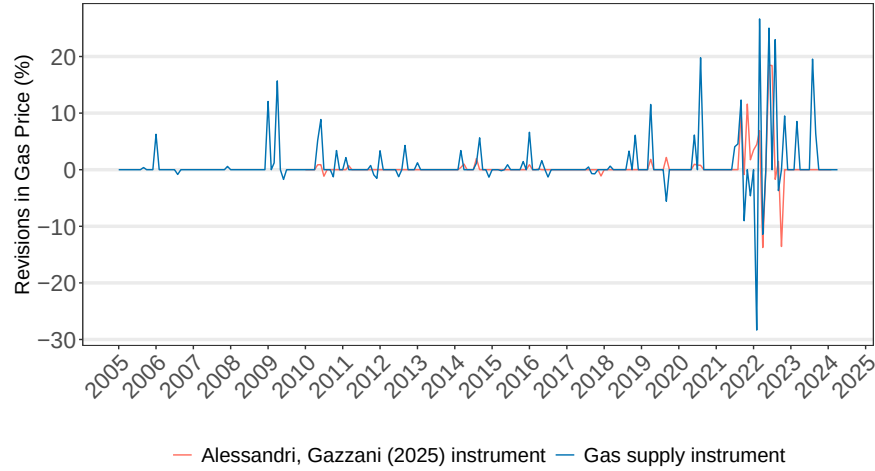


Figure H52: *Comparison of the two supply instruments for Europe*

Notes: The correlation between the two instruments is 0.30.

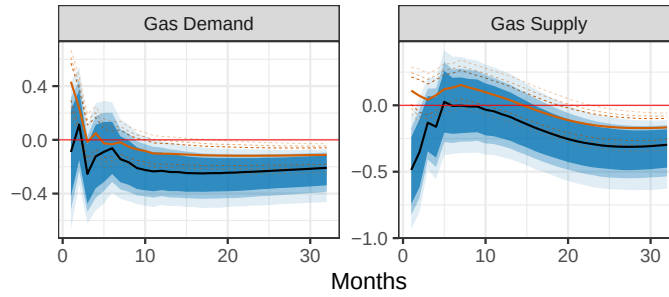


Figure H53: *Responses of gas demand obtained by using the Alessandri and Gazzani (2025) gas supply instrument in our baseline specification*

Notes: The Fstat is 50.8 and the Robust F is 7.9. The black solid line with blue shaded credible bands represents our baseline responses, and the red solid and dashed lines the results obtained with the alternative instrument.

To further investigate the source of these results, we examine the correlation between their instrument and a range of established shocks from the literature (see Table H11), following a similar diagnostic approach to that used for our instrument in Appendix C.3. The analysis reveals that their instrument is primarily correlated with demand-side indicators, including the oil demand shock identified by Baumeister and Hamilton (2019), the FF4 monetary policy shock from Gertler and Karadi (2015)—with which the correlation is highly significant—and our own temperature-based gas demand shock.

In addition, Table H12 indicates that, while we can reject the null hypothesis that their instrument is Granger-caused by a broader set of macroeconomic variables, there is evidence of Granger causality from its own lags and the real spot price of gas.

Source	Shock	ρ	p-value	n
This study	Temperature demand shock	-0.10	0.23	155
Kilian (2009)**	Oil supply	-0.04	0.61	155
Kilian (2009)**	Aggregate demand	-0.09	0.24	155
Kilian (2009)**	Oil-specific demand	0.01	0.92	155
Baumeister and Hamilton (2019)*	Oil supply	-0.02	0.80	155
Baumeister and Hamilton (2019)*	Oil demand	0.13	0.11	155
Känzig (2021)**	Oil supply expectations	-0.11	0.17	155
Caldara et al. (2019)*	CCI oil supply	-0.23	0.05	72
Altavilla et al. (2019)*	Target monetary policy (EA)	0.01	0.88	155
Miranda-Agrippino and Nenova (2022)	Target monetary policy (EA)	0.01	0.90	135
Jarociński and Karadi (2020)	Information median monetary policy (EA)	0.19	0.02	155
Gertler and Karadi (2015)	FF4 monetary policy (US)	-0.19	0.31	30
Miranda-Agrippino and Nenova (2022)	Target monetary policy (US)	-0.04	0.71	114
Bloom (2009)**	VXO-VIX	0.03	0.75	155
Gilchrist and Zakrajšek (2012)*	Corporate credit spread index	-0.09	0.27	155
Caldara and Iacoviello (2022)*	Geopolitical risk index	0.08	0.33	155

Table H11: *Correlation of Alessandri and Gazzani (2025) instrument with other shocks*

Notes: This table presents the correlation coefficients (ρ) and p-values for the gas supply instrument in relation to a variety of economic shocks from the literature. The p-values correspond to two-sided tests for the null hypothesis of no correlation.

*Extended by the original authors beyond the original sample used in the published paper.

**Extended by the authors of this study.

Variable	p-value
Instrument Lags	0.00
Gas price	0.01
Oil price	0.78
Gas demand	0.80
Gas inventories	0.35
Headline inflation	0.53
Industrial production	0.45
Financial volatility	0.16
Interest rate	0.99
Nominal exchange rate	0.74
Stock market (STOXX50E/SP500)	0.29
Supply Chain Bottlenecks (GSCPI)	0.71
Real economic activity	0.64
Joint Test	0.33

Table H12: *Granger causality tests*

Notes: The table presents the p-values obtained from Granger's causality tests of the gas supply surprise series using the set of variables included in our baseline specification, expanded with financial and real activity variables. To conduct standard inference, the series are rendered stationary by taking first or second differences as required. The analysis includes 6 lags and a constant term.

These findings suggest that the instrument proposed by Alessandri and Gazzani (2025) may reflect some demand-side components of gas price variability, rather than isolating a purely supply-driven shock. This could arise from the selection of gas-market news used to construct the instrument, some of which may be directly related to demand developments or confounded by concurrent demand-side factors.

For instance, news from October 21, 2022, refers to Italy’s approval of a new LNG terminal and the start of LNG exports from Mozambique to the EU. This coincided with an unseasonal heatwave in Europe. Further analysis of gas-market news around that date suggests that price movements were primarily driven by changes in demand due to weather conditions.

Another example is the news from May 16, 2022, reporting on Norway’s potential to expand gas exports to the EU and the approval of new LNG projects in the United States. In our view, this news lacks clear evidence of exogeneity and may instead reflect supply responses to rising demand, rather than an exogenous movement in supply.

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